
POTHOLE DETECTION AND ROAD CONDITION REPORTING AND MAPPING WITH IMAGE PROCESSING USING YOLOV11

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DEDICATION

We, the researchers, humbly dedicate this thesis to our beloved family, whose unwavering love, support, sacrifices, and constant encouragement have guide us throughout this journey. To our mentors, we extend our deepest gratitude for your guidance which have inspired us to pursue this study. We also dedicate this work to our friends, who stood by us with understanding and motivation during the sleepless nights of coding and revisions.

Above all, we offer this success to God Almighty, for granting us wisdom and perseverance to conquer this milestone.

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ABSTRACT

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Road infrastructure faced significant challenges, particularly in maintaining road quality and ensuring the safety of motorists and pedestrians. Traditional manual inspection method of assessing the road condition, were time-consuming and often inefficient, particularly in rural areas where resources and manpower for infrastructure maintenance were limited. This study aimed to develop a real-time pothole detection and road condition mapping system using image processing and the YOLOv11 algorithm. A dataset of 856 road images manually annotated with potholes classified as minor, medium, and major was used, with data augmentation techniques applied to improve model robustness. The YOLOv11 model was trained using a staged transfer learning approach and evaluated using mAP, precision, recall, and F1-score. The trained model was converted to TensorFlow Lite and deployed on a mobile platform, with GPS integration for road condition mapping. The results showed that the system achieved an mAP@0.5 of 39.99%, precision of 42.0%, recall of 47.7%, and an F1-score of 44.6%, with better detection performance for medium and major potholes. The successful implementation of the YOLOv11 algorithm emphasized the potential of deep learning models to achieve balance between accuracy and processing efficiency.

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CHAPTER 1

INTRODUCTION

This chapter discussed the background of the problem, problem statement, scope and limitations, objectives, the significance of the study, and the project dictionary. This chapter aims to give a general description of the entire project report, enabling the readers to have a brief understanding of the importance and goal of the study.

Background of the Problem

Road infrastructure faced significant challenges, particularly in maintaining road quality and ensuring the safety of motorists and pedestrians. The World Health Organization (WHO) revealed that the leading cause of death globally are road traffic incidents [8]. To support this, the Philippine Statistic Authority (PSA) stated that in 2022 road traffic accidents caused up to 12,000 fatalities and one of the contributing factors for this was the poor road condition [3]. Aside from safety, this issue also affected the country's economy as the Department of Transportation estimates that road accidents cost the Philippine economy approximately 105 billion pesos annually [3]. Included to this are the healthcare costs, damaged property, and productivity loss. This proved that roads play a crucial role in growth of a country which aligns to the World Bank's definition of roads as the "arteries through which the economy pulses" [11].

Potholes were common issues across the country, especially in rural or less developed areas. There are a combination of factors that caused these road defects, including heavy rainfall, poor drainage systems, and the constant passage of overloaded vehicles. The consequences of these road conditions were severe as potholes does not only caused discomfort to the drivers but led to damaged vehicles, higher maintenance costs for public and private

transportation but more importantly, it contributed to traffic accidents which raised a significant risk to public [11]. This may not be noticed by everyone, but road conditions in a community affected people's lifestyle as it brought people home to their families and individuals to where they wanted to go.

Despite the need for regular road maintenance, there were still challenges in addressing these issues effectively since traditional methods of assessing the road condition, which is manual inspection, were time consuming and often inefficient particularly in rural areas where resources and manpower for infrastructure maintenance were limited. There was also a lack of real-time data on road conditions which made it difficult for the Department of Public Works and Highways (DPWH) to strategically prioritize and allocate resources for road repairs. The advancement of technology in the fields of computer vision and artificial intelligence have opened new opportunities for addressing these challenges as YOLOv11 the latest version of You Only Look Once (YOLO) algorithm showed remarkable performance and speed in real-time object detection [12]. By utilizing this, a system was able to detect potholes in real-time using a mobile phone's camera. In addition, integrating GPS data with image processing for creating a visual road condition map proved to be a valuable tool for the authorities to conduct more targeted and timely road maintenance. Compared to the traditional manual inspection method, this approach offered a more efficient solution.

Considering the impact of poor road conditions on the public safety, the economy, and the infrastructure of the community, this study aimed to develop a real-time pothole detection that can enhanced driver safety and a road condition mapping system, which helped the authorities locate potholes quickly. By utilizing YOLOv11 for image processing and integrating GPS data, the proposed system sought to provide a cost-effective solution to monitor road conditions, improve road safety, and assist local authorities to make a data-driven decision for road maintenance, contributing to better infrastructure and public safety.

Statement of the Problem

Road conditions played a critical role in ensuring the safety and effectiveness of transportation systems. However, many roads, especially in rural areas, experienced neglect due to oversight and insufficient monitoring. Potholes caused by factors such as extreme weather conditions, poor drainage systems, and high vehicle loads, continued to be a persistent problem. These road defects not only led to vehicle damage and costly repairs, but also contributed to traffic congestion and increased accident risks.

Regardless of these dangers, current methods for assessing road conditions relied heavily on manual inspection, which was labor-intensive, time-consuming, and evidently, inefficient. The lack of technology to road condition data made it strenuous for the authorities to promptly identify and issue the necessary repairs. Additionally, limited resources in rural areas made it difficult for local authorities to conduct frequent and systematic inspections.

Objectives of the Study

The following sections outlined the key objectives of this study.

General Objective

The objective of this study is to develop a pothole detection system using YOLOv11 algorithm and generate a road condition map that will marked the locations of the potholes identified by the system.

Specific Objectives

To achieve the goal of developing a real-time pothole detection and road condition mapping system, this study had the following specific objectives:

1. To develop a real-time pothole detection system using YOLOv11.

2. To evaluate the performance of the proposed detection model using mAP, F1-score, and latency metrics.
3. To integrate GPS data with the pothole detection system to generate a road condition map.

Significance of the Study

This part of the study discussed the importance and purpose of the study. This study held significant value for various stakeholders such as:

Camarines Sur Polytechnic Colleges (CSPC): As an academic institution, CSPC benefited from this research by integrating its findings into relevant courses and fostering innovation in the field of image processing and object detection, providing valuable learning opportunities for the students and faculty. The study can also served as a basis for future research and development efforts in related areas.

Vehicle Owners and Commuters: This study was beneficial to the motorists and commuters providing information on road conditions. The system can helped drivers stay informed on route choices, adjust speed, avoid potholes, and reduce the risk of accidents fostering safer driving experience.

Local Government Units (LGUs): The developed system directly benefited the government providing information for road maintenance and resource allocation. The system assisted LGUs in identifying and prioritizing road maintenance efforts, leading to enhanced road safety.

Department of Public Works and Highways (DPWH): The study aligned with the department's goals streamlining the workflow of the DPWH road maintenance teams. By providing precise location data of the potholes and road damages, the system helped maintenance staff immediately locate and repair potholes, improving the efficiency of maintenance operations.

Researchers: The researchers involved in this study enhanced their research skills in written, oral, data collection, and model training, as well as gained valuable experience and refined their abilities in developing and deploying real-time image processing systems.

Future Researchers: The result of this study provided a foundation for future researchers and the growing body of knowledge on computer vision applications in road condition monitoring, particularly in the application of deep learning algorithms and mobile technologies. This study is helpful as this can be used as a resource for future research. The findings of this study also serve as a reference to those who are interested in engaging in a similar problem. The insights gained from this provided them with similar aspects for future investigations.

Scope and Limitation

The study aimed to develop a system that can detect potholes and upload it for mapping system. The research team looked for a way to address the challenges caused by the road conditions, especially potholes, which are usually because of the factors such as intense rainfalls, extreme heat, and poor maintenance. The system utilized a mobile phone's camera to capture road footage. The YOLOv11 (You Only Look Once) object detection algorithm then analyzed the captured potholes. By combining the GPS data with the processed images, the system generated a visual road condition map that marks the locations of the detected potholes in the area. The users can also report potholes by uploading them, enhancing the accuracy and coverage especially in areas where automatic detection might have been limited.

This research have several limitations that influenced the accuracy and use in the real-world of the proposed system. The pothole identification of the system depends on several environmental factors, such as lighting, weather condition, and the quality of the road surface. Moreover, this study focused solely in detecting potholes and does not extend to

assessing other issues such as erosion, bumps, or debris. Furthermore, the data collection depended on mobile phone cameras (Android and IOS), which might result to lower accuracy compared to high-end devices. The latency of the image capturing and GPS alignment may have also reduced the precision of the generated road condition maps. Lastly, the design of the system highlights the offline functionality of the compact model ensuring the usability in areas with poor internet access.

Project Dictionary

The Project Dictionary contained the technical terms that defined the conceptual and operation of this study:

- **GPS (Global Positioning System):** A satellite-based system that provides location data, enabling the mapping of road defects by integrating spatial information with detected potholes [6]. Its data was vital for creating detailed road condition maps, as it precisely located and prioritized repair.
- **Image Processing:** The use of a digital computer to process digital images using computer algorithms and mathematical models [4]. The specifics of image processing included resizing the road footage into smaller snapshots, applying noise rejection for a clearer image, and improving contrast for accurate detection of the pothole.
- **Major Potholes:** Potholes classified as minor or non-threatening to both pedestrians and vehicles [1]. Used to describe that when a small pothole, when driven over, causes little to no shaking in the vehicle.
- **Medium Potholes:** Potholes that pose a potential threat to pedestrians but are minor or non-threatening to vehicles [1]. This describes that this pothole makes the vehicle noticeably jolt or shake when passed, but the vehicle can still pass safely without damage.

- **Minor Potholes:** Potholes that present a significant risk to both pedestrians and vehicles [1]. It is used to describe the severity of a large pothole that might make a driver stop upon seeing it, or if driven over, could cause scratches or damage to the vehicle.
- **Pothole:** A hole in a road surface that resulted from gradual damage caused by traffic and/or weather. Potholes typically form when water infiltrated the underlying soil structure, weakening it, and traffic loads stress the pavement beyond its breaking point, eventually leading to hole formation [5]. In this study, potholes are the main object our system aimed to detect using the YOLOv11 object detection algorithm.
- **Pothole Detection:** Identification of potholes using image processing and object detection techniques. This process employed algorithms to analyze images or video streams to locate and classify potholes [7]. We refer to pothole detection as the process of capturing and the road condition using mobile phone's camera and integrating an object detection algorithm to identify potholes.
- **Real-time:** Refers to a process that occurs in a short period of time, and the system's capability to provide immediate responsiveness [2]. In this study, it is the ability of the application to capture input data, process it using the model, and provide feedback to have an immediate result during an actual operation.
- **Road Condition:** The status or condition of road surface, with consideration of weather conditions, traffic maintenance, and construction matters [13]. It includes the overall quality and condition of roadways and the usability of the roadway for drivers and pedestrians. This is associated with this study through images and data captured via mobile phone cameras, that evaluated the road conditions and classified sections of the presence of defects on the road surface.

-
- **Road Condition Mapping:** Visualizing road defects on a digital map using image processing and GPS data. This process created a comprehensive visual representation of road conditions, including the location and severity of potholes and other defects [10]. In this study, road condition mapping is defined as producing a georeferenced map using GPS data and processed road video, visualizing the detected potholes.
 - **Road Condition Reporting:** A process of documenting the current condition of the road that includes any damages, irregularities, and potential hazards on the drivers and pedestrian. This system generates automated reports by analyzing the road images and/or videos, as well as the GPS coordinates to provide users with information about the detected potholes and defects.
 - **YOLOv11:** Current latest version of You Only Look Once(YOLO) algorithm. A deep learning algorithm for real-time object detection. It was known for its speed and accuracy in identifying objects with image or video streams. Its ability to process data quickly is essential for applications requiring immediate feedback [9]. YOLOv11 was an algorithm that was implemented to the proposed system to detect potholes and process the data in order to provide real-time feedback regarding the detected pothole.

NOTES

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CHAPTER 2

RELATED LITERATURE AND STUDIES

This chapter contained reviews of finished research papers, methodologies, conclusions, and ideas relevant to the present study. The information and papers included in this chapter contribute in familiarizing relevant details and acted as the roadmap for the researchers as we develop and improve the current study.

Review of Related Literature and Studies

Presented in the succeeding sections are a comprehensive review of related studies and articles thoughtfully organized into specific topics. The related studies and articles highlighted the evolution of pothole detection, road mapping, and object detection algorithms, demonstrating how automation and improved technologies are transforming road maintenance and safety.

Pothole Detection

Potholes were clear signs of neglected roads, revealing a deeper structural problem. When a vehicle hits a pothole, it not only disrupted the journey but also caused serious damage to your vehicle's wheels, tires, and suspension system, all at the cost of repairing it [6]. In an early attempt at automation of pothole and object detection relied on manual visual inspections by certified engineers or road maintenance personnel were used, with the use of hand-crafted features and rule-based algorithms. Though effective in detecting defects, manual methods were inefficient and labor intensive, particularly when applied to large-scale road networks [16, 10]. Such approaches were also time-consuming and being subjective, produced inconsistent results, which becomes especially important when

the area to be evaluated was large and the infrastructure is dense and complex and thus requires more efficient and objective evaluation techniques.

In line with this, the study by Bidve et al. [2024] stated that traditional methods of pothole detection were ineffective, and there was a need for a new solution [3]. The transformational solution proposed in this study integrated cutting-edge technologies specifically computer vision and machine learning, intending to enhance roads, increase safety, and streamline maintenance operations.

Ma et al. [2022] paper stated that road imaging with a focus on pothole detection had progressed over the last twenty years, moving from classical 2-D image processing techniques combining edge detection, and thresholding approaches, to move advanced 3-D point models to sophisticated depth information approaches as laser scanners and Microsoft Kinect (depth cameras) for developing 3-D point clouds modeling. Early approaches were limited in robustness and real-time processing, but the advent of machine learning and deep learning, especially Convolutional Neural Networks (CNNs), has transformed the field, paving the way for accurate pothole detection utilizing multi-modal semantic segmentation. Even with these advancements, there are still several challenges that remain, such as the reliance on large amounts of labeled data for training and the constraints on processing in real-time. This highlights the transition from traditional algorithms to AI-driven solutions, presenting CNNs as the best choice for the future road condition evaluation systems [18].

Moreover, as current techniques for identifying potholes typically rely on labor-intensive manual image processing and vibration-based sensors, which were often slow. To address this issue, Addanki and Lin [2023] proposed the use of recent advancements in deep learning particularly Convolutional Neural Network (CNN) leveraging YOLOv8 for evaluation of advanced model detection. With the focus on real-time performance and computational efficiency led to impressive results, its findings achieved a 91.1% mAP with an inference speed of 8.8 milliseconds per image, highlighting the effective transition from manual in-

spections to AI-powered real-time detection that marked a significant enhancement to road safety technology [1].

Ahmed [2021] paper incorporated a dilated convolution-based deep learning model for smart pothole detection to address the limitations of the standard CNNs that failed to capture a wider context without leading to a substantial increase in computational cost [2]. The study demonstrated that dilated convolutions are used to increase the size of the receptive field, which resulted in the model being able to detect potholes of different sizes with higher accuracy.

Consequently, Guerreiri et al. [2020], conducted a case study concerning several urban roads in Palermo, Italy, and the different distress types, extensions, and proliferation of such highways, and road areas crossed by railway lines. With the various Deep Learning techniques used, specifically the YOLOv3 to inspect the pavement distress system the model obtained a 98.7% for classifying distress on different types of pavement areas successfully implemented a real-time and efficient detection system [9].

As per Xu et al. [2023] the recent advancements in the realm of automated road-condition monitoring, particularly emphasizing sophisticated systems designed for sensor-based pothole detection. This study emphasized how important the VIDAR (Visual and Depth-Aware Radar) technology in fixing the present challenges related to infrastructure maintenance. It found that the modified VGG16 networks, improved with particularly optimized dilation rates, achieving an excellent accuracy in pothole detection with a success rate of around 94.8% [29]. By thoroughly evaluating both traditional and modern methods closely, it shows a big gaps that proved why the new VIDAR approach works well, highlighting its potential to improve road safety and make the repairs faster and better.

In the local literature aspect, Fortin et al. [2024] proposed a novel solution called the 'EyeWay' system, which employed advanced computer vision, AI-on-edge technology, and visualization map techniques, as well as the prototype of AIoT focused on pothole detection and road infrastructure monitoring. The study also employed the CRISP-DM framework

to develop, deploy, and evaluate the system. With an inference time of 1.76 milliseconds using YOLOv8m, the model achieved 91.8% mAP and 82.95% for testing which expects substantial improvements for a sustainable transportation infrastructure [7].

Road Mapping

Road mapping had become a critical component of transportation systems nowadays. Integrating the Global Positioning System (GPS) and Geographic Information System (GIS) enabled the development of an accurate and dynamic road network by combining real-time geospatial data with advanced spatial analysis. GPS provides precise geospatial data while GIS facilitates data visualization, pattern identification, and decision making for infrastructure management, which are used in several studies to address challenges such as signal obstruction in urban areas, and limited infrastructures in rural areas. Using technologies such as GPS and GIS for road mapping of potholes helped with having an accurate detection and timely repairs, which reduced the road accidents and helped the authorities enhance the road quality.

The performance of GPS technology in terms of mapping has significantly improved over time. The article of Jiang and Zhao [2023] analyzed the application of GPS in Engineering Surveying and Mapping. It reveals that GPS technology provided precise positioning data that is essential for accurate mapping. By connecting with Real Time Kinematic enables GPS to obtain real-time and effective information. Moreover, the article highlighted the relatively simple operation of GPS that reduces the proficiency requirements for surveying and mapping making it more accessible and efficient [13].

Agricultural tractors were not able to maximize their power due to mismatched tools and poor driving skills which led to wasted energy. To address this problem, Kushwah et al. [2024] developed a low-cost digital system that monitored and displayed the tractor's real-time performance with accurate location data. The developed system utilized the U-blox Neo-7 series GPS module that offered 10Hz signal reception rate which ensured the

accurate positioning of the field equipment. Moreover, the module's compact design was ideal for deployment in mobile applications particularly in fields where space and power efficiency are important. The integration of GPS-assisted GIS mapping in their study successfully created detailed spatial maps, facilitating site-specific analyses and improving the operation of agricultural tractors [17].

Dwiandharu et. al [2024] developed a system that detected and mapped road damage using an accelerometer to record acceleration values and integrate GPS to visualize the mapping of road damage. In their study, GPS was evaluated by visualizing the reading results with an application and comparing the actual location with the position shown by the GPS based on the given data. Furthermore, their study revealed that the average distance between the actual hole position with GPS coordinates reached 5.7 meters [20].

Desai and Zala [2024] conducted a study that addressed the problem of road safety by developing an IoT based system that was capable of identifying potholes in real-time and notifying the drivers and authorities in order to take precautions. GPS is integrated to collect data on road conditions, provide real-time location and ensure the accuracy of the marked road irregularities in the map. The study concluded that through the GPS data real-time monitoring is achieved within a predefined range [5].

Smartphones were widely used by almost every individual, may it be for work, studying or just mere entertainment. However, the article of Kersten [2020] introduced a different perspective beyond just being perceived as a personal tool, it explored the use of smartphones for professional mapping. Since modern smartphones were already equipped with various sensors, high resolution cameras, GNSS receivers, accelerometers, gyroscopes, magnetometers, and barometers which are the requirements to enable data acquisition for mapping purposes. In addition, the article stated that smartphones are already being used as a supporting mapping tool and as the technological development of smartphones continues, "Therefore, the use of smartphones as a professional mapping tool is undoubtedly set to increase." [14].

To fill in the gaps and limitations of Google Maps features, Ghifari and Sutarman [2024] developed an application named “StellarPath”. It was a mobile-based digital map location marker that used Location Based Service (LBS) method as an alternative mapping solution. The application focused on developing offline friendly mapping and manual route creation for users. The StellaPath’s testing rate is 92.5% highlighting its effectiveness in allowing the users to create and save a route, as well as managing location markers offline. Their application offered a flexible and personalized navigation tool compared to other existing navigation applications [8].

Mapping of potholes and pavement distress are important for the authorities who were responsible for maintaining the roadways. To improve the time consuming manual classification and monitoring of the potholes, Ravi et al. [2020] proposed a methodology that mapped the potholes using ultra-high accuracy LiDAR-based 3D point cloud units on a wheel-based mapping system that is driven at a speed of 40-50 mph. Using the high-grade system shows an accuracy of $\pm 1-2$ cm, this indicated that the high-grade mobile mapping system significantly improved pothole detection and the software for mapping helped in having a better visualization, reducing the inspection time and enabling proactive road maintenance [22].

Advancements in Object Detection Algorithms

Object detection has undergone a rapid change over the years, with advancements in machine learning (ML) and computer vision (CV). The concept of object detection, however, was based earlier on traditional methods such as edge detection, template matching, and haar-like features, and thus was not entirely accurate nor efficient [27]. These methods were mostly hand-crafted based on low-level image features to identify objects in images.

The YOLO(You Only Look Once) algorithm was introduced in June 2015 by Joseph Redmon et al. This algorithm superseded Faster R-CNN, one of the leading object detection algorithms at the time. The introduction of YOLO was groundbreaking. It became popular

because it performed object detection as a single regression problem; this made it faster even when detecting multiple objects at once, achieving real-time speed even on regular processing power. Each successive version improved speed, accuracy, and efficiency.

YOLOv1 offered a single-shot object detection that easily achieved higher speeds than region-based methods. However, small-object detection and localization errors were its Achilles' heel. Redmon et al. [2016] proved that YOLOv1 could process images in real time, but it fell short with its precision in complex scenes [23]. YOLOv2, also known as YOLO9000, fixed that problem by adding batch normalization, anchor boxes, and multi-scale training, making it more accurate and robust in real-world applications. This updated version uses Darknet-19 for its backbone, a variant of the VGGNet architecture with simple progressive convolution and pooling layers. YOLOv2 scored a better mean average precision (mAP) of 76.8 mAP at a pace of 67 FPS on the PASCAL VOC 2007 dataset. Its mAP increased to 78.6 at a pace of 40 FPS, breaking previous state-of-the-art methods by speed as well as accuracy, even outperforming Faster R-CNN with ResNet and SSD, as described in the research by Redmon and Farhadi [2016] [24].

The introduction of YOLOv3 further improved multi-scale detection with the aid of a feature pyramid network, thus improving the detection of small, medium, and large objects even more. This version incorporated changes that switched from softmax to independent logistic classifiers to maximize flexibility in multi-label tasks. YOLOv4 had its architecture optimized with CSPDarknet53 as the backbone, PANet as the neck, and YOLOv3 as the detection head, a significant change detailed in the documentation of Ultralytics [2020] [26]. Additionally, enhanced data augmentation was employed, proposing to combine bag of freebies and bag of specials to acquire more accuracy but not cause delay in inference time. The research of Bochkovskiy et al. [2020] discovered that in terms of accuracy and speed, YOLOv4 is better compared to Faster R-CNN and RetinaNet [4].

Next YOLO versions, from YOLOv5 to YOLOv8 aimed at developing lightweight designs, better detection accuracy and faster real-time performance. Its speed improved and

made it easier to use on smaller devices such as edge computers. For instance, unlike the original YOLO, YOLOv5 used a smarter setup which is the EfficientDet. This helped identifying things more accurately and has better generalization across many categories. To have better object localization and scene understanding YOLOv9 and YOLOv10 added transformer-based features. Adaptive feature is also added in this version to make detection more robust in cluttered environments. The huge step in YOLOv10 was the introduction of C3k2 block, this improved feature combining while reducing the computational overhead. This makes the YOLOv10 still accurate even in resource-constrained environments, which is ideal for edge use, as stated by Wang et al. [2024] [28].

The most recent version, YOLOv11, managed to combine the efficiency with previous versions and the latest deep learning implementations of anchor-free detections and attention methods. According to Khanam and Hussuain [2024], the version further developed the framework with the integration of C2PSA (Cross Stage Partial with Spatial Attention) blocks, this greatly improved its spatial awareness by allowing the model to focus more on significant parts within an image [15].

Recent researches demonstrated YOLOv 11's advancements across several domains. Some key innovations in addition to the previously described advancements, YOLOv11 features a transformer backbone, large kernel convolution, dual label assignment, dynamic head design, NMS-free training, and modules for partial self-attention (PSA). These developments significantly improved performance and effectiveness, allowing the model to be applicable in various kinds of applications [19].

In terms of performance, YOLOv11 has a mean average precision (mAP) of 61.5% at 60 frames per second (FPS) with a relatively low parameter count of 40 million, outperforming its predecessors in terms of speed and accuracy. As such, YOLOv11 becomes a highly opportunistic approach for several applications, including autonomous vehicles, healthcare, retail, and more. By leveraging these capabilities, the model stood out as a versatile and effective solution across various industries [21].

Jegham et al. [2024] carried out a thorough benchmark research comparing YOLOv3 and YOLOv11 across multiple datasets such as traffic signs, wildlife, and maritime vessels. With an average inference time of just 2.4 ms and an accuracy of 0.81 mAP on the African Wildlife dataset, the YOLOv11 consistently remains the best in terms of speed and accuracy [12]. The study highlighted its effectiveness and compactness in real-time environments, which makes it ideal for both cloud and edge deployment [12].

The effectiveness of YOLOv5, YOLOv8, YOLOv9, YOLOv10, and YOLOv11 in terms of power equipment object detection was a study conducted by He et al. [2024]. The results showed that the mean average precision (mAP) for power equipment on a public dataset was 54.4%, 55.5%, 43.8%, 48.0%, and 57.2%, respectively. The evaluation was dominated by YOLOv11 because of its improved performance in detection, higher recall rates, and fewer false positives [11].

The application of YOLOv11 in recognizing fruitlets in orchard environments, which area is often disrupted by occlusion and various lighting conditions was explored by Sapkota et al. [2024]. In order to identify immature green apples, their research compared YOLOv8, YOLOv9, YOLOv10, YOLOv11, and YOLOv12. The results revealed that YOLOv12 had the highest recall rate of 0.90, while the YOLOv11 had the fastest inference speed with 2.4ms, leaving the other models behind having 4.1ms, 11.5ms, 5.5ms, and 4.6ms respectively [25]. These improvements suggest that any further advancements is vital in applying real-time object detection.

Evaluation of YOLOv11

Based from He et al. [2024] study they used Mean Average Precision (mAP) to systematically evaluate the performance of YOLOv5, YOLOv8, YOLOv9, YOLOv10, and YOLOv11 on power equipment object detection. The usage of public dataset for the experiment returns demonstrated a mAP of 54.4%, 55.5%, 43.8%, 48.0%, and 57.2% with YOLOv11 outperforming the other methods by achieving the highest detection perfor-

mance. These findings highlighted the model's great detection performance when it comes to accuracy [11].

Jegham et al. [2024] examined various integration objects, including mapping and spatial datasets, in multi-domain environments. YOLOv11 achieved consistent mAP across domains and maintained an inference speed of up to 2.4ms, striking an optimal balance between accuracy, efficiency, and model size. These benchmarks indicated us that YOLOv11 was best used in a GPS-integrated road mapping system when fast detection and geolocation are needed to update road condition maps in real time [12].

YOLOv11 managed to outperform the previous versions in several performance metrics, particularly latency and accuracy, revealing significant improvements in accuracy with 61.5% mean average precision (mAP), and speed by reducing 25-40% latency compared to YOLOv10, making it ideal for real-time applications [21].

Synthesis of the State-of-the-art

From the research reviewed, there was a definite transition from traditional pothole detection methods to those of AI- driven, focusing on automation and possibly real-time monitoring. Traditional methods, especially human visual inspections that is carried out by local road maintenance personnel, have turned out to be inefficient and inapplicable especially in large areas. Modern AI technologies, in particular, CNNs, have now contributed greatly in terms of the improvement in the areas of accuracy and speed in detection. Examples given include **Addanki et al. [2023]** and **Guerreiri et al. [2020]**, noting that the transition from traditional methods to real-time detection techniques using deep learning models like YOLO, improves both efficiency and accuracy in pothole detection. The transition from manual inspection to AI-powered systems was commonly mentioned in several studies as they emphasized the basic necessity for systems with increased speed and accuracy.

Despite the shared adoption of AI-powered solutions, variations naturally arose in different methodologies employed in utilizing the technologies. **Addanki et al. [2023]** analyzed YOLOv8 with real-time performance in mind, having attained astonishing results of 8.8 ms/image inference time. **Ahmed [2021]**, based on the detection of dilated convolutions with a specific emphasis on detection of potholes of different sizes without incurring a huge computational cost, highlighted a contrasting approach. These differences highlighted the focus of each study on different aspects of AI models — speed and efficiency in the case of Addanki and Lin, and accuracy and computational efficiency in Ahmed’s work.

GPS played a pivotal role in modern pothole detection and road mapping. Various studies have brought GPS into the picture to map the precise position of potholes for faster repair, such as **Dwiandharu et al. [2024]** and **Desai & Zala [2024]**. With the integration of GPS, road mapping systems have been improved, making these systems more accurate and efficient. However, even though LiDAR systems have been conducted in another study by **Ravi et al. [2020]** that give better results with $\pm 1\text{-}2$ cm, this is far better than GPS. LiDAR’s ability to construct 3D point clouds constituted a more detailed and accurate map of road conditions compared to GPS, which is better employed for mapping general locations.

A key difference between the systems reviewed is the focus on high-precision systems versus mobile-based solutions. Applications like StellarPath, **Ghifari & Sutarman [2024]**, provided a flexible and offline-friendly mapping solution for potholes, and large-scale road monitoring is brought about by high-precision systems such as LiDAR, **Ravi et al. [2020]**. The latter tried to provide mobile solutions for public accessibility and flexibility through user-mapping, whereas the LiDAR systems allowed for ultra-high precision on large infrastructures needing proactive maintenance. These approaches rendered a fundamental trade-off between accessibility and precision in road monitoring systems.

The YOLO model has seen wide use in various studies, indicating the growing prominence of real-time object detection in pothole detection and road condition surveys. The

succeeding versions of YOLO have demonstrated significant improvements in detection speed and accuracy of particularly small objects such as potholes. Significantly, YOLOv11 marked a new milestone in the line of models as it combined formerly the best models with modern implementation such as C2PSA blocks (Cross-Stage Partial with Spatial Attention), transformer backbones, large kernel convolutions, dynamic heads, and PSA (Partial Self-Attention). It provided enhanced spatial attention, context understanding, and training strategies with respect to NMS-free methods aimed at improving detection robustness and effectiveness.

Validation of real-world applicability for this version of YOLO was found in various domains. **Jegham et al. [2024]** compared YOLOv11 to previous versions on datasets from traffic signs, wildlife, and maritime vessels. YOLOv11 proved faster and more accurate, scoring an mAP of 0.81 against the African Wildlife dataset with an inference time of just 2.4 ms. In a similar study conducted by **He et al. [2024]**, YOLOv11 outperformed previous iterations (for instance, YOLOv5) based on proposed mAP and fewer false-positive results during equipment detection in the electrical-power domain, while in orchard conditions — which usually had effects from occlusion and variability in illumination. **Sapkota et al. [2024]** found YOLOv11 to be the fastest in inference speed (2.4 ms), very promising for real-time deployment.

Finally, while AI and GPS topped the field, there are onlookers to a slew of new sensor integration mechanisms, such as VIDAR, demonstrated by **Xu et al. [2023]**, which combined visual and depth-aware radar technologies for more accurate and reliable detection of road conditions. This system is notable since this system seeks to counter the limitations of purely visual-based approaches like YOLO, particularly in complex environment where lighting variations and weather conditions often reduce detection performance. These advancements pointed to a suggestion that sensor fusion and AI solutions would soon enter the next stage of innovation, leading to enhanced reliability and efficiency in road condition monitoring.

Gap Bridge by this Study

The gathered related literature and studies provided a comprehensive foundation that highlighted the methods in developing a pothole detection system with integration of GPS for road condition mapping. The clear transition from traditional, labor intensive pothole detection methods to advanced AI driven solutions highlighted the need for real-time and automated systems. Previous studies noticed the inefficiency and inconsistency of manual inspection especially in urban road networks while emphasizing the advantage of utilizing deep learning models such as CNNs and YOLO variants in acquiring accurate and real-time detection. This study was based on and improves the past works by adopting to the advancements of the latest version of YOLO algorithm, the YOLOv11, which incorporated features like anchor free detection and spatial attention mechanism to improve performance even under challenging environmental conditions.

Moreover, the past studies about mapping have underscored the importance and benefits of integrating GPS technology for accurate and reliable geospatial mapping. These studies highlighted the critical need for real-time road condition maps, combining the GPS with image processing, which not only enhanced travel efficiency but also significantly improved safety for drivers. Additionally, mobile-based systems designed for rural communities offered a cost-effective solution. By addressing these challenges related to scalability and localized applicability, this study introduced an innovative pothole detection system alongside a road condition mapping solution, providing a practical tool for underserved areas.

NOTES

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CHAPTER 3

METHODOLOGY

This chapter outlined the methodology used to develop a real-time pothole detection and road condition mapping system. Present in this chapter were the research design, algorithms, theoretical and conceptual framework to provide a clear roadmap for achieving the objectives of the study. The materials, statistical tools, and evaluation methods used are also discussed to assess the system performance.

Research Design

This study employed a constructive method to serve the major problems arising from the poor road conditions in rural areas. These various problems are associated with road conditions in rural areas, including traffic accidents, damaged vehicles, and economic losses because of compounded heavy rain, improper drainage, and overloaded vehicles. This study is intended to develop a real-time pothole detection system and road condition mapping using image processing with YOLOv11 model and GPS data to map roads accurately [1]. The term “construct” refers to the integration of the pothole detection system that was designed and developed to test its validity in the form of bench marking activities and was pioneering work providing an automated road condition monitoring system. In addition, the constructive design is selected because of its suitability to provide technical solutions designed to create and test an actual functional prototype. Moreover, this type of research provides flexibility to test validity in the form of actual performance measurements rather than rigid empirical experiments.

The system development followed a structured workflow. First, data collection involved capturing road images/videos using mobile phone cameras, alongside gathering

corresponding GPS coordinates. Next, data preprocessing refined the collected data by preprocessing images into a uniform format by resizing, normalization, and augmentation, as well as annotating potholes using bounding boxes, splitting the dataset into training, validation, and testing sets, and aligning GPS coordinates with the images to ensure accurate georeference. Once the data is prepared, it was followed by the model training focused on preprocessing images and training YOLOv11 on annotated datasets, optimizing for pothole detection while validating performance under varying conditions based on mean Average Precision (mAP), F1 scores, Latency Evaluation, Accuracy, and Confusion Matrix to understand detection reliability, integration of the trained model with the camera for pothole detection in real-time, and synchronizing GPS data to produce georeferenced road condition maps. In the system integration phase develop a mobile-based system to capture real-time road footage and GPS data, integrates YOLOv11 for on-device/cloud-based detection, and generate dynamic road condition maps by fusing detection results with GPS coordinates. Finally, the evaluation benchmarked system accuracy against manual inspections and existing tools, and tests latency and scalability. This process offered a cost-effective solution that improves road safety, maintains optimal management, and supports the development of public infrastructure.

Theorems, Algorithms, and Mathematical Framework

This section presented the core computational engine of the proposed system: the real-time pothole detection algorithm, along with theoretical grounds and formulas that substantiated its performance evaluation.

Algorithm Used: YOLOv11 (You Only Look Once version 11)

The primary algorithm used in this study was YOLOv11, which belonged to the family of single-shot object detectors. YOLO-based models have become the gold standard for

real-time object detection applications because it localized and classified the objects, all within a single pass through the network.

YOLOv11 introduces key architectural innovations that enhanced its performance [7]:

1. **C3k2 Block (Cross-Stage Partial with kernel size 2):** YOLOv11 employed C3k2 block, an improved version of the C3 block. This component enhanced depth and feature learning efficiency by conserving important information while reducing redundancy in computation.
2. **C2PSA (Parallel Spatial Attention):** Introduced a more recent attention mechanism by which the model gives focus to relevant spatial information (e.g., road textures, edges of potholes), thus aiding detection capability in congested or noisy backgrounds.
3. **SPPF (Spatial Pyramid Pooling-Fast):** SPPF was a retained feature block from previous YOLO models, this allowed for the extraction of multi-scale features by pooling from different receptive fields, which is especially effective for detection.

In Figure 1 below, the YOLOv11 architecture was illustrated showcasing its modular structure. It was designed to optimize both speed and accuracy compared to previous versions of YOLO. Its single-shot design allows for high frame-rate inference, making it an ideal for real-time detection task where it requires low latency. The updated building blocks and anchor-free prediction techniques, follows a conventional three-part architecture consist of backbone, neck, and detection head, that is designed for dense, multi-scale situations such as road-surface monitoring [7].

The backbone serves as the primary feature extractor, converting the input image into hierarchical feature maps that represents the edges, textures, and more complex shapes. YOLOv11 modifies the prior CSP-style modules with the C3k2 block, which is a more efficient Cross-Stage Partial module that deepend the network by using smaller kernels

and split-transform-merge operations, reducing an unnecessary processing [11]. The architectural enhancements, composed of a modular transformer-based backbone and neck components enables the model to capture both the global context (an overall road layout) and fine-grained data (pothole edges) without incurring extensive latency.

After the backbone, the neck combines the features from numerous scales to detect things of varying sizes reliably. The YOLOv11 maintains the Spatial Pyramid Pooling Fast (SPPF) neck module, in which pools input across various receptive-field at the same time allowing the network to merge local features with wide contextual cues at a low cost. Following SPPF, the C2PSA (Parallel Spatial Attention) block improves the neck by learning spatial attention maps across the feature channels and highlighting the regions that likely contains distinctive structure such as damage surface or road markings, even in a visually noisy or congested environment [7, 11]

The data is then fed into the anchor-free detection head, that operates with multi-scale feature maps (often representing small, medium, and large objects). Rather than depending on predetermined anchor boxes, in every cell in these feature maps directly predicts the center, dimensions, confidence, and class scores of objects whose centers are included within that cell. This anchor-free approach enhances adaptability across object sizes and shapes, simplifies hyperparameter tuning, and minimizes mismatches when the distributions of bounding boxes target, diverge from conventional datasets, which is beneficial to use in specific fields like road-detection monitoring.

Across all three components, YOLOv11 is the one designed for high frame-rate inference while still maintaining its strong accuracy. The integration of efficient C3k2 blocks, lightweight yet expressive SPPF and C2PSA modules, and an anchor-free head reduces the computing cost compared to prior YOLO models while at the same time boosting localization and classification quality. Overall, the refined version of YOLOv11 of the YOLO family, strikes a balance between architectural intricacy and efficiency for practical real-time applications.

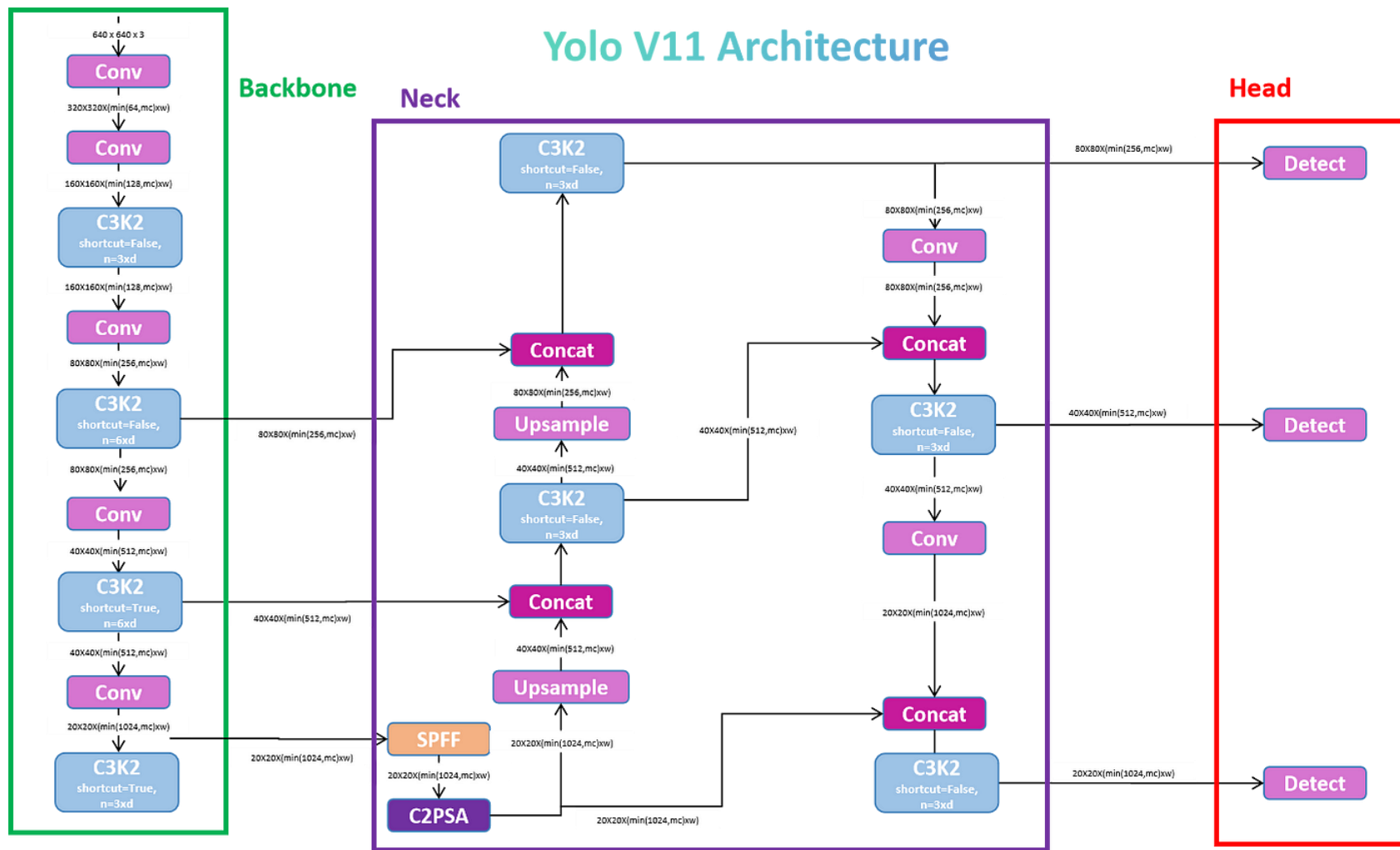


Figure 1: The YOLOv11 Architecture [6]

Basic Working Principle

The YOLOv11 model was based under the principle of single-pass object detection, where recognition of the objects is treated as a regression problem that predicted the bounding box coordinates and class probabilities directly from full images. The algorithmic implementation, as shown in Figure 2, was explained below through the following comprehensive steps:

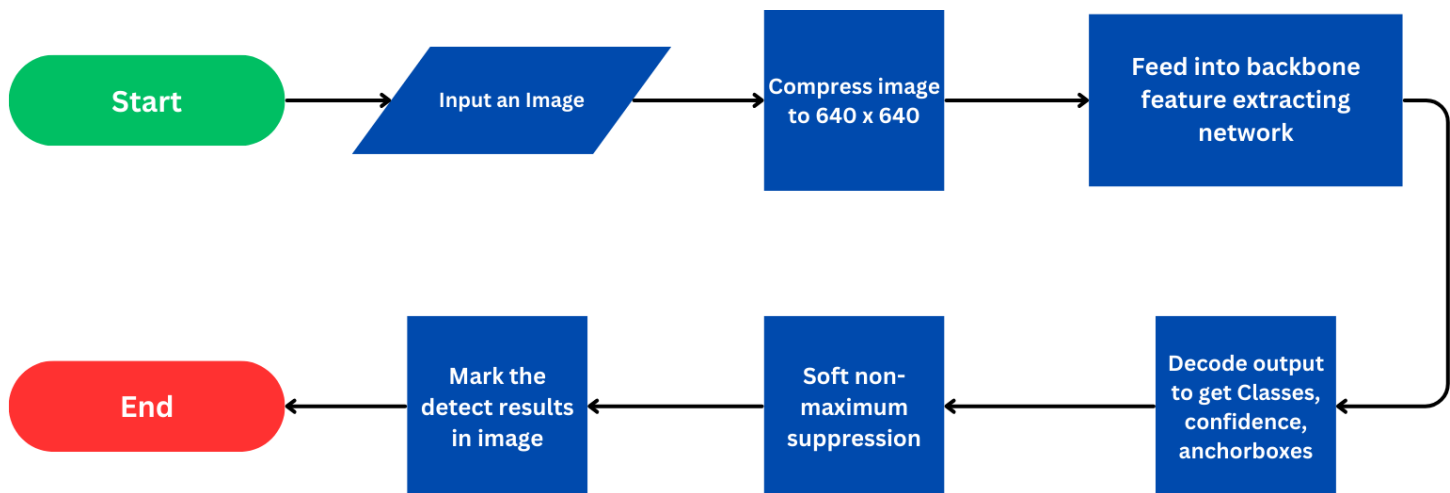


Figure 2: YOLO Inference Flow Chart [10]

Image Input and Preprocessing

The unprocessed images or video frames were resized to a fixed input size (typically 640x640 pixels) and normalized according to input conditions, with size normalization subduing computation cost.

Backbone Network — C3k2 and CSP Modules

The backbone network of YOLOv11 extracted rich hierarchical features from the input image. The C3k2 block improved on traditional cross-stage partial connections using

smaller kernels to discriminate better against small-scale features such as small potholes or surface cracks.

Neck Layer — SPPF and C2f Blocks

This layer collated and adjusted features from various scales. The Spatial Pyramid Pooling Fast (SPPF) module pooled spatial features at multiple scales via multiple parallel pooling operations. C2f blocks facilitated gradient flow and information reuse, hence increasing the efficiency and robustness of the model.

Attention Mechanism — C2PSA (Parallel Spatial Attention)

These modules implemented a kind of visual attention, which directed the model's resources to the informative areas of the image (such as road surfaces) while down-weighting features that would have been considered as noise (for instance, road signs, shadow, or vegetation).

Detection Head — EfficientHead

This layer predicted the class labels, bounding box dimensions, and objectness confidences scores of the predicted object. YOLOv11 head was anchor-free, meaning that it was very convenient in scenarios where the object shapes are arbitrary or variable, such as the unpredictable shape of potholes.

Materials and Statistical Tools/Evaluation Methods

This section presented the materials, software tools, and technologies needed for the development of the pothole detection and road condition mapping system. Statistical tools, methods, and evaluation metrics were applied to assess the accuracy, reliability, and effectiveness of the model.

Materials

The development of the system for this study relied on a combination of hardware and software tools to effectively detect potholes and map road conditions. For data collection, the required hardware was mobile phones with high resolution camera and GPS capabilities to capture road footage/images and geospatial coordinates. The mobile phone's camera capture real-time footage of the road which was processed to detect potholes by using the YOLOv11 object detection algorithm, while the device's integrated GPS recorded the location coordinates to ensure precise tracking of the detected potholes which were then used on generating the road condition map. Additionally, to handle computational workloads in assessing the model training, data processing, and system integration, a standard laptop and desktop computers were used.

The YOLOv11 model for real-time pothole recognition in this study was trained using dataset with annotated images, made specifically for pothole detection. The dataset contains a total of 856 images that were classified into three categories which are minor, medium, and major potholes. These categories are only used during the training of the model in order to identify potholes with different sizes and forms. Using this dataset, the model was trained to generalize the detection of potholes on different type of roads and conditions.

The development of the system was also supported by different software tools and libraries such as deep learning frameworks, image processing libraries, and mapping tools for preprocessing datasets, applying augmentation techniques, evaluating the model, and displaying the detected potholes. In order to integrate the trained model in the prototype application and ensure a smooth execution, testing, and refinement, the development environments and coding tools are essential as these materials guarantee a completely operational pipeline for the system, allowing the system to acquire precise and real-time pothole identification.

Table 1
Tools Used

Tools	Description
Python	Primary language used, known for libraries supporting CV and ML.
PyTorch	Framework for implementing YOLOv11; flexible for DL tasks.
OpenCV	Library for real-time image/video processing; aids pothole detection.
Flask	Web framework to deploy the system and integrate UI/APIs.
OpenStreetMap (OSM)	Open-source map tool to display detected potholes.
ONNX (Open Neural Network Exchange)	Open format built to represent ML models to run across different platforms and frameworks.
TensorFlow Lite	On-device inference for pothole detection and tracking with image recognition.
Dart	Programming language and runtimes used to build cross-platform mobile apps with Flutter.
Hive / Hive Flutter SDK	Lightweight key-value database for Flutter to store detection results locally.
Firebase	Cloud-based backend platform for real-time database, storage, and user authentication.
Cloudinary	Cloud service for storing, managing, and delivering road images or video data.
GitHub	Version control and collaboration platform for managing source code.
Flutter SDK	UI toolkit for building cross-platform mobile interfaces to display detected potholes.
Android Studio SDK	App builder for development and testing of the mobile application.

On the software side, this study utilized the tools presented in Table 1 for the development and deployment of the pothole detection and road condition mapping system. Python served as the primary programming language because of its extensive libraries for computer vision and machine learning. Key libraries include OpenCV and PyTorch, PyTorch was for implementing the YOLOv11 leveraging its flexibility for deep learning tasks [16] while OpenCV was for real-time image and video processing as it allows frame by frame analysis for pothole detection [8]. For deployment of the system, Flask served as the lightweight web framework that facilitated the integration of the detection model with user interface and APIs. To visualize the road map and the detected potholes, the system integrated OpenStreetMap (OSM) as it is open-source and provides a flexible way to create interactive maps that displays the detected potholes and its location [14]. This method saved costs, works for large-scale use, and still accurately shows where potholes are located.

Additionally, other tools and technologies were added to further strengthen the systems features and performance. The trained YOLOv11 model was represented and converted to ONNX format using ONNX(Open Neural Network Exchange) for platform interoperability [13]. TensorFlow Lite was also incorporated into on-device inference to execute pothole detection on the device [2], meanwhile Dart and Flutter SDK were used in creating a cross-platform mobile application. On the other hand, Hive for Flutter SDK was employed as a lightweight local database to retain the detection results and user data [20]. For database synchronization and user authentication, Firebase provided the support for cloud-based backend functionality [17], whereas Cloudinary is responsible for managing the image storage and handling the road data. Meanwhile Github served as the version control platform for collaborative use, as well as organizing and tracking of project. For the integration development of environment for the testing and deploying of the mobile application [4], Android Studio SDK was employed. These enhancements the efficiency and scalability of the system across multiple platforms.

Dataset

This study utilized an annotated image dataset specifically designed for pothole detection to train the YOLOv11 model for real-time pothole identification. The dataset encompasses a nnotated potholes labeled with three categories namely minor pothole, medium pothole, and major pothole. These labels are used solely to improve the model's learning during training to ensure that it can recognize potholes of varying shapes and sizes. The data was split into training, validation, and testing sets, ensuring proper training and evaluation of the model.

Data Collection

The dataset used for this study was obtained from Kaggle repository, these are composed of 700 road pothole images that are captured under varying weather conditions such as daylight, and rainy weather while the other 156 pothole images are captured and collected manually using mobile phone. There is a total of 856 image datasets with corresponding 856 labels that were divided into training, validation, and testing subsets.

Table 2
Dataset Split by Subset

Split	Images	Labels
Train	505	508
Validate	282	282
Test	69	69
Total	856	856

Table 2 presents the splitting of the dataset with a total of 505 images for training set to ensure that there is enough data for the model to classify potholes. The validation set has 282 images for fine-tuning the model's parameters and monitor overfitting while

training. Lastly, the testing set has a total of 69 images reserved for evaluating the model's performance on unseen data. The allocation followed the 59-33-8 distribution approach to ensure that the model will both learn from a large sample size and generalize well when given new and unseen road images.



Figure 3: Sample Dataset

To support the development of automated road condition detection and monitoring, the compiled large amount of datasets consisting of pictures captured of potholes, collected from a public roads under varied environmental conditions. Typically, these datasets were available through repositories like Kaggle that includes wide range of images, including asphalt, wet pavement especially after rainfall and roadways that is partially covered with water, which represent an actual scenarios that are prevalent in road maintenance.

In Figure 3, presents the sample images of potholes from the dataset gathered from the Kaggle repository, that is used in this study. It shows the road conditions with potholes collected under various road conditions, including dry, wet, rainy, and partially flooded roadways. These demonstrates the variety of potholes available in different forms, sizes, and surroundings in order for the model to learn to recognize potholes even in congested, poor or damaged road conditions.

Data Annotation

In order to develop an accurate and reliable dataset for training and evaluating the proposed detection model, this illustrates how pothole images were systematically labeled to guarantee reliability and accurate performance in actual road conditions.



Figure 4: Annotated Potholes

The annotation process involved labeling each pothole present in the image dataset using bounding boxes to clearly define their location within the images presented in Figure 4. The identified potholes were then be classified as one of the three categories which are minor, medium, and major depending on its severity. This allowed the model to distinguish different levels of road damage. Minor potholes were shallow depressions with perceived little impact on driving. Medium potholes may affect comfort or handling. Major potholes are deep or wide potholes posing safety risks. Potholes were labeled based on visual depth estimation and face-value damage, considering how severe they appeared at first glance.

Data Preprocessing

Before using the dataset for training there are several preprocessing steps applied in order to ensure the consistency and improve model's performance. Every images will be resized to 640x640 pixels matching the input requirements of the YOLOv11 algorithm, then normalization was performed to standardize the pixel values of the dataset. In terms of the issue of imbalance and variability, it was addressed using data augmentation techniques such as random flipping rotation, scaling, and brightness adjustments. This helped in learning more robust features, reduced the risk of overfitting, and improved the model's ability to generalize when used in real-world settings.

Evaluation and Statistical Methods

To evaluate the performance of the system, several performance metrics were used to assess the speed, accuracy, and efficiency of the system. The metrics include Mean Average Precision (mAP) for assessing the system's accuracy, F1-score to evaluate the detection quality, Latency evaluation, which showed how quickly the model processes every frame, and the Accuracy for the system's overall effectiveness. In addition, Confusion Matrix was used to analyze the prediction of the model by identifying the false positives and negatives as well as the true positives.

Mean Average Precision (mAP): The primary metric to evaluate the model. It measured the model's accuracy based on how well it can detect potholes across different confidence levels. A higher AP value indicated a better performing detection model [3].

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (3.1)$$

F1-score: This metric was defined as the “harmonic mean of Precision and Recall” [3] as it assessed the balance between the correctly detected potholes and missed potholes.

The combination of these two provided a comprehensive evaluation of the model's effectiveness, especially if there is an imbalance.

$$\text{F1 Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \quad (3.2)$$

Latency Evaluation: The system's real-time performance through a frame processing latency, measured how long it took to process each video frame. This metric was crucial as it evaluated the YOLOv11 model's inference speed by measuring the time it takes from frame processing each frame to showing the final mapped result. The system's speed was assessed in various conditions such as day/night and on different types of roads, to ensure consistent real-time and reliable operation.

Latency(Single Image) in Milliseconds

$$\text{Latency}_i(\text{ms}) = (t_{\text{end}} - t_{\text{start}}) \times 1000 \quad (3.3)$$

Average Latency

$$\text{Average Latency} = \frac{1}{N} \sum_{i=1}^N \text{Latency}_i \quad (3.4)$$

Confusion Matrix: The model's output was classified as true positive, false positive, and false negative to evaluate and visualize the detection performance of the system. This metric provided a detailed description of how well the model detected potholes while minimizing the incorrect detection [3].

Complete Intersection over Union (CIoU) Loss: This served as the primary localization loss function which enhanced the accuracy of bounding box predictions by considering the normalized distance between bounding box centroids, aspect ratio consistency, and overall geometric alignment [19]. This addressed the key limitation in pothole detection as it allowed the model to learn more accurate spatial relationships, particularly when it appeared with varying sizes, shapes, and in cluttered road environments.

$$CIoU = S(A, B) + D(A, B) + V(A, B) \quad (3.5)$$

Where:

- S — Overlap Area
- D — Normalized Central Point Distance
- V — Aspect Ratio

Binary Cross Entropy (BCE): This loss function was applied to both objectness score and classification. It measured how confidently the model detects whether a pothole exists within the predicted box. For classification, it guided the model to better distinguish potholes from background. Minimizing the BCE loss reduced the false positive and improved the reliability of the system's prediction [5].

$$BCE = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] \quad (3.6)$$

Model testing was done under different environmental conditions, such as low light, daylight and rainy weather, in order to assess the model's robustness and ensure its adaptability in real-life scenarios particularly in rural areas where road conditions are unpredictable. For clear performance visualization, Matplotlib and Seaborn are utilized to provide detailed graphs of evaluation results. The results obtained from these metrics should refined the system and optimized it for practical application.

Mean Score: The mean score was used for determining the overall usability perception of each system feature. By averaging the participants ratings, the study identifies how positively or negatively users generally evaluated the system. This helps in summarizing

the central tendencies of the user feedback and provides a clear indicator of the system's overall usability performance.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad (3.7)$$

Standard Deviation: This equation is included to show how consistent were the participants responses. Low standard deviation indicates that users mostly agreed in their ratings, which suggest a stable and reliable user experience. However, high SD shows diverse opinion, suggesting that the system may have inconsistencies and require improvement. This makes the equation essential for understanding the spread of user perceptions.

$$s = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2 \quad (3.8)$$

Likert Scale: This evaluates participants' agreement with each statement with regard to the system's functionality, usability, and overall efficacy. The responses allows the study to quantitatively capture user opinions and attitudes about the system. It provided a structured method for summarizing user feedback, with higher ratings indicating better agreement and a more positive evaluation of the pothole detection system [12].

Detection Survey Evaluation Procedure

To further assess the real-world performance, reliability, and usability of the mobile-based pothole detection and road condition mapping system, the researchers conducted a structured Detection Survey. This instrument required gathering evaluations of the respondents wherein they will be comparing the model's predicted outputs with the characteristics of the visible potholes under actual mobile deployment conditions based on the provided videos and images.

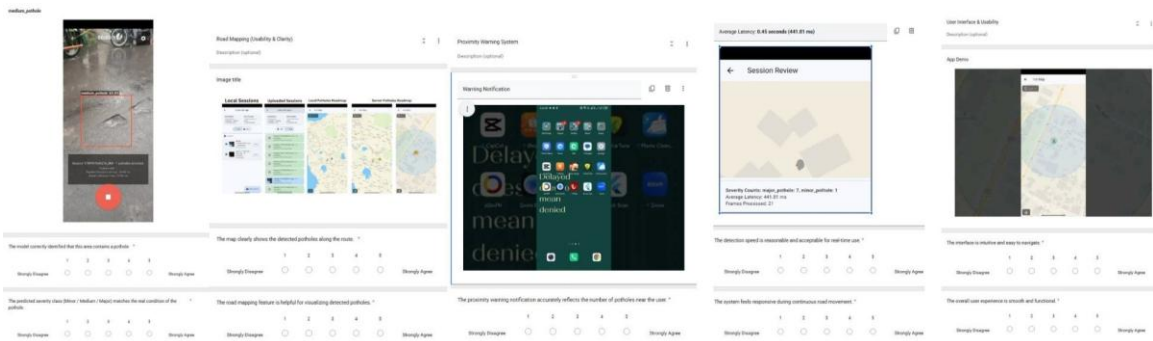


Figure 5: Survey Sample

The survey was administered using a Google Forms questionnaire and included images and videos captured directly from the Mobile Application. Due to preprocessing and conversion differences across the PyTorch → ONNX → TFLite pipeline, minor bounding box misalignments may appear. Respondents were therefore instructed *not* to evaluate bounding box placement and instead focus exclusively on: (1) the visual appearance of the pothole as shown in the image, and (2) the correctness of the model’s predicted labels for pothole presence and severity. Participants rated each statement using a five-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree).

Table 3
Likert Scale Interpretation

Rate	Verbal Interpretation	Description
1	Strongly Disagree	Participants strongly disagree with the statement.
2	Disagree	Participants generally disagree with the statement.
3	Neutral	Participants neither agree nor disagree with the statement.
4	Agree	Participants generally agree with the statement.
5	Strongly Agree	Participants strongly agree with the statement.

The survey evaluated the following core areas of the system:

1. **Pothole Presence Validation** — determines whether the model correctly identified the presence or absence of a pothole in each displayed image.

2. **Severity Classification Accuracy** — evaluates whether the predicted severity label (Minor, Medium, Major) matches the face-value damage perceived visually, based on apparent depth, width, and overall road deterioration.
3. **Road Mapping Clarity** — assesses whether the generated map clearly and accurately displays the detected potholes along the user's route.
4. **Proximity Warning Accuracy** — validates whether the pothole proximity notification reflects the correct number of detected potholes near the user.
5. **Latency and Real-Time Responsiveness** — measures the perceived responsiveness of the system given its recorded average latency of 0.45 seconds (441.81 ms) during continuous road movement.
6. **User Interface and Usability** — examines the intuitiveness of the interface, ease of navigation, and overall smoothness of the user experience based on the application demonstration.

The survey results served as an additional qualitative technique of evaluation, as well as a quantitative evaluation approach, complementing the system's technical performance metrics. The mean and standard deviation were used to determine the survey result evaluation and provide an added insight into the data analysis of user experiences. Adding a human-centered feedback/perspective to the data collection of this study is essential in gaining an in-depth understanding of the models usability and reliability in real-world applications.

Research Process

This section explored the foundational theories, scientific principles, and technical concepts that underpin the development of the proposed system.

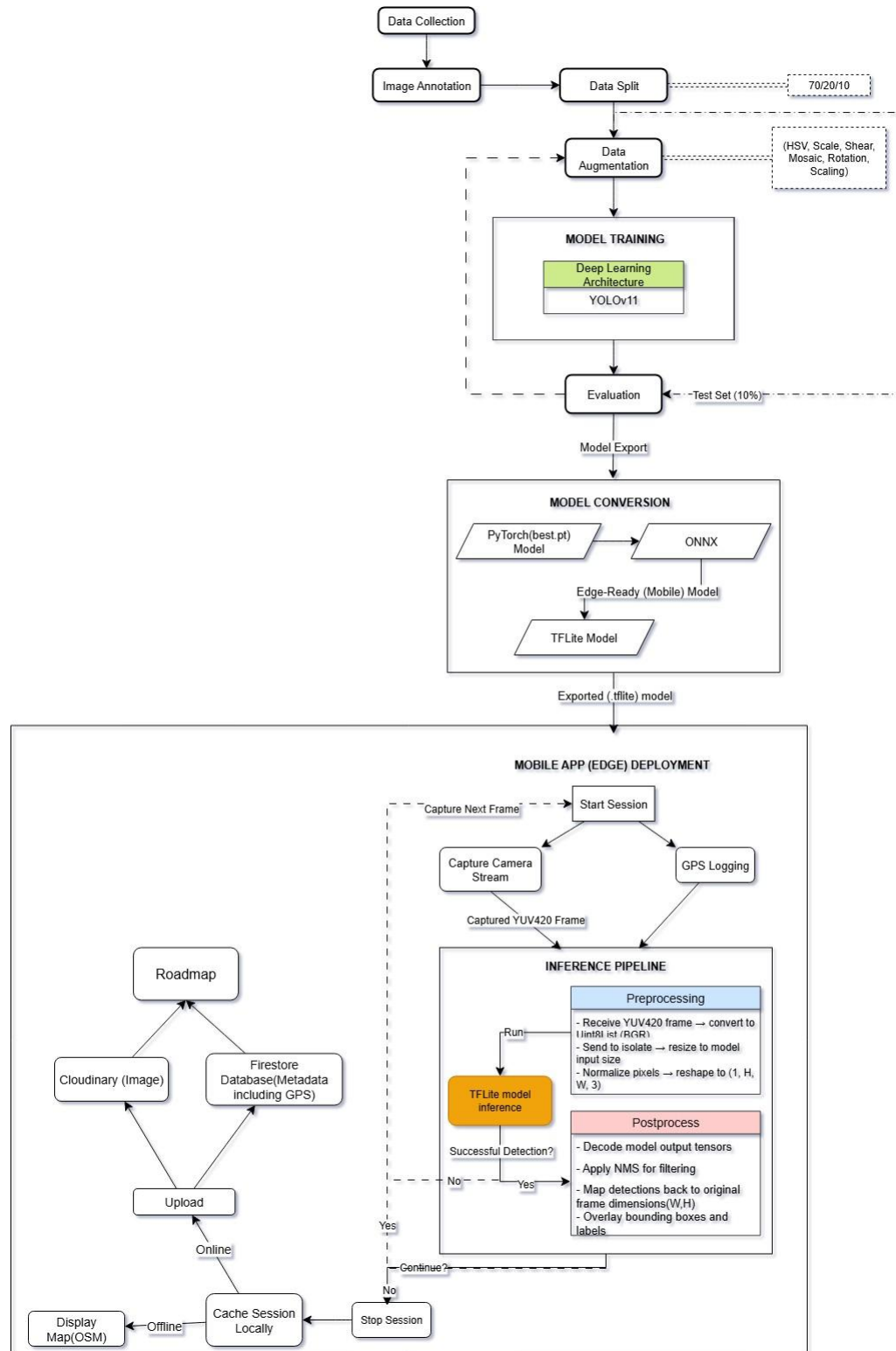


Figure 6: Research Process Flowchart

Figure 6 illustrates the diagram showing that this research employed a comprehensive methodology for pothole detection, beginning with data collection from both online sources and manual smartphone captures of real road surfaces, with images labeled according to severity classifications (minor, medium, major) based primarily on impact assessment for two-wheeled vehicles like motorcycles, which represent the predominant transportation mode in the study region, while also considering four-wheeled vehicles for balanced evaluation. The labeled data, stored in XML and YOLO formats, was divided into training, validation, and test sets (60/30/10 split) with data augmentation applied to enhance the training dataset, followed by model training using YOLOv11 deep learning architecture that produced an optimized PyTorch checkpoint subsequently converted to ONNX and then TensorFlow Lite (TFLite) format with float16 quantization for mobile deployment. The edge deployment system operates through real-time detection on mobile devices, initializing GPS and camera systems to capture frames at approximately 30fps in YUV420 format, converting them to Uint8List (BGR) for processing at 1fps inference rate through the TFLite model, with post-processing steps including decoding, Non-Maximum Suppression (NMS), confidence thresholding, and coordinate mapping back to original frames along with GPS metadata, ultimately saving detected potholes locally using Hive caching before presenting results to users through a session reviewer interface that enables optional online uploading to Cloudinary for image storage and Firestore Database for metadata, incorporating duplicate filtering mechanisms to ensure data integrity.

With the problem identified and reviewed, the research process truly begins at data collection. Where samples are collected both via online sources, and manual captures using smartphones aimed at potholes on real road surfaces. Captured images are then labeled according to severity classification — minor, medium, or major — based on the perceived degree of impact on vehicles, primarily referencing the experience of two-wheeled vehicles such as a motorcycle, which are the most common mode of transportation in the country. Four-wheeled vehicles were also taken into consideration for balanced assessment. The

labels were saved in XML and YOLO format, split into training, validation, and test sets of 60/30/10 and data augmentation techniques were also applied to the training set.

The model training utilized the deep learning architecture, YOLOv11. The best-performing checkpoint produced a PyTorch tensor file, which was converted to ONNX, then to TensorFlow Lite(TFLite) with a quantization of float32.

The converted TFLite model will be used for Edge Deployment using a Mobile Phone. The real-time detection starts with initializing the GPS and camera which captures frames at 30fps in YUV420 format and converting them to Uint8List(BGR). Each frame is resized, normalized, and reshaped for inference using the TFLite model at 1 fps. Post-process includes decoding, Non-Maximum Suppression(NMS), Confidence Threshold, and mapping detections back to original frame coordinates with the metadata which includes the GPS. Detected potholes are saved into a session and cached locally using Hive and is sent to a session reviewer screen. The user can opt to upload their sessions online which saves the images to a Cloudinary asset folder, and the metadata to Firestore Database, with ensured data integrity through duplicate filtering.

Theoretical Framework

The theoretical framework integrated the following foundational theories and concepts:

Object Recognition and Detection in Computer Vision

Object recognition and detection in computer vision involved the identification and localization of objects within images or videos using computational models. Unlike traditional image processing techniques reliant on handcrafted features, CNNs automatically extracted spatial and semantic information, leading to superior accuracy and robustness.

Among CNN-based methods, object detection algorithms were generally categorized into two types which are two-stage and one-stage approaches. Two-stage detectors like R-CNN and its variants (Fast R-CNN and Faster R-CNN) first generated region proposals

and then classified objects within these regions, offering high accuracy but lower speed. In contrast, one-stage detectors such as YOLO (You only Look Once) and SSD (Single Shot MultiBox Detector) treated detection as a regression problem, achieving real-time performance with acceptable precision, making them suitable for applications such as video surveillance, and traffic monitoring.

In object detection, loss function, especially in bounding box regression is an essential component. Developing the YOLOv4 model, Salem et al.'s paper [2023] recognized the limitations of current existing functions such as IoU, GIoU, and DIoU. An enhanced CIoU called Area-CIoU is then introduced. Extra geometric factor is added, which is the area difference between the predicted and ground truth bounding boxes. This updated version with improved localization prevented performance reduction in situations where aspect ratios are similar as standard CIoU may regress to DIoU and lose localization precision [15].

To strengthen the model's sensitivity to spatial differences, the study focused on the theoretical integration of geometric attributes like area difference into the loss function. Overall, Saleem et al. [2023] theory emphasized the importance of loss function design in CNN-based models for object detection and recognition in this study. To have more accurate object detection frameworks, YOLOv4 integrates the ACIoU function [15].

The Theory of Intelligent Transportation Systems (ITS)

The theory of Intelligent Transportation System (ITS) described by Maimaris and Papeorgiou [2016] combined communication and information technology with transportation infrastructure and vehicles for increased road safety, traffic efficiency, and infrastructure management. Automatic road condition monitoring, including pothole detection and mapping are essential parts of ITS. Through real-time collection and analysis of data, ITS sought to provide timely information for repair, streamline traffic, and improve overall transportation network safety and reliability [9].

Risk-Based Infrastructure Management Theory

Risk-Based Management Theories proposed by Talebi et al. (2024) provided a strategic framework that is applied to infrastructure, prioritize repair according to severity of the defect and threat to road users. Unlike the traditional methods of maintenance approach that often relies on scheduled inspection this theory emphasized a systematic assessment on the potential hazards posed by road defects. Integrating pothole detection in risk assessment models facilitated decision-making on resource distribution, hazard analysis on their basis of their preference but also the severity, size, depth, and location. For potholes, this encompassed the characteristics like depth, width, and frequency, directly reliant on the perceive danger presented by vehicles and users. Automated detection with YOLOv11 integration with methods presented an effective tool set for road improvement, extending pavement lifespan [18].

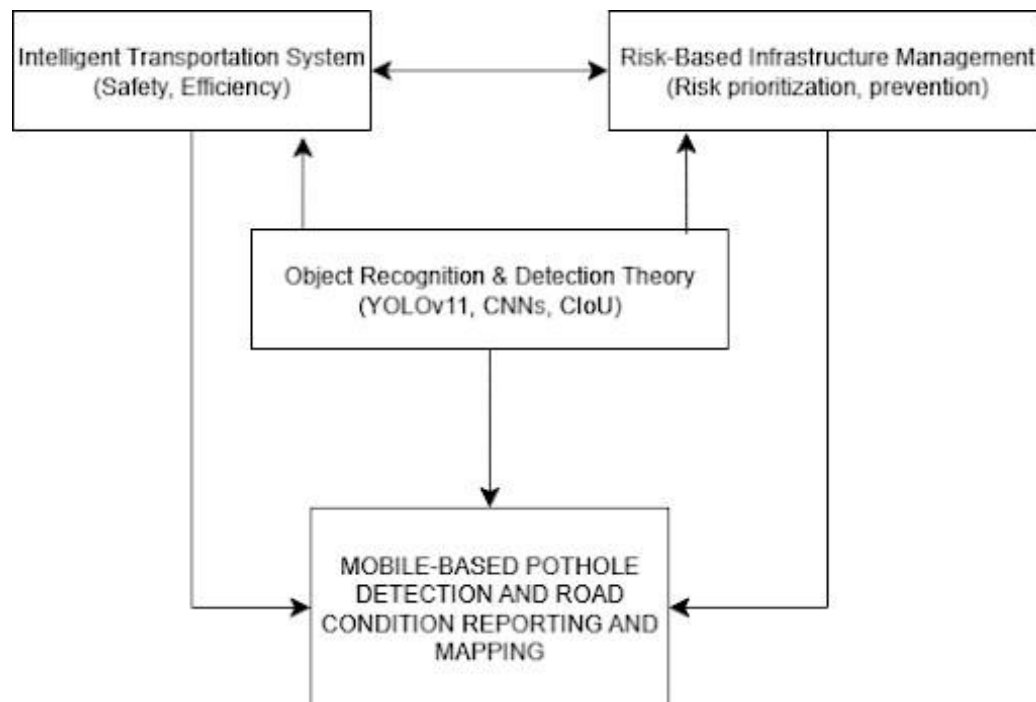


Figure 7: Theoretical Paradigm

The relationship among the three theories is interdependent, with multiple flows of influence converging into the system, as illustrated on Figure 7. First, Object Recognition and Detection Learning provides the core capability of detecting potholes in real time. With the YOLOv11 algorithm, not only does it merely detect the existence of potholes but also generates bounding box data, and confidence scores. Of which contains characteristics such as the pothole image, class label, size, and frequency. These attributes are fundamental indicators of pothole severity, which directly informs the prioritization strategies described in Risk-Based Infrastructure Management Theory. In this sense, Object Recognition is both a detection mechanism and a data supplier for risk assessment.

Second, the results of Object Recognition are also linked to the Intelligent Transportation System (ITS) Theory. ITS emphasizes the integration of technological innovations for road monitoring, safety, and efficiency. The detections generated by the system in real time, coupled with its GPS tagging, enable continuous mapping of road conditions, thereby aligning with ITS objectives of enhancing transportation safety and infrastructure management

Third, ITS and Risk-Based Infrastructure Management(RBIM) exist in a feedback relationship. ITS ensures that road map data is collected and shared efficiently and systematically, while RBIM uses this data to determine which potholes require urgency based on severity and risk to road users. In return, risk-based prioritization can inform how ITS platforms should flag or highlight high-risk areas for local authorities and drivers.

Ultimately, the Mobile-Based Pothole Detection and Road Condition Reporting and Mapping System integrates all three theories. Object Detection contributes accuracy and measurable severity indicators while ITS situates these detections within a real-time transportation safety context; RBIM ensures actionable outcomes by its maintenance priorities. This integration ensures that the system not only detect road damage but also converts these detections into structured information that can guide real-world maintenance decisions and safety interventions.

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CHAPTER 4

RESULTS AND DISCUSSION

This chapter presented, analyzed, and discussed the results obtained from the evaluation and implementation of the study. Findings, observations, and insights are organized by research process.

The initial version of the model employed used 700 pothole images gathered from Kaggle repositories. This created a baseline for the pothole detection system that demonstrated capabilities to recognize potholes using the YOLOv11 architecture, however it limits the optimization and its generalization. Given that the previous model was trained on a less varied and smaller pothole dataset, it primarily concentrated on detection without implementing additional improvements to boost its detection accuracy and its real-time responsiveness. Considering the constraints, this served as the guide on which enhancements and improvements for dataset preprocessing and model tuning.

Data Collection

This section outlines the process of collecting image data, annotation of pothole categories, splitting of the dataset into training, validation, and testing sets, and the use of data augmentation techniques in enhancing the model's robustness.

The dataset used in this study is a combination of images retrieved from the Kaggle repository containing annotated images, categorized into minor, medium, and major potholes and manually captured images to add more localized and realistic scenarios, enhancing dataset diversity. As seen in Table 2 a total of 856 dataset images was then organized and compiled before preprocessing to maintain consistency and quality. These combined

dataset will serve as the foundation for training and evaluating the performance of the pothole detection model.

Image Annotation

To precisely mark the location and boundaries of potholes in each image, ensuring that the training dataset contains reliable reference data for the detection model.

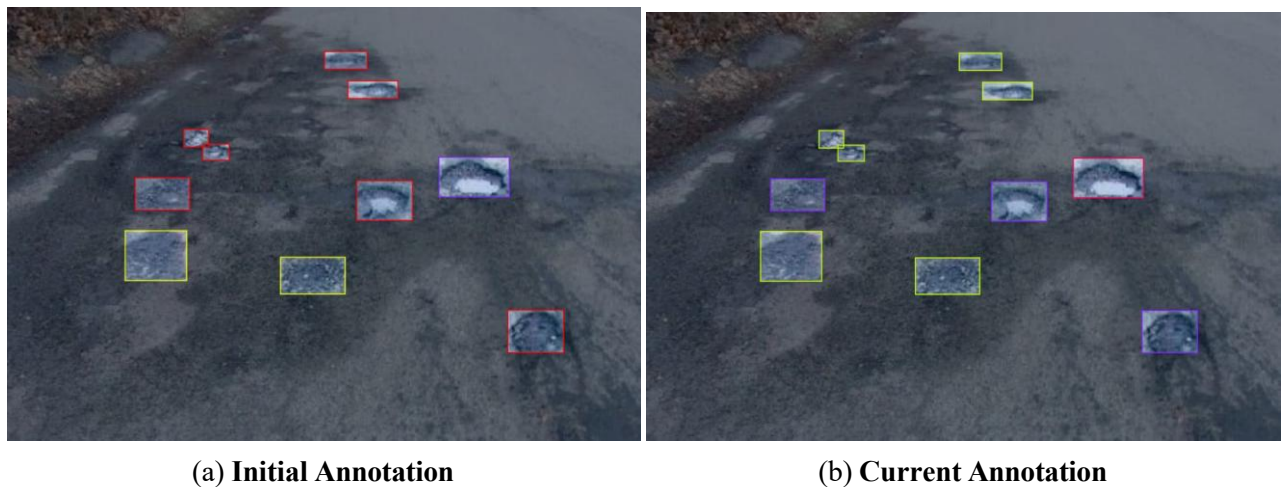


Figure 8: **Previous vs. Current Annotation Strategy**

After capturing the pothole images, each sample was then manually annotated to identify the class and label all the potholes present within each image, this is presented in Figure 8. Moreover, although the dataset from Kaggle was initially annotated and already categorized, the images were still re-annotated manually in order to address inconsistencies in class labels and bounding boxes. The annotation process was performed using Roboflow, where bounding boxes are drawn around potholes and a predefined classes will be assigned, which are minor, medium, and major. Manual annotation was performed to ensure accuracy and consistency, this allows the model to learn the right visual pattern for each category. The annotated images was then exported in a YOLO-compatible format to use for training.

Data Split

This stage shows the annotated pothole dataset that was divided into training, validation, and testing, as well as the resulting sample distribution for each of the severity class. It presents an overview of the data utilized at each stage of model development, providing the basis for evaluating the models performance on minor, medium, and major potholes.

Table 4
Annotated Pothole Count

Class	Training	Validation	Testing	Total
Minor	438	330	42	810
Medium	482	253	60	795
Major	438	192	45	810
Total	1,358	775	147	2,280

The model was trained on a dataset of 856 valid images, which included 2,280 annotated pothole instances. The dataset was split into training (1,358 annotations), validation (775 annotations), and testing (147 annotations) sets. Two sequential training runs were performed: the first with 100 epochs without data augmentation. This division ensures that the model is assessed on unseen data, providing an accurate measurement of its real-world performance in Table 4.

The distribution of minor, medium, and major classes is generally even, which helps avoid the model from overfitting to a single severity level. At the same time, the broader training set provides sufficient data for the model to learn discriminative features, and the validation and testing sets provide dependable checkpoints for monitoring and final evaluation.

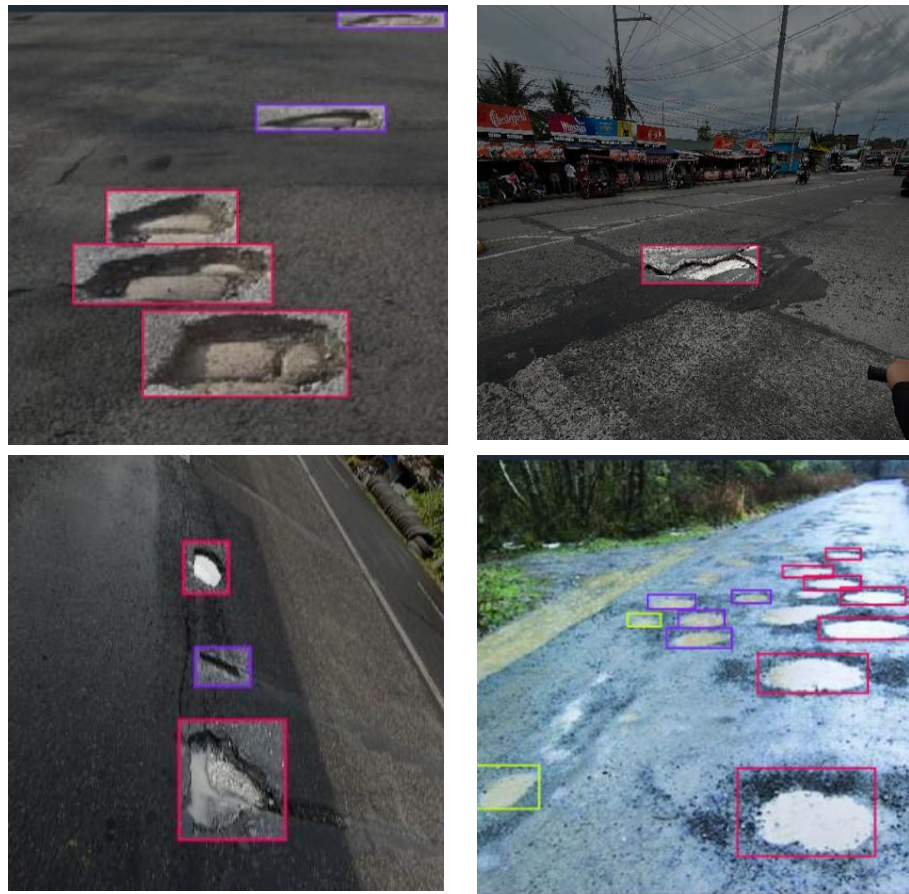


Figure 9: Annotated Dataset

Present in Figure 9 is the sample dataset used in this study illustrating the annotating process of the potholes present in an image using bounding boxes to visually represent the areas where potholes are identified. A raw image of a road surface containing multiple potholes with varying sizes and shapes are manually annotated using Roboflow annotation tool with boxes assigned with specific classes namely minor, medium, and major with different colors representing these classes.

Although manual annotation improves the model's accuracy, the process of performing this is time consuming and may still involve human inconsistencies, particularly if pothole boundaries are partially obstructed. This ensures consistencies across the dataset, correcting mislabeled and missing annotations from the existing sources.

Additionally, environmental conditions like water filled potholes or shadows may also cause uncertainties in placing bounding boxes. Figure 9 provides a clear example of the dataset's annotation quality. By manually labeling each pothole, the dataset achieves greater consistency and reliability making it a strong foundation for training the detection model.

Data Augmentation

The variation of data and it being exposed to a range of data showed that the model was trained to identify potholes. Its second run of data augmentation improved the models capability for generalization. That means it can identify potholes even if it's from a new set of images or it was not seen before. Acquiring a bigger training set than validation test set prevents overfitting and increases detection accuracy even in a variety of environments. Although potholes visualization may change, the model's adaptability after applying the initial phase of the data augmentation was improved. The evaluation of metrics between the division from training and testing, showcases how well the model becomes reliable when it comes to evaluation even outside the original dataset.

Even though the size of the overall dataset was feasible, the model may pose biased judgement towards dominant class particularly medium sized potholes. Additional dataset enhancements such as adding more minor potholes sample or augmentation techniques might resolve the class imbalance problem and improve detection accuracy.

In sum, the structured approach of combining sequential learning with data augmentation and systematic dataset splitting provides a solid basis for robust pothole detection models.

Data Augmentation Settings for Second Run:

To further enhance the model's learning capability several data augmentation techniques were implemented during the second training phase of the YOLOv11 model.

Table 5
Data Augmentation Techniques

Augmentation Technique	Parameter/Value	Description
Image Size	640x640	Training dimension of each image
HSV Adjustment	Hue: 0.015, Saturation: 0.7, Value: 0.4	Adjusts color tone, intensity, and brightness.
Horizontal Flip	50% Random left-right flipping	
Mosaic	True(Closes at 10)	Combines 4 images into 1
Random Augment	True	Applies heavy randomized transformations
Random Erasing	0.4	Removes small patches to improve robustness
Copy-Paste	0.6	Inserts small pothole patches into new environments
Mixup	0.2 (20%) probability	Blends two images and their labels to reduce overfitting.

In the second training sessions, multiple data augmentation techniques were utilized to strengthen the YOLOv11 models robustness and capability to generalized varied road conditions, presented in Table 4. The input images were resized to 640×640 pixels, and several targeted transformations were implemented, including HSV adjustments (hue = 0.015, saturation = 0.3, value = 0.2) and horizontal flipping at a 50% probability. Advanced augmentation strategies were also incorporated, such as Mosaic augmentation (enabled and scheduled to close during the final 10 epochs), RandAugment for strong stochastic transformations, Random Erasing (0.4), Copy-Paste augmentation (0.6), and Mixup blending (20% probability). These augmentations emulated real-world variations such as lighting, color, angle, and object scale, allowing the model to handle environmental diversity and reduce overfitting.

Model Training

As the model progressed throughout the several epochs by tracking significant training and validation metrics. This outlines the hyperparameters, optimization strategy, and criteria that demonstrate convergence behavior, possibility of overfitting, and the overall stability of the training process.

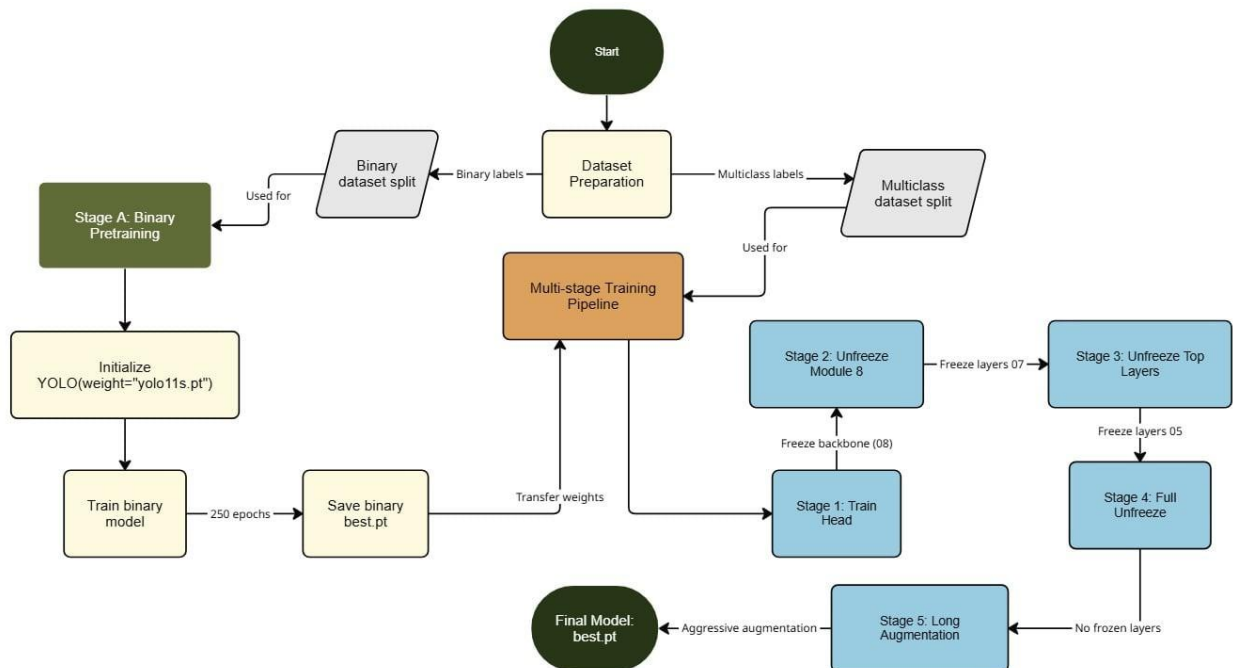


Figure 10: **Progressive Transfer Learning with Staged Unfreezing**

To effectively tailor the model for domain-specific pothole detection, a progressive transfer learning strategy was implemented. The process began with head-only fine-tuning, in which the backbone was frozen to preserve pretrained feature representations while allowing the detection head to adapt to the pothole dataset. In following phases, the network's increasingly deeper layers were unfrozen, which allowed controlled feature adaptation and at the same time reduced the risk of catastrophic forgetting. The learning rate was gradually reduced at each stage to provide stability in the weight updates as more parameters were trained, illustrated in Figure 10.

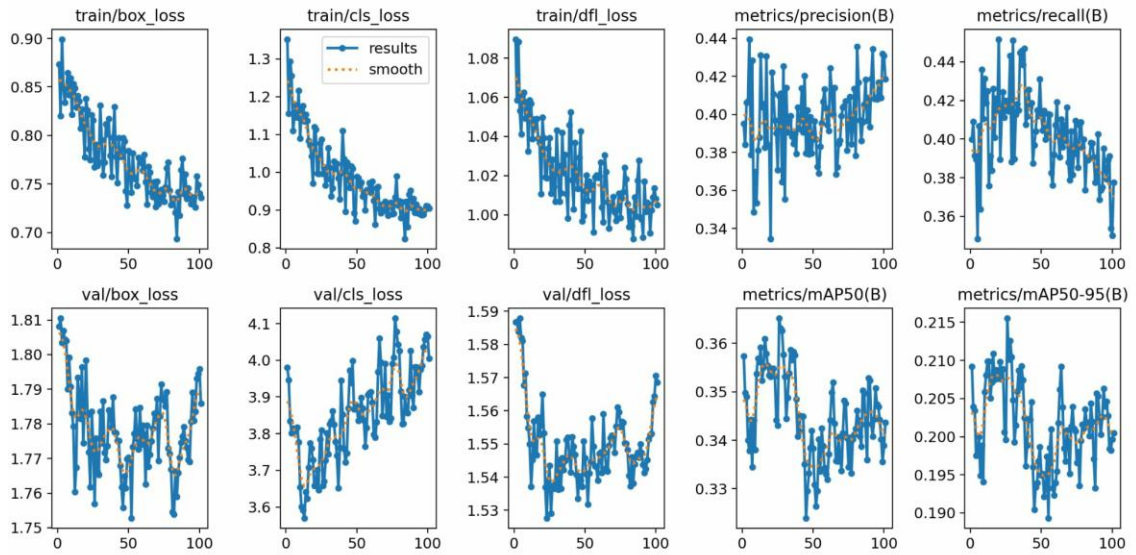


Figure 11: Model Metric Trend

The model demonstrated in Figure 11 showed a relatively consistent improvement across training and validation metrics over 100 epochs, with epoch 18 as the best epoch. Loss curves indicated stable optimization: training box loss decreased from 0.90 to 0.72, class loss from 1.30 to 1.05, and DFL/object loss from 1.08 to 0.98. Validation box loss decreased from 1.81 to 1.76, class loss from 4.0 to 3.6, and DFL loss from 1.59 to 1.53. Precision started around 0.34 in early epochs and reached 0.44. Recall improved from 0.38 to 0.42. mAP@0.5 increased from 0.33 to 0.36, while mAP@0.5:0.95 improved from 0.20 to 0.21.

The progressive reduction in training and validation losses indicated that the model's learning process was effective and consistent throughout the epochs. The near-symmetrical patterns in training and validation losses indicate minimal overfitting, implying that the model was well-adapted to the unseen data. The upward trajectory of precision, recall, and mAP metrics confirms that the YOLOv11 model's ability to precisely localize and classify potholes improved over time. The plateau near epoch 90 represents a suitable convergence point where the model performed its best trade-off between accurate detection and generalization.

The findings indicated that the model is an appropriate use for deployment where the medium and major potholes are the primary focus. The consistent optimization trend and improved evaluation metrics show that the YOLOv11 has effectively learned to distinguish features needed for detecting potholes depending on their severity. Given the overall improvement in the model, the more gradual the convergence of recall and the lower the $mAP@0.5:0.95$ values indicated a difficulty in detecting minor potholes or the less visible potholes. This problem may stem from dataset imbalance, when smaller potholes are underrepresented, as well as the visual noise induced by different road surfaces and lighting conditions.

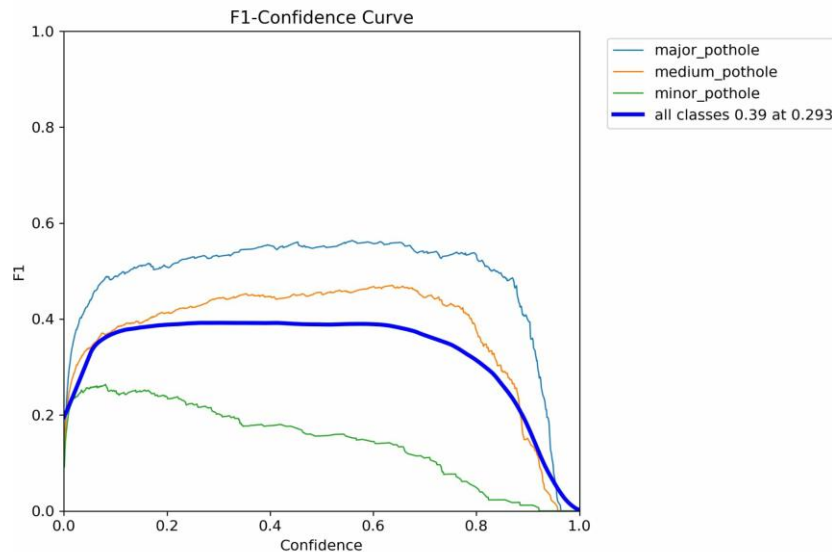


Figure 12: F1-score vs. Confidence Threshold Curve

The evaluation of the model illustrated on Figure 12 revealed an optimal F1-score of approximately 0.327 across confidence thresholds from 0.0 (0%) to 1.0 (100%). Lower thresholds yielded high recall but increased false positives, while higher thresholds reduced false positives at the cost of missed detections. Major potholes were consistently detected even at 0.1 to 0.7 confidence threshold. Medium potholes thrived at around 0.4 threshold, but minor potholes showed low F1 scores peaking at 0.2 confidence threshold. A 0.327 confidence threshold is recommended to maximize detection performance.

The model training demonstrates a progressive improvements across performance metrics, the results shows that the YOLOv11 architecture effectively learned to detect potholes in various conditions. Over successive epochs, the model was able to achieve an increasing precision, recall, and mAP scores, implying successful learning and optimization during training. The incorporation of data augmentation techniques contributed to more stable learning curves and reduced overfitting, especially in detecting medium and major potholes.

These findings suggest that YOLOv11 efficiently identifies and differentiate road surface defects based on their visual characteristics. The improvement in performance metrics and steady decline in loss values reflects the model's effective distinguishing features such as edges, contours, and surface irregularities of potholes. However it is noticeable that minor potholes are relatively harder to classify because of its subtle visual difference from medium potholes and dataset imbalance.

The study's YOLOv11 training resulted to in mAP at 0.5 of 47.0% after 150 periods with significant class imbalances (392 minor, 825 medium, 554 major potholes) this align closely with challenges documented in current computer vision literature. The study by He et al. [2024] demonstrated that YOLO v11 achieved a superior performance metrics of 57.2% mAP in power equipment detection, while the lower performance of this pothole study reflected the built-in complexity of road surface defect detection compared to more structured objects [4]. The progressive improvement in loss functions (training box loss from 2.0 to 1.166) and minimal overfitting patterns validated YOLOv11's architectural robustness that is consistent with the findings from recent literature that emphasized CNN to have "transformed the field, paving the way for accurate pothole detection". The observed dataset imbalance challenges findings from Bucko et al. [2022], who noted that "labelling in bad light and weather conditions was difficult" and that "data annotation and post-processing had to be carefully performed" for effective pothole detection systems [1].

The successful training results show the potential of YOLOv11 when applied in road condition monitoring systems, demonstrating its capability to operate effectively within automated detection frameworks. Its ability to identify different pothole severities can support the local government units in prioritizing maintenance schedules and allocating resources more efficiently, thereby improving response time and reducing potential road hazards. Moreover, the model's performance provides a baseline for future researchers who are interested on enhancing road defect detection using advanced architecture or multimodal data such as depth sensing or infrared imaging, which may further improve robustness and detection accuracy.

Despite acquiring an encouraging results, the model's accuracy was affected by the limited diversity of the dataset, particularly the underrepresented categories like minor potholes. Environmental factors such as shadows, and lighting inconsistencies also raise challenges during training. Furthermore, hardware limitations constrained the training duration and the potential for hyperparameter optimization.

Overall, the model development and training phase confirmed that YOLOv11 is a suitable algorithm for real-time pothole detection. This development clears the path for its integration into a comprehensive road condition mapping platform, resulting in safer and more efficient navigation. YOLOv11's adaptability allows for potential modifications to changing environmental conditions, enhancing its reliability in various kinds of urban and rural locations. Furthermore, with steady performance improvement and reliable convergence, it established a solid foundation for evaluating the system's overall detection performance.

Model Evaluation

This section provides a quantitative evaluation of the trained YOLOv11 pothole detection model using conventional object detection metrics on unseen data. Through evaluating measures, it determined the systems performance on accuracy and efficiency within the defined experimental state.

Table 6
Model Performance Metrics

Metric	Result
mAP@0.5	0.399 (39.99%)
mAP@0.5:0.95	0.236 (23.6%)
Precision	0.42 (42.0%)
Recall	0.477 (47.7%)
F1-score	0.446 (44.6%)
Average Speed	2.5 ms per image (0.1 ms preprocess, 1.3 ms inference, 1.1 ms postprocess)

The overall mAP@0.5 for the model reached 39.99%, while mAP@0.5:0.95 was 23.6%, meaning that detection performance is somewhere between poor and moderate for all severity classes. Precision was 42.0%, recall 47.7%, resulting in an F1-score of 44.6%. The model demonstrated fast performance with an average processing time of 2.5 ms per image, including 0.1 ms for preprocessing, 1.3 ms for inference, and 1.1 ms for postprocessing on an NVIDIA RTX 5000 GPU at a 640×640 input size.

Despite the model’s moderate performance, the results presented in Table 6 remain acceptable for a preliminary implementation of a real-time pothole detection system. Although the precision, recall, and F1-score are all below 50%, these values are reasonable given the challenges present in the dataset—particularly the heavy imbalance of up-close images, inconsistent lighting, and significant variations in road texture, which make pothole features difficult to generalize. The model’s ability to detect potholes with nearly 40% mAP@0.5 demonstrates that it still learns meaningful visual patterns, and its extremely fast processing speed (2.5 ms per image) confirms that it is efficient enough for on-device or near real-time deployment. These findings indicate that while the model is not yet highly accurate, it establishes a solid baseline for future improvements, showing clear potential to become reliable once supported by a more diverse dataset, refined annotations, and further optimization.

Evaluation Result

Table 7
System Usability Scale (SUS) Result

Aspect	Mean	Standard Deviation
Detection Accuracy	4.22	0.68
Severity Prediction	4.08	0.85
Map Visualization	4.4	0.88
Road Mapping Feature	4.6	0.52
Proximity Warning	4.8	0.42
Detection Speed	4.6	0.52
System Responsiveness	4.4	0.88
Interface	4.3	0.92
Overall UX	4.5	0.53

Table 7 presents a System Usability Scale (SUS)-based survey results. This assessment was conducted to evaluate how well the system can support tasks like detecting potholes, categorizing severity levels, and accurate mapping. The survey focused on key features such as detection accuracy, severity prediction, mapping, speed, and user interface. The respondents of the survey used the pothole detection and road condition mapping application and rated each feature on a 5-point scale with 5 as "Strongly Agree" and 1 as "Strongly Disagree". With this, mean scores and standard deviation were calculated for each usability aspect.

The results show that the system is highly usable, as it acquire all mean scores above 4.0. The highest rated features were the Proximity Warning with 4.8 mean score and Detection Speed of 4.6 mean score, implying strong performance in real-time alerts. Road mapping feature and Overall UX also performed well with 4.6 and 4.5 mean score. However, the Detection Accuracy and Interface obtained a slightly lower mean score, suggesting

refinement for minor areas such improving visual clarity. Overall, the SUS results shows that the system delivers a high level of usability, particularly in alert responsiveness and mapping features.

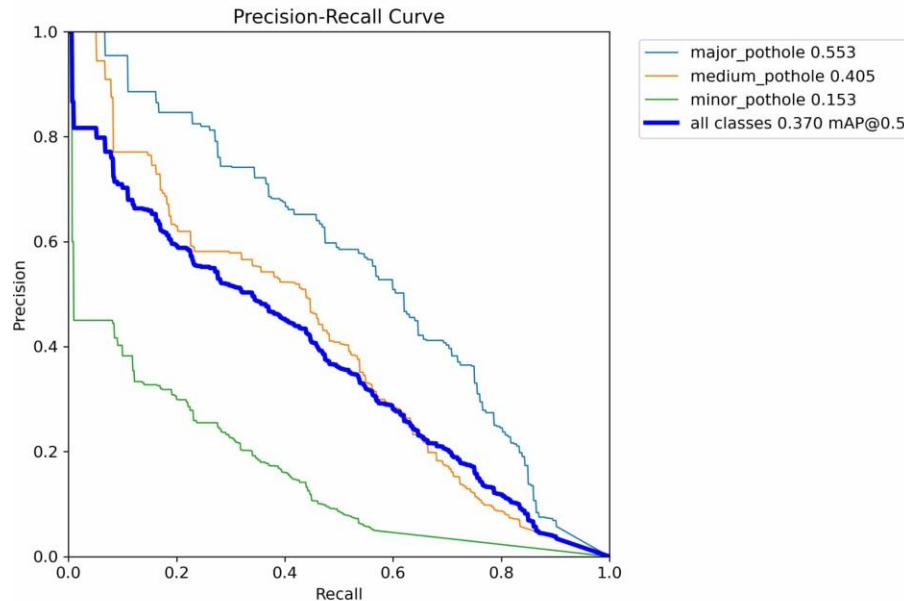


Figure 13: Precision-Recall Curve for Yolov11

The Figure 13 shows that the model achieved an overall mAP@0.5 of 0.370 across all pothole classes. Minor potholes obtained a lower mAP@0.5 of 0.153, reflecting the challenge of detecting small, less prominent potholes. Medium potholes achieved 0.405, while major potholes were detected most reliably with an mAP@0.5 of 0.553, indicating the model performs best on larger, more visually distinct potholes.

The evaluation results show the model has modest overall performance, with significant performance inconsistencies across the different classes. To begin, the high AP score of 0.553(55.3%) for major potholes indicates that the model has the ability to accurately identify larger road damages. This is because major potholes are easier to detect. They have clearer edges, deeper shadows, and more pronounced textures that the model is able to capture. Practically, this means the system can accurately identify the most severe road hazards, which are important for ensuring road safety and prioritized maintenance.

In contrast, the performance for medium potholes corresponds to a lower AP score of 0.405(40.5%), which shows a reasonable and less consistent performance. This is because medium potholes share visual characteristics with both the minor and major classes, which makes them more susceptible to being misclassified. This also demonstrates the problem of intra-class variability, which is the variability within the class.

The minor class has the lowest AP of 0.153(15.3%) and, as the confusion matrix Figure 14 shows, was often confused with the background. This weakness stems from two issues: One, the visual subtlety of minor potholes. They are often misclassified as shallow road cracks or surface irregularities that blend in with the road. Two, the data imbalance problem, which is due to training the model on fewer clear examples of minor potholes compared to major/medium. From an application perspective, this means the system will face difficulties in early detection of minor road damage, potentially limiting its usefulness for preventive maintenance strategies.

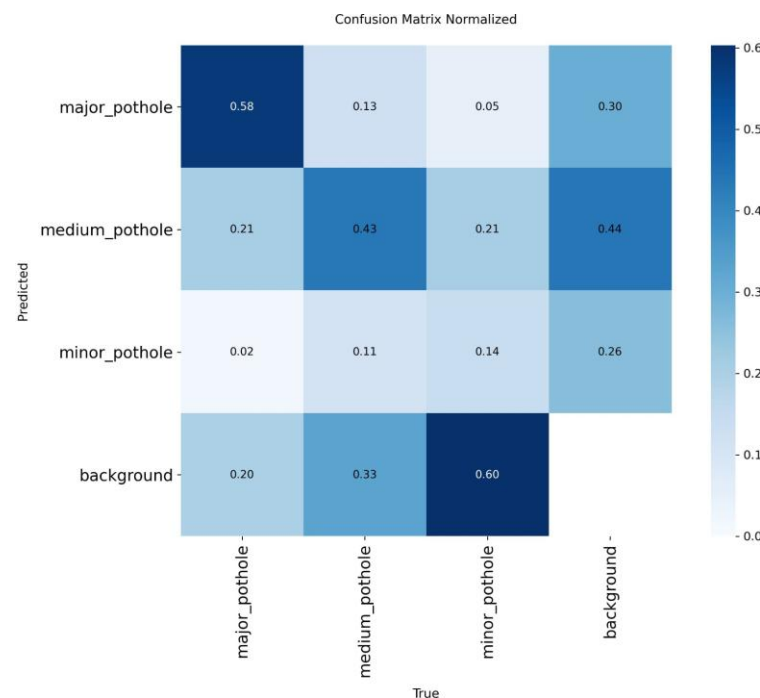


Figure 14: Confusion Matrix of Predicted Pothole Classes

As shown in Figure 14 above, the confusion matrix of predicted pothole classes shows the distribution of correct and incorrect predictions across different categories. For minor potholes, 2% of instances were correctly classified, while 20% were misclassified as background, 21% as medium potholes, and 58% as major potholes. Medium potholes achieved a correct classification rate with 43%, with misclassifications as background 33%, major 13%, and minor 11%. Major potholes had the highest accuracy at 61%, though 20% were misclassified as background, and 18% as medium. Background instances were frequently misidentified as major at 34%, medium 39%, or minor 26%.

These results revealed that the YOLOv11 model performs better in detecting larger potholes, such as the medium and major potholes, but showing a significant confusion with minor potholes and background regions. The low accuracy for minor potholes indicates that it is hard to distinguish subtle defects from the surrounding road environment, most likely due to their smaller size and less distinctive visual features. The model's frequent misclassification of the background areas as potholes indicated that the model tends to over-detect when a road pattern or shadow resembles pothole characteristics.

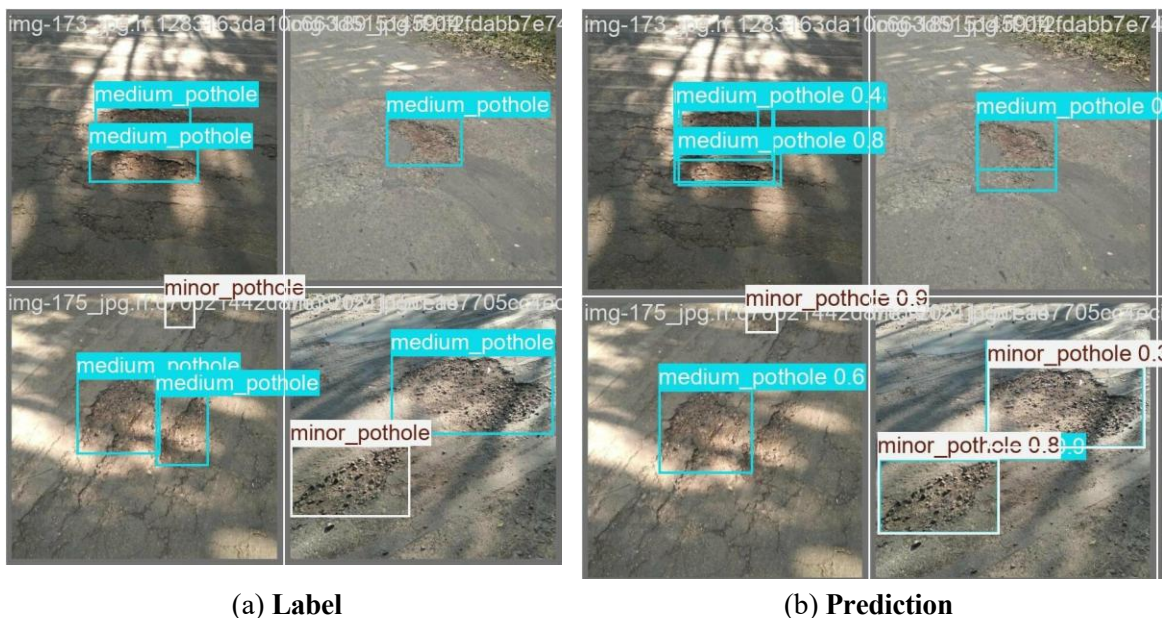


Figure 15: Validation Sample

Figure 15 illustrates the detection pattern of the model, indicating that it can effectively identify larger and more pronounced potholes that often pose significant safety risks to vehicles and drivers. The trained model achieved a satisfactory performance showing a mean average precision(mAP@0.5) of 47.0%, with mAP@0.5:0.95 of 28.9%, precision of 52.4%, recall of 47.3% and an average latency of 7.0 milliseconds per frame, this indicates that the model is capable of performing rapid detections which is suitable for real-time deployment. Among the three classes, major potholes were the most accurately detected, followed by medium, while minor potholes remained the most difficult to identify acquiring the lowest precision due to their less distinctive visual features. The lower accuracy for minor potholes suggests that the model occasionally missed smaller or less visible potholes, primarily because of their visual similarity to normal road textures. Nonetheless, the model's processing speed and consistent performance across various conditions demonstrates its readiness for integration into mobile based detection and mapping systems.

The study's detection performance demonstrates both the capabilities and limitations of the YOLOv11 when it comes to real-world pothole detection scenarios that are documented in current literature. The significant performance variation across classes wherein major potholes have average precision of 0.553 vs. minor that has 0.153, reflects a well-documented challenge in computer vision research. Frinda et al [2024] discovered that the current computer vision methods outperformed its predecessors in correctly detecting potholes on the road, especially in advertised weather conditions [3]. Confusion matrix shows a result with 20 minor potholes being misclassified as the background aligns with the literature findings that "small cracks and dark spots on the road might be identified as pothole in image with low visibility.". The achieved processing speed of the model was 143 FPS equivalent which surpassed the benchmarks from recent studies that acquired 8.8ms inference time using YOLOv8. This proves the YOLOv11's computational efficiency improvements and real-time deployment viability intended for mobile road monitoring applications.

Model Conversion

This section illustrates how the trained YOLOv11 model is translated into deployment-ready formats with integration of a mobile application. It describes the procedures taken to export the model, optimize it for on-device inference, and assure compliance with the target runtime environment while maintaining detection accuracy and real-time performance.

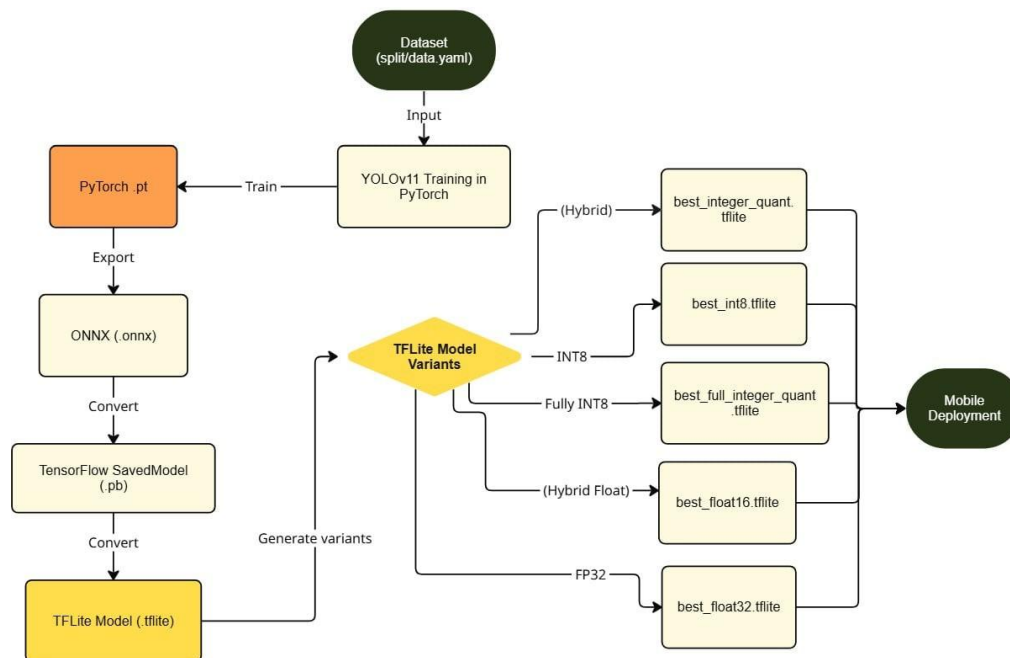


Figure 16: PyTorch to TFLite Model Conversion

After the completion of training using YOLOv11 on the pothole dataset using 640x640 pixel images, the PyTorch weights (best.pt) were exported to ONNX format to enable framework interoperability. The ONNX model was then converted to a TensorFlow Saved-Model and subsequently to TensorFlow Lite (TFLite) for mobile deployment. During conversion, the input size was adjusted to 320×320 pixels to reduce model size and improve inference speed on mobile devices. In the final deployment, the FP16 TFLite variant was used to leverage reduced memory footprint and enable GPU use on qualified devices for faster inference.

This conversion pipeline highlights the significance of model optimization for mobile and edge platforms with limited computational resources. The system strikes a balance between detection precision and real-time performance by lowering input resolution and using FP16 quantization. These changes are designed to ensure that the pothole detection model stays responsive during on-road usage scenarios. Overall, the approach promotes fast deployment while preserving adequate accuracy for practical mobile pothole detection.

Mobile App Deployment

This outlines the deployment of pothole detection systems as a mobile application. It emphasizes essential aspects such as on-device caching, cloud storage, and external image hosting, which all of which collectively ensure consistent functionality throughout various kinds of network conditions while providing users with a scalable road condition monitoring.

<pre>49 static const Map<InferenceSize, String> _modelAssets = { 50 InferenceSize.s320: 'assets/models/float16_320.tflite', 51 }; 52 String get currentModelAsset { 53 return _modelAssets[_inputSize]!; 54 }</pre>	<pre>59 Future<void> loadModel() async { 60 if (_initialized) return; 61 final modelBytes = (await rootBundle.load(62 DetectionConfig.instance.currentModelAsset, 63)).buffer.asUint8List(); 64 final labels = await DetectionConfig.instance.loadLabels(); 65 await loadModelFromBuffer(modelBytes, labels); 66 }</pre>
(a) Deploy and Set Model	(b) Load and Init Model

Figure 17: **Model Deployment and Initialization for Edge Use**

The initialization of the pothole detection system at the edge begins with mapping specific model assets to corresponding inference sizes. By statically assigning optimized TFLite files, such as the float16_320.tflite model for the 320-pixel inference size, the system ensures that memory footprint and compute requirements are tailored for mid-range mobile devices. This arrangement allows the detector pipeline to seamlessly select the appropriate model configuration depending on the input resolution, balancing accuracy and performance without user intervention, shown in Figure 17.

Upon calling the `loadModel()` function, the model is asynchronously loaded into the device's runtime environment. This pipeline handles the essential preprocessing steps, including input normalization, tensor conversion, and channel mapping, preparing the incoming video frames for inference. The detector then iterates through candidate bounding boxes, generating predictions for each class—minor, medium, or major potholes—and calculates confidence scores, intersection-over-union metrics, and non-max suppression filtering. This structured flow from input acquisition to prediction ensures that even in motion, with variable frame quality and GPS tagging, the model consistently outputs reliable detection results.

The technical impact of this pipeline is notable, as it integrates both model flexibility and operational efficiency. By decoupling the inference asset selection from the actual detection routine, the system supports rapid adaptation to different device capabilities while maintaining a consistent detection structure. While the pipeline effectively prioritizes real-time responsiveness, it is still constrained by the input resolution and processing bandwidth, highlighting the potential benefit of dynamic quantization, batch accumulation, or lightweight backbone architectures. Future iterations could further optimize detection latency and class-wise confidence calibration, enhancing both robustness and accuracy for in-field pothole detection.

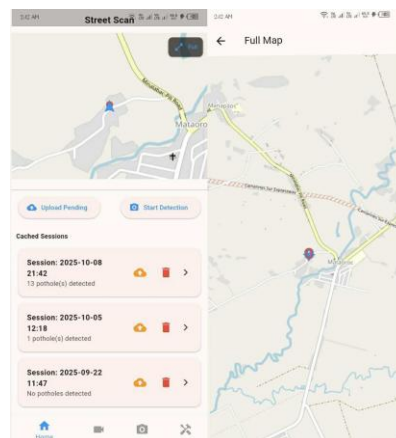


Figure 18: **Cached GPS-anchored Detections**

The mobile application efficiently stores detected potholes locally on the device, along with GPS coordinates, timestamp and associated image, presented in Figure 18. This feature ensures that logging continues even in areas with poor or no network coverage. This prevents the loss of detection events while also retaining all contextual information for further synchronization and analysis. In each cached entry implies that the mobile application can split up detection from upload, allowing the main detection function to be free from constraints of real-time internet connectivity. Its comprehensive metadata associated with each cache data enables traceability, allowing users and road maintenance administrations to examine, validate, and prioritize the reported pothole conditions once the internet connection has been restored.

Similar studies deployed YOLO-based models that have shown the significance of real-time, location-anchored data collection that enhanced road safety and manage road maintenance efficiently. The application of GPS-tagged, as well as the image-backed offline data, on the other hand, emphasized the importance of accurate localization and visualization of images to ensure continuous pothole logging [2]. Although it is beneficial, relying extensively on this may impose limitations on low-end devices, particularly if a lot of high-resolution images are cached before the upload.

Ultimately, the mobile application presents a robust and purpose-driven solution for real-time pothole detection and data management. Its architecture is specifically tailored in order to facilitate reliable and continuous data collection, even in environments with inconsistent or limited network connectivity. By locally caching detections on the device, the system mitigates the risk of data loss during periods where there is no network connection or temporary offline status. Each detection event is systematically annotated with GPS coordinates, timestamps, and confidence scores, thereby providing essential spatial and contextual metadata. This thorough approach ensures the reliability and integrity of the dataset, which is indispensable for researchers and municipal authorities engaged in mapping road conditions.

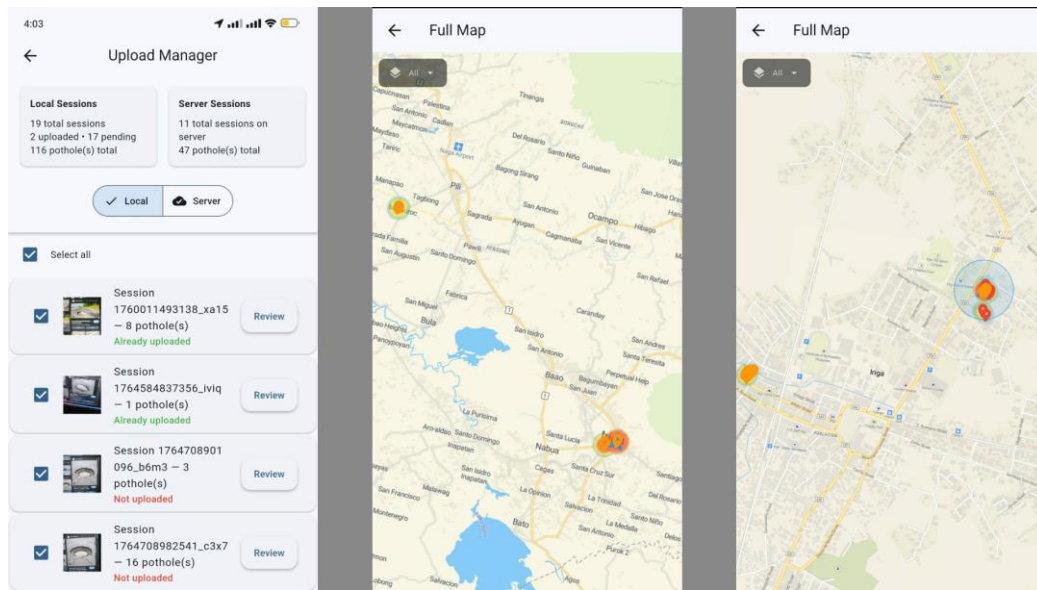


Figure 19: Global Detections Preview for Reporting

The interface displayed in Figure 19 shows the global detection preview in the application, giving users both local and server-side information about the occurrence of potholes. The upload manager panel gives a very brief overview of the local detection sessions along with the server-synced sessions and allows the users to access and check the detections that have been contributed by other users. This feature implicitly draws attention to the collaborative reporting feature where data from various sources is combined to provide a comprehensive picture of the state of the roads.

The full-screen map views give more importance to local and server-synced detections distinction by means of color-coded markers—red for local detections and orange for server detections—so that users can instantly tell the sources of information apart. The use of heatmaps also increases the interpretability of the application by indicating through visuals the density and frequency of pothole occurrences, which are very important for deciding on the maintenance or rigorous monitoring of road sections.

The app, from the technical point of view, utilizes geospatial tagging of detection outputs so that each pothole, whether it is detected locally or retrieved from the server, gets

mapped correctly on the map interface. This geospatial tagging requires a structured data output from the detection pipeline consisting of the GPS coordinates, the severity classification, and the timestamped frames, which are then aggregated and visualized in a user-friendly manner. Thus, the feature displays data synchronization and visualization in an efficient way, which is also very helpful in road monitoring.

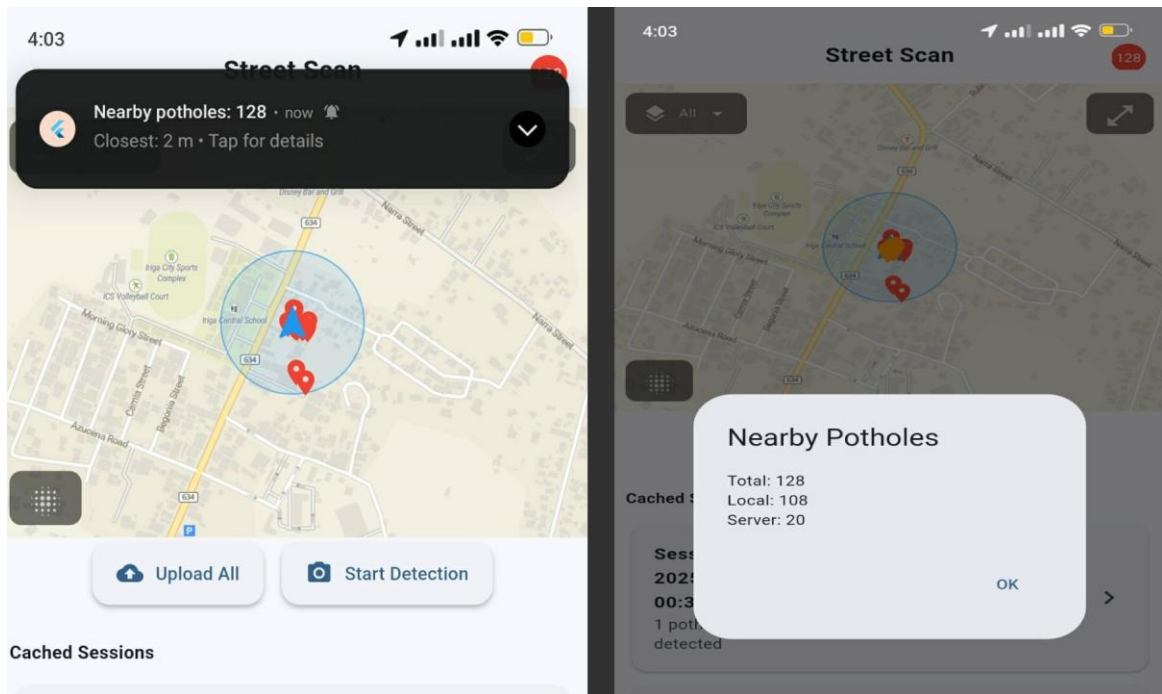


Figure 20: Nearby Pothole Proximity Warning Notification

Figure 20 demonstrates the application's real-time proximity alert system, highlighting how nearby potholes are detected, aggregated, and communicated to the user. In order to update pertinent metrics like total nearby count, local count, server count, and nearest distance, the system continuously tracks the device's coordinates and calculates the distance to both locally recorded and server reported potholes. The pipeline maintains the synchronization between offline detection and cloud sourced data, at the same time ensuring timely and persistent notifications by using ProximityService and NotificationService.

In theory, every position update triggers the computation of Haversine distance. It is then fed into reactive ValueNotifiers that operate the user interface and alerts. Separation of

detection, aggregation, and warning presentation made modularity and scalability possible, which enables the system to effectively manage several data sources and maintain precise proximity information. The efficient alignment of the detection pipeline and notification system is illustrated by integrating geographic computation, real-time data handling, and user facing alert.

Figure 20 emphasizes the application's ability to provide actionable proximity information, supporting user awareness and interaction with both locally and globally detected potholes.

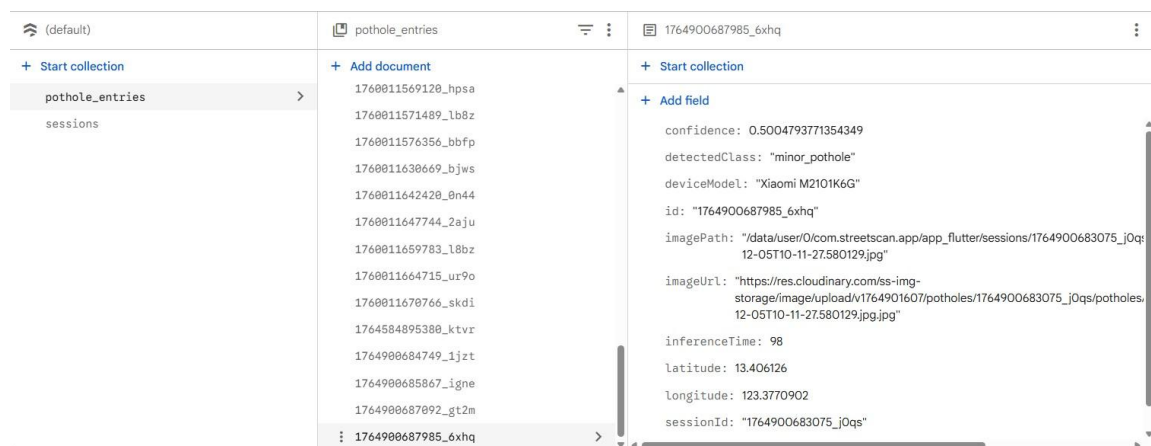


Figure 21: Structured Firebase Schema

The application employed a structured, hierarchical session-based schema within Firebase. Each session compiles multiple pothole entries alongside a comprehensive metadata. This metadata encompasses parameters such as session duration, average inference latency, total frames processed, and categorized counts of pothole severity. Individual entries store image URLs, detected severity level, confidence scores, precise GPS coordinates, inference time, and the assigned session ID.

As illustrated in Figure 21, its database is organized into two main collections: 'pothole entries' and 'sessions'. Wherein in each session aggregated multiple pothole entries with essential metadata. These structured data organizations enabled easier monitoring of batch detection activities and support analytics.

Moreover, the use of GPS and timestamp data allows precise spatial and temporal mapping to detect potholes. Regardless of its efficiency, the Firebase structure may still struggle to handle a large datasets or high-frequency uploads if the indexing and optimization are not properly implemented. The dependency of network may also have an impact on the real-time synchronization, especially in those cases where low-connectivity environments are involved. Beyond that, the reliance on external storage image URLs may result in latency or storage cost if not managed properly through an optimized caching and compression algorithms.

Overall, it effectively manage the structured data for the pothole detecting application using an structuring entries hierarchically and incorporation crucial key metadata variables such as the inference time, confidence scores, and GPS coordinates, designed to ensure robustness in traceability, scalability, and analytical flexibility. Although there were still room for improvement regarding the handling of large amount of data, the structure in general aligns well with the modern cloud-based system schema, enabling efficient and reliable monitoring of road conditions.

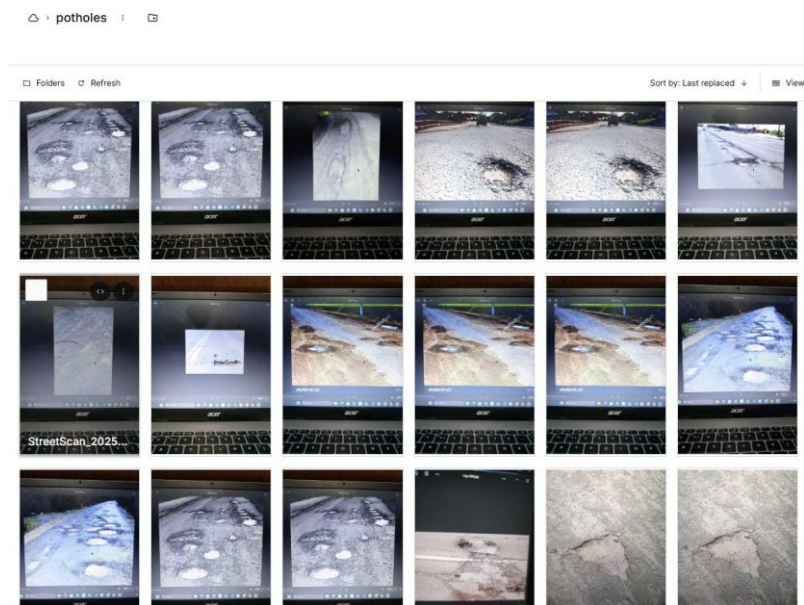


Figure 22: Cloudinary Asset Folder

Figure 22 presented a notable aspect of the application is its seamless integration of local storage and cloud services. Hive operates as the offline storage engine, maintaining structured records of detection sessions, inclusive of multiple pothole entries and their associated metadata. Upon restoration of network connectivity, the data is synchronized with Firebase, which employs a hierarchical schema to support centralized, cloud-based management. Images linked to detection entries are stored in Cloudinary and referenced via unique URLs within Firebase, creating a coherent linkage between local, cloud, and media data. This architecture effectively bridges offline functionality and cloud accessibility, ensuring that no segment of data is overlooked in the synchronization process.

The practical impact of this architecture is particularly significant in rural or underdeveloped regions, where network reliability is often a barrier to real-time data transmission. Users are empowered to collect and queue data instances confidently, with the assurance that the system will handle synchronization autonomously when connectivity permits. The preservation of both local and cloud-based datasets not only minimizes the risk of data loss but also supports long-term monitoring initiatives and collaborative data sharing among multiple users or research teams.

Central to the system's innovation is its offline-first design, combined with automated cloud synchronization. Unlike conventional mobile solutions that depend on persistent connectivity, this application leverages Hive and a session-based schema to efficiently organize and store detections. Each session aggregates multiple entries and metadata, such as processed frame counts, average inference latency, and categorized pothole severity data. The automation of image uploads through Cloudinary, in conjunction with the structured Firebase schema, establishes a scalable and maintainable workflow that accommodates substantial data volumes without overburdening device storage.

Notes

- [1] Boris Buc'ko, Eva Lieskovská, Katarína Zábovská, and Michal Zábovský. 2022. Computer vision based pothole detection under challenging conditions. *Sensors*, 22, (Jan. 2022), 8878. DOI: 10.3390/s22228878.
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CHAPTER 5

CONCLUSION

Summary

This study aimed to design and develop an automated system that detects, classifies, and maps the potholes based on severity using deep learning and image processing. This study is conducted to address the lack of an efficient road damage monitoring system by introducing a computer vision-based solution that is capable of classifying minor, medium, and major potholes in real-time. The study was guided by three main objectives: first, to develop a real-time pothole detection system using YOLOv11 which is trained using a labeled image dataset, second, to evaluate the performance of the proposed detection model using mAP, F1-score, and latency metrics, and lastly, to integrate GPS data with the pothole detection system to generate a road condition map in order to visualize the detected potholes.

The study was limited to real-time image-based detection as well as the additional classification of potholes utilizing pre-collected datasets without hardware deployment. A supervised learning approach was employed, wherein YOLOv11 served as the primary detection architecture. In order to improve the training performance, several training strategies were employed, this includes dataset annotation, image preprocessing, and augmentation, gradual backbone freezing, fine-tuning, optimizer selection, and learning rate adjustment. The model was then evaluated using the standard metrics, which are Mean Average Precision (mAP), precision, recall, F1-score, and latency, and further assessed through a survey where respondents reviewed the predicted detections, severity classifications, and other related features to provide a qualitative feedback on the system that can be used for further enhancement.

Results revealed that the YOLOv11 model achieved a mean average precision (mAP) of 39.99%, with inference latency of 2.5 ms per image, signifying real-time detection capability. The model was observed to performed best in identifying major potholes however showed reduced accuracy for minor ones. When deployed, in the mobile application, the system processed detections at approximately 450-550ms on low end devices, correctly logging GPS-tagged pothole detections and displaying them on the interactive road map.

Findings

The following findings are presented based on the study results to highlight the systems development, evaluation of performance, and overall accomplishment of the specified research objectives:

1. The development phase successfully produced a working YOLOv11-based detection model, trained using 865 image datasets and 2,280 labeled pothole instances. The model demonstrates consistent learning progress across training epochs, with loss functions converging smoothly and steady improvement in detection metrics. Application of data augmentation techniques significantly contributed in enhancing the robustness and generalization of the model, allowing it to recognize a wide variety of pothole shapes and surface textures, indicating that the architecture effectively learned to differentiate between different categories of potholes, meeting the first objective of the study.
2. When evaluated based on accuracy, precision, recall, F1-score, and latency, the model achieved an mAP@0.5 of 39.99%, a precision of 42.0%, a recall of 47.7%, and an F1-score of 44.6%, while the inference speed average was 2.5 milliseconds per image. This validates the model's real-time performance capability. Major potholes acquired the highest accuracy, followed by medium, while minor classes have the

lowest accuracy. These findings indicate that the model performed with moderate precision and strong computational efficiency, fulfilling the second objective.

3. The final objective, which focused on integrating GPS data with the pothole detection system in order to generate a road condition map was successfully achieved as the system utilized the built-in GPS functionality of a mobile device to automatically capture the coordinates of each detected pothole. These coordinates were then used to mark the potholes on a digital road condition map, providing the users with a clear visualization of the exact location of road defects. This integration evolved the prototype into a functional road monitoring tool that provides both detection results and spatial data that support more efficient tracking and assessment of road conditions.

Conclusions

Based on the findings and outcomes of this study, the following conclusions are summarized to the systems overall performance, constraints, potential contribution, and application.

1. The study successfully developed and trained the pothole detection model capable of identifying and classifying road surface damage with acceptable performance, fulfilling the first objective. This concludes that YOLOv11 is a suitable algorithm for pothole detection and road condition monitoring task. The result shows that the combination of deep learning and image processing could effectively automate the detection of potholes, minimizing the reliance on manual road inspections. Pothole detection and eliminated the need for manual road inspections by utilizing deep learning and image processing. Data augmentation improved the model's robustness and generalization capabilities, allowing it to recognize a wide range of pothole varying from forms and surface textures, indicating that YOLOv11 is an appropriate approach for pothole detection and road condition monitoring.

2. The model achieved the intended objective of this study by accurately detecting and classifying potholes, displaying them on the map interface and operating efficiently in real-time. When evaluated using mAP@0.5, precision, recall, F1-score, and latency, the model demonstrated average detection performance but high computational efficiency, attaining real-time inference in 2.5 milliseconds per image. Despite the model's limited accuracy for minor potholes due to dataset imbalance and visual complexity, the overall detection capability for medium and major potholes proves the system's reliability.
3. The integration of GPS data with the pothole detecting system was effective in turning the prototype into a working road monitoring system, thus meeting the final objective. The system allowed clear visualization of road defects as well as more effective tracking and assessment of road conditions by automatically capturing and mapping the coordinates of identified potholes on a road condition map. This mapping feature boosts its utility in maintenance planning and infrastructure management.

The results of this study carry several implications for the advancement of computer vision and studies about infrastructure monitoring. The successful implementation of the YOLOv11 algorithm emphasized the potential of deep learning models to achieve balance between accuracy and processing efficiency, this encourages future studies to explore model optimization, data augmentation, and adaptive learning strategies. Additionally, the findings of this study also highlights the importance of dataset quality and diversity to improve detection consistency.

Recommendation

With the limitations and findings of this study, several recommendations are presented to help future researchers to enhance the system's performance, reliability, and real-world applicability.

1. Future iterations of this system should focus on improving detection sensitivity toward distant and, especially, minor potholes, which were found to be difficult to identify accurately. This can be achieved by enriching the dataset with more samples of small, shallow, and partially visible potholes captured from varying distances and angles. Using higher-resolution training images, advanced data augmentation strategies—such as blur, brightness variation, multi-scale, and possibly random erasing—could strengthen the stability of the model’s ability to generalize to subtle potholes. Transfer learning from a broader road anomaly dataset may also help in enhancing detection capability without extensive data collection.
2. It is recommended to conduct training and testing in varied conditional contexts such as rainy, foggy, and low-light environments. Glare, shadows, and wet surfaces should be explicitly represented in the dataset to improve robustness under real driving scenarios. Dynamic histogram equalization, or shadow-removal filters, may also be integrated in the system’s preprocessing to mitigate the effect of inconsistencies before inference. These enhancements would ensure that the model performs consistently regardless of external factors like sunlight intensity or vehicle speed.
3. The model’s relatively low mean average precision(mAP) and precision scores indicate room for hyperparameter optimization. Fine-tuning parameters such as learning rate, anchor box dimensions, confidence thresholds, and other related parameters could help improve performance.
4. Using model pruning or quantization by converting to int8 instead of float32 reduces latency, ensuring smoother inference on low-mid range mobile phones. For further work, deploying the model using optimized backends such as OpenVINO, NCNN, or Apple Core ML may further accelerate inference time and minimize the frame delays.

-
5. From a system perspective, it is recommended to further develop the mobile application's handling of motion scenarios. Poor detection during vehicle travel, or shaky detections caused by sudden movements, can be minimized by implementing frame stabilization. Light software-based stabilization or motion-aware preprocessing can minimize frame distortion and improve detection consistency even at moving vehicles.
 6. To improve usability, visual tools such as heatmaps may also help authorities identify high-density pothole areas. Further, connecting the app with a city's GIS system can streamline the use of collected data for road maintenance planning and resource allocation.
 7. Further implementations of this study at a high level should consider using high-end smartphones or alternative edge devices that handle heavier computational workloads required for real-time detection. Devices equipped with dedicated NPUs (Neural Processing Units), GPUs (Graphics Processing Units), and AI accelerators, such as Google Pixel Phones (Pixel 6+), which has built-in TPUs, iPhones (A11 Bionic+), which contains a Neural Engine, Samsung Galaxy Phones (S21+), and embedded edge-AI platforms like the NVIDIA Jetson Nano that can significantly improve inference speed and accuracy.

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APPENDICES

APPENDIX A

RELEVANT SOURCE CODE

Listing A.1: Environment Setup and Dependency Initialization

```
import sys
import subprocess
import os
import random
import yaml
from pathlib import Path

def pip_install(package):
    subprocess.check_call(
        [sys.executable, "-m", "pip", "install", package]
    )

try:
    import ultralytics
except Exception:
    pip_install("ultralytics>=8.0.0")
    import ultralytics

import torch
from ultralytics import YOLO
from ultralytics.data.utils import visualize_image_annotations
```

Listing A.2: CUDA and GPU Device Configuration

```
os.environ["CUDA_VISIBLE_DEVICES"] = "3"
os.environ["TORCH_USE_CUDA_DSA"] = "1"

torch.cuda.empty_cache()
torch.backends.cudnn.benchmark = True

DEVICE = torch.device(
    "cuda:2" if torch.cuda.is_available() else "cpu"
)
```

Listing A.3: Dataset Loading and Annotation Visualization

```
def load_yaml(path):
    with open(path) as file:
        return yaml.safe_load(file)

def visualize_dataset_split(
    data_yaml,
    splits=("train", "val", "test"),
    samples=2
):
    data = load_yaml(data_yaml)
    label_map = data["names"]
    base_dir = data_yaml.parent

    for split in splits:
        images_dir = (base_dir / data[split]).resolve()
        labels_dir = images_dir.parent / "labels"
```

```
images = list(images_dir.glob("*.jpg"))
for image in random.sample(
    images, min(samples, len(images))
):
    label_path = labels_dir / f"{image.stem}.txt"
    if label_path.exists():
        visualize_image_annotations(
            image,
            label_path,
            label_map=label_map
        )
```

Listing A.4: Binary Pothole Detection Model Training

```
binary_yaml = Path("split_binary/data.yaml")
binary_project = "binary_runs/train"
binary_run = "Stage0_BinaryDetection_phase1"

model_binary = YOLO("yolo11s.pt")

model_binary.train(
    data=str(binary_yaml),
    epochs=250,
    imgsz=800,
    batch=32,
    device=DEVICE,
    project=binary_project,
    name=binary_run,
```

```
optimizer="SGD",  
lr0=0.005,  
lrf=0.2,  
momentum=0.937,  
weight_decay=0.0005,  
patience=40,  
save_period=5,  
augment=True,  
mosaic=0.4,  
mixup=0.0,  
fliplr=0.4,  
hsv_h=0.015,  
hsv_s=0.52,  
hsv_v=0.35  
)
```

Listing A.5: Multi-Class Dataset Initialization and Configuration

```
multi_yaml = Path("split/data.yaml")  
  
imgsz = 640  
batch = 32  
workers = 8  
  
run_base = "Phase1_FreezeUnfreeze"
```

Listing A.6: Stage 1 Training with Frozen Backbone

```
model_stage1 = YOLO(  
    "binary_runs/train/"  
    "Stage0_BinaryDetection_phase1/"
```

```
        "weights/best.pt"
    )

model_stage1.train(
    data=str(multi_yaml),
    epochs=40,
    imgsz=imgsz,
    batch=batch,
    device=DEVICE,
    project="runs/train",
    name="Phase1_FreezeUnfreeze_head",
    optimizer="SGD",
    lr0=0.01,
    freeze=list(range(0, 9)),
    patience=10,
    save_period=5,
    workers=workers
)
```

Listing A.7: Stage 2 Training with Partial Backbone Unfreezing

```
model_stage2 = YOLO(
    "runs/train/"
    "Phase1_FreezeUnfreeze_head/"
    "weights/best.pt"
)

model_stage2.train(
    data=str(multi_yaml),
```



```
epochs=40,  
imgsz=imgsz,  
batch=batch,  
device=DEVICE,  
project="runs/train",  
name="Phase1_FreezeUnfreeze_head_mod8",  
optimizer="SGD",  
lr0=0.001,  
freeze=list(range(0, 8)),  
patience=10,  
save_period=5,  
workers=workers  
)
```

Listing A.8: Stage 3 Training with Upper Backbone Unfreezing

```
model_stage3 = YOLO(  
    "runs/train/"  
    "Phase1_FreezeUnfreeze_head_mod8/"  
    "weights/best.pt"  
)
```

```
model_stage3.train(  
    data=str(multi_yaml),  
    epochs=40,  
    imgsz=imgsz,  
    batch=batch,  
    device=DEVICE,  
    project="runs/train",
```

```
name="Phase1_FreezeUnfreeze_head_modhalf",  
optimizer="SGD",  
lr0=0.0005,  
freeze=list(range(0, 6)),  
patience=12,  
save_period=5,  
workers=workers  
)
```

Listing A.9: Stage 4 Full Network Fine-Tuning

```
model_stage4 = YOLO(  
    "runs/train/"  
    "Phase1_FreezeUnfreeze_head_modhalf/"  
    "weights/best.pt"  
)  
  
model_stage4.train(  
    data=str(multi_yaml),  
    epochs=40,  
    imgsz=imgsz,  
    batch=batch,  
    device=DEVICE,  
    project="runs/train",  
    name="Phase1_FreezeUnfreeze_full_finetune",  
    optimizer="SGD",  
    lr0=0.00025,  
    freeze=[],  
    patience=9,
```

```
        save_period=5,  
        workers=workers  
    )
```

Listing A.10: Stage 5 Long-Run Training with Aggressive Augmentation

```
model_stage5 = YOLO(  
    "runs/train/"  
    "Phase1_FreezeUnfreeze_full_finetune/"  
    "weights/best.pt"  
)
```

```
model_stage5.train(  
    data=str(multi_yaml),  
    epochs=200,  
    imgsz=imgsz,  
    batch=batch,  
    device=DEVICE,  
    project="runs/train",  
    name="Phase1_Augmented_LongTrain",  
    optimizer="SGD",  
    lr0=0.0005,  
    lrf=0.2,  
    patience=50,  
    save_period=10,  
    augment=True,  
    auto_augment="randaugment",  
    mixup=0.1,  
    copy_paste=0.6,
```

```
erasing=0.4,  
fliplr=0.5,  
close_mosaic=10,  
plots=True,  
save=True  
)
```

Listing A.11: Model Inference and Visualization of Detection Results

```
from ultralytics import YOLO  
import matplotlib.pyplot as plt  
  
model = YOLO(  
    "runs/train/"  
    "Phase1_Augmented_LongTrain/"  
    "weights/best.pt"  
)  
  
results = model("test_image.jpg", conf=0.25)  
image = results[0].plot()  
  
plt.imshow(image)  
plt.axis("off")  
plt.show()
```

APPENDIX B

DOCUMENTATIONS





APPENDIX C

DATASET



APPENDIX D

MODEL EVALUATION SURVEY FORMS



Pothole Detection Model Evaluation Survey

(Predicted Output vs. Ground Truth Assessment)

Instructions:
Please evaluate each model prediction by rating your level of agreement with the following statements.

Use the scale below:

1 - Strongly Disagree
2 - Disagree
3 - Neutral
4 - Agree
5 - Strongly Agree

kmamala@my.caps.edu.ph [Switch account](#)

* Indicates required question

Email *

☒ Record kmamala@my.caps.edu.ph as the email to be included with my response

Notice to Respondents

Please be informed that all images shown in this evaluation are taken from the **TFLite mobile deployment of the pothole detection model**. While the original PyTorch model produces correctly aligned bounding boxes, the **TFLite + Flutter implementation may display slight misalignments**. This occurs due to technical factors including differences in **image preprocessing, aspect-ratio handling, tensor coordinate scaling, and conversion** from PyTorch → ONNX → TFLite.

For this prototype, the researchers did not prioritize perfect bounding box alignment, as the main objective was to evaluate the model's real-world detection and severity prediction under mobile deployment.

Because of these deployment-level inconsistencies, **respondents should not evaluate the model based on bounding box positions**. Instead, please assess the predictions solely based on:

1. the visual appearance of the pothole in the image, and
2. the model's predicted labels (pothole presence and severity).

Your responses will help measure the **real-world accuracy, reliability, and usability** of the system under mobile conditions.

Next

Page 1 of 6

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Pothole Presence Validation

Notice to Respondents

Please be informed that all images shown in this evaluation are taken from the **TFLite mobile deployment of the pothole detection model**. While the original PyTorch model produces correctly aligned bounding boxes, the **TFLite + Flutter implementation may display slight misalignments**. This occurs due to technical factors including differences in **image preprocessing, aspect-ratio handling, tensor coordinate scaling, and conversion from PyTorch → ONNX → TFLite**.

For this prototype, the researchers **did not prioritize perfect bounding box alignment**, as the main objective was to evaluate the model's real-world detection and severity prediction under mobile deployment.

Because of these deployment-level inaccuracies, **respondents should not evaluate the model based on bounding box positions**. Instead, please assess the predictions solely based on:

1. the visual appearance of the pothole in the image, and
2. the model's predicted labels (pothole presence and severity).

Pothole Severity Labels (based on visual perception / face-value damage)

- **Minor:** Appears as a shallow surface depression with limited impact on vehicle movement.
- **Medium:** Appears moderately deep or irregular, potentially affecting driving comfort or handling.
- **Major:** Appears deep, wide, or heavily damaged, posing potential safety hazards.

Your responses will help measure the **real-world accuracy, reliability, and usability** of the system under mobile conditions.

major_pothole



The model correctly identified that this area contains a pothole. *

1 2 3 4 5
Strongly Disagree ☐ ☐ ☐ ☐ ☒ Strongly Agree

The predicted severity class (Minor / Medium / Major) matches the real condition * of the pothole.

major_pothole



The model correctly identified that this area contains a pothole. *

1 2 3 4 5
 Strongly Disagree ☐ ☐ ☐ ☐ ☒ Strongly Agree

The predicted severity class (Minor / Medium / Major) matches the real condition * of the pothole.

1 2 3 4 5
 Strongly Disagree ☐ ☐ ☐ ☐ ☒ Strongly Agree

major_pothole





Pothole Detection Model Evaluation Survey

lomanale@my.caps.edu.ph [Switch account](#)

Your email will be recorded when you submit this form

* Indicates required question

Road Mapping (Usability & Clarity)

Local Sessions



Uploaded Sessions



Local Potholes Road Map



Server Potholes Road Map



Server Potholes Road Map



The map clearly shows the detected potholes along the route. *

12345

Strongly Disagree

☐

☐

☐

☐

☒

Strongly Agree

The road mapping feature is helpful for visualizing detected potholes. *

12345

Strongly Disagree

☐

☐

☐

☐

☒

Strongly Agree

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Pothole Detection Model Evaluation Survey

lonanella@my.cspc.edu.ph [Switch account](#)

Your email will be recorded when you submit this form

* Indicates required question

Proximity Warning System

Warning Notification



The screenshot shows a mobile phone screen with a green background. At the top, it says "December 11, 2025" and "Christine Kyle Delacruz". Below this, there are several app icons. In the center, there is a large red text overlay that reads "Delayed mean denied". At the bottom, there are more app icons.

The proximity warning notification accurately reflects the number of potholes near the user.

1 2 3 4 5

Strongly Disagree ☐ ☐ ☐ ☐ ☒ Strongly Agree

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Pothole Detection Model Evaluation Survey

kmwale@msc.caps.edu.ph [Switch account](#)

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* Indicates required question

Latency & Real-Time Performance

Average Latency: 0.45 seconds (441.81 ms)



← Session Review



Severity Counts: major_pothole: 7, minor_pothole: 1
Average Latency: 441.81 ms
Frames Processed: 21

The detection speed is reasonable and acceptable for real-time use. *

1 2 3 4 5

Strongly Disagree ☐ ☐ ☐ ☐ ☒ Strongly Agree

The system feels responsive during continuous road movement. *

1 2 3 4 5

Strongly Disagree ☐ ☐ ☐ ☐ ☒ Strongly Agree

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Pothole Detection Model Evaluation Survey

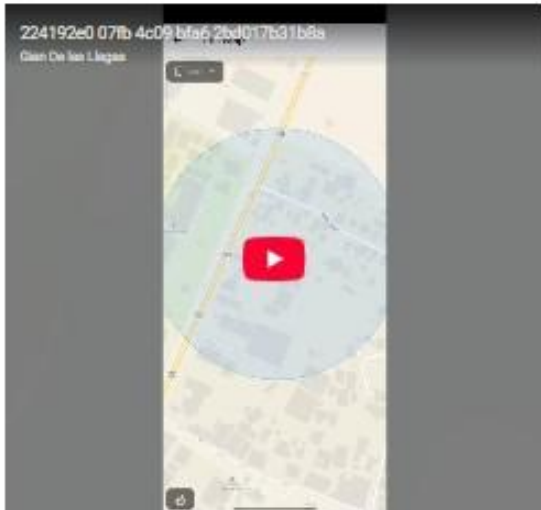
lmanale@my.caps.edu.ph [Switch account](#)

Your email will be recorded when you submit this form

* Indicates required question

User interface & Usability

App Demo



The interface is intuitive and easy to navigate. *

1 2 3 4 5

Strongly Disagree ☐ ☐ ☐ ☐ ☒ Strongly Agree

The overall user experience is smooth and functional. *

1 2 3 4 5

Strongly Disagree ☐ ☐ ☐ ☐ ☒ Strongly Agree

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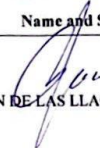
Google Forms

APPENDIX E

PROJECT TEAM ASSIGNMENT FORM

PROJECT TEAM ASSIGNMENTS FORM

Team Alias	APATOSAURUS
Course Code	CS - 3214
Subject adviser/ TSA	ROSEL O. ONESA

Name and Signature	Project Role	Email address / mobile#(s)
 GIAN DE LAS LLAGAS	PROJECT HEAD/ PROGRAMMER	gidelasllagas@my.cspc.edu.ph 09613730229
 CHRISTINE RYLA O. BELANO	UI/UX DESIGNER / DOCUMENTATION WRITER	chbelano@my.cspc.edu.ph 09922826260
 KRIZIA BELLE L. NANALE	QA TESTER/DOCUMENTATION WRITER	krnanale@my.cspc.edu.ph 09501645392

APPENDIX F

PRE-PROPOSAL TITLE FORM

PRE PROPOSAL TITLE

Team Alias	Team Apatosaurus
Project Title:	Pothole Detection and Road Condition Mapping With Image Processing Using YOLOv11 in Rinconada
Proponents/Researchers:	1. Gian De Las Llagas 2. Christine Kyla Belano 3. Krizia Belle Nanale
Scope of the Study:	The study aims to develop an offline, pothole detection and road condition mapping system for Iriga City or Nabua, Camarines Sur. The system will take advantage of a cellphone's camera to collect road footage whilst driving which will include the use of an Object Detection Technology such as YOLOv9 for image processing. The road footage and the GPS location of the potholes will be fed into the system to create a mapping of the road condition in Rinconada.
Limitations of the Study:	The study will focus solely on pothole and crack detection and will not extend to assessing surface conditions such as erosion. The accuracy of this system depends on various factors, such as lighting conditions, quality of the road surface, and weather. The system's detection capabilities are limited by equipments, such as the quality of the camera footage and the Video-GPS latency. Additionally, the coverage area of the road condition mapping will be limited due to deficit in manpower.
Project Design/Development Plan(Software Methodology):	Program Specification Module 1: Video and GPS Data Collection - This allows the user to record road footage using a phone's camera while driving through specific areas. The GPS module will log the user's location constantly. Footage will be stored locally for later processing Module 2: Pothole Detection - This module will process the recorded video to detect potholes using YOLOv9 object detection. The system will analyze the video to identify potholes and cracks on the road. Detected anomalies will be annotated with bounding boxes to mark their location within each frame. Module 3: GPS Data Synchronization and Mapping - This module will synchronize the pothole detection results with the corresponding GPS coordinates logged during video recording. The system will match the timestamps of the GPS data with the video frames. Finally, it will generate a map with the potholes' GPS coordinates, showing the exact detected pothole. Module 4: Road Condition Visualization - This module will visualize the road condition by displaying the GPS coordinates of the detected potholes on an interactive map. The system will use geopandas or folium to create a map that shows the locations of the potholes and mark areas with poor road conditions.

	Software Specification Operating System: Windows 10, Windows 11, macOS Programming Language: Python Libraries and Frameworks: - You Only Look Once(YOLO) - for object detection - OpenCV - for video frame processing - Google Maps or Mapbox SDK - for pothole locations visualization - Geopandas - for map visualization Algorithm: YOLOv9 object detection algorithm Hardware Specification Processor: Intel Core i5 or higher Memory: 8gb RAM or higher Storage: Atleast 200gb SSD/HDD or higher Graphics: Dedicated GPU(e.g NVidia GTX 750 Ti) or higher
☒ Recommended  ROSEL O. ONESA, MIT Thesis Subject Adviser	☐ Rejected ROSEL O. ONESA, MIT Thesis Subject Adviser

APPENDIX G

ROLE ACCEPTANCE FORM (ADVISER, CONSULTANT, GRAMMARIAN)

ROLE ACCEPTANCE FORM College of Computer Studies
Date: February 04, 2025 To: ROSEL O. ONESA We, the third year students of Camarines Sur Polytechnic Colleges pursuing a degree in BACHELOR OF SCIENCE IN COMPUTER SCIENCE, are currently enrolled in Thesis1. We are writing to humbly request your service and expertise to serve as our Adviser for our thesis. We believe that your knowledge and experience will be essential to greatly enrich our work. Attached are our thesis tentative title proposals for your kind reference. Thank you and looking forward to your favorable response of our request. Respectfully, Gian De Las Llagas Christine Kyla Belano Krizia Belle Nanale
To: <u>ROSEL O. ONESA, MIT</u> Dean, CCS This formally signifies that I <u>ACCEPT</u>/ REJECT the request to serve as Adviser of the team APATOSAURUS. As Adviser, I agree to perform my duties and responsibilities stipulated in Section 2.6 of the TCP Guidebook from Thesis 1 until Thesis 2. Furthermore, I agree to set the schedule for advising or consultation to help the students and ensure the success of the thesis. Conformed:  <u>ROSEL O. ONESA, MIT</u> Name and Signature

ROLE ACCEPTANCE FORM College of Computer Studies
Date: February 04, 2025 To: ALLAN IBO JR. We, the third year students of Camarines Sur Polytechnic Colleges pursuing a degree in BACHELOR OF SCIENCE IN COMPUTER SCIENCE, are currently enrolled in Thesis1. We are writing to humbly request your service and expertise to serve as our Consultant for our thesis. We believe that your knowledge and experience will be essential to greatly enrich our work. Attached are our thesis tentative title proposals for your kind reference. Thank you and looking forward to your favorable response of our request. Respectfully, Glan De Las Llagas Christine Kyla Belano Krizla Belle Nanale
To: <u>ROSEL O. ONESA, MIT</u> Dean, CCS This formally signifies that I ACCEPT/ REJECT the request to serve as Consultant of the team APATOSAURUS. As Consultant, I agree to perform my duties and responsibilities stipulated in Section 2.6 of the TCP Guidebook from Thesis 1 until Thesis 2. Furthermore, I agree to set the schedule for advising or consultation to help the students and ensure the success of the thesis. Conformed:  <u>ALLAN IBO JR., MSc.</u> <u>Name and Signature</u>

ROLE ACCEPTANCE FORM College of Computer Studies
Date: March 10, 2025 To: ELLAMAE S. ALAMO We, the third year students of Camarines Sur Polytechnic Colleges pursuing a degree in BACHELOR OF SCIENCE IN COMPUTER SCIENCE, are currently enrolled in Thesis1. We are writing to humbly request your service and expertise to serve as our Grammarian for our thesis. We believe that your knowledge and experience will be essential to greatly enrich our work. Attached are our thesis tentative title proposals for your kind reference. Thank you and looking forward to your favorable response of our request. Respectfully, Gian De Las Llagas Christine Kyla Belano Krizia Belle Nanale
To: <u>ROSEL O. ONESA, MIT</u> Dean, CCS This formally signifies that I ACCEPT/ REJECT the request to serve as Grammarian of the team APATOSAURUS. As Grammarian, I agree to perform my duties and responsibilities stipulated in Section 2.6 of the TCP Guidebook from Thesis 1 until Thesis 2. Furthermore, I agree to set the schedule for advising or consultation to help the students and ensure the success of the thesis. Conformed:  ELLAMAE S. ALAMO Name and Signature

APPENDIX H

FINAL PROJECT TITLE FORM

FINAL PROJECT TITLE FORM

Team Alias: APATOSAURUS
Proponents/Researchers:

- | |
|-----------------------------|
| 1) Gian B. De las Llagas |
| 2) Christine Kyla O. Belano |
| 3) Krizia Belle L. Nanale |

Proposed Thesis/ Capstone Project Title:

Pothole Detection and Road Condition Reporting and Mapping with
Image Processing Using YOLOv11

Submitted by:  GIAN B. DELAS LLAGAS (Signature of Project Head over printed name) Date: <u>May 16, 2025</u>	Noted:  ROSEL O. ONESA, MIT (Signature of Subject Adviser over printed name) Date: <u>May 16, 2025</u>
Recommending Approval:  ROSEL O. ONESA, MIT (Signature of Thesis Adviser over printed name) Date: <u>May 16, 2025</u>	Approved:  ROSEL O. ONESA, MIT DEAN, CCS Date: <u>May 16, 2025</u>

*** Accomplished in 3 copies

APPENDIX I

NON-DISCLOSURE AGREEMENT FORMS

Non-Disclosure Agreement Form

EFFECTIVE DATE: May 5, 2025
(TD submission date)

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and Joseph Jessie S. Oñate, MSc hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:

JOSEPH JESSIE S. OÑATE, MSc – Panel Chair
Name and Title

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:

ROSEL O. ONESA, MIT - Dean
Name and Title

2. The confidential information disclosed under this Agreement is described as:
Contents of the TCP by:

Gian B. De las Llagas, Christine Kyla O. Belano, Krizia Belle L. Nanale

which is entitled:

Pothole Detection and Road Condition Mapping with Image Processing using YOLOv11 in Rinconada

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
- marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
 - if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
- which is or becomes publicly available without breach of this Agreement;
 - which is already known to the Recipient prior to its disclosure hereunder;
 - which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property

identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special, or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.

9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
 - a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
 - b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

**CAMARINES SUR
POLYTECHNIC COLLEGES**


JOSEPH JESSIE S. OÑATE, MSc

Expert
(Print name and Signature)

DATE: May 5, 2025


ROSEL O. ONESA, MIT

CSPC Representative
(Print name and Signature)

DATE: May 5, 2025

Non-Disclosure Agreement Form

EFFECTIVE DATE: May 5, 2025
(TD submission date)

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies- Camarines Sur Polytechnic Colleges Kaela Marie N. Fortuno, MIT hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:

KAELA MARIE N. FORTUNO, MIT- Panel Member
Name and Title

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:

ROSEL O. ONESA, MIT - Dean
Name and Title

2. The confidential information disclosed under this Agreement is described as:
Contents of the TCP by:

Gian B. De las Llagas, Christine Kyla O. Belano, Krizia Belle L. Nanale

which is entitled:

Pothole Detection and Road Condition Reporting and Mapping with Image Processing using YOLOv11

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
- marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
 - if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
- which is or becomes publicly available without breach of this Agreement;
 - which is already known to the Recipient prior to its disclosure hereunder;
 - which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the

Expert shall not be liable for any direct, indirect, special, or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.

9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
 - a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
 - b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

CAMARINES SUR POLYTECHNIC
COLLEGES


KAELA MARIE N. FORTUNO, MIT
Expert
(Print name and Signature)


ROSEL O. ONESA, MIT
CSPC Representative
(Print name and Signature)

DATE: May 5, 2025

DATE: May 5, 2025

Non-Disclosure Agreement Form

EFFECTIVE DATE: May 5, 2025
(TD submission date)

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and Ian P. Benitez, DIT hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

15. The Expert authorized to disclose and receive the confidential information is:

IAN P. BENITEZ, DIT – Panel Member
Name and Title

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:

ROSEL O. ONESA, MIT - Dean
Name and Title

16. The confidential information disclosed under this Agreement is described as:
Contents of the TCP by:

Gian B. De las Llagas, Christine Kyla O. Belano, Krizia Belle L. Nanale

which is entitled:

Pothole Detection and Road Condition Mapping with Image Processing using YOLOv11 in Rinconada

17. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
18. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
19. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
- marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
 - if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
20. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
21. The obligations imposed upon an Expert hereunder do not apply to information:
- which is or becomes publicly available without breach of this Agreement;
 - which is already known to the Recipient prior to its disclosure hereunder;
 - which is independently developed by the Expert.
22. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis

without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special, or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.

23. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
24. Neither Party:
- a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
 - b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
25. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
26. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
27. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
28. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:


IAN P. BENITEZ, DIT
Expert
(Print name and Signature)

DATE: May 5, 2025

**CAMARINES SUR
POLYTECHNIC COLLEGES**


ROSEL O. ONESA, MIT
CSPC Representative
(Print name and Signature)

DATE: May 5, 2025

Non-Disclosure Agreement Form

EFFECTIVE DATE: May 5, 2025
(TD submission date)

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies- Camarines Sur Polytechnic Colleges Rosel O. Onesa, MIT hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

15. The Expert authorized to disclose and receive the confidential information is:

ROSEL O. ONESA, MIT – Subject Adviser
Name and Title

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:

ROSEL O. ONESA, MIT - Dean
Name and Title

16. The confidential information disclosed under this Agreement is described as:

Contents of the TCP by:

Gian B. De las Llagas, Christine Kyla O. Belano, Krizia Belle L. Nanale

which is entitled:

Pothole Detection and Road Condition Reporting and Mapping with Image Processing using YOLOv11

17. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
18. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
19. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
- marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
 - if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
20. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
21. The obligations imposed upon an Expert hereunder do not apply to information:
- which is or becomes publicly available without breach of this Agreement;
 - which is already known to the Recipient prior to its disclosure hereunder;
 - which is independently developed by the Expert.
22. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the

- Expert shall not be liable for any direct, indirect, special, or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.
23. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
24. Neither Party:
- a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
 - b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
25. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
26. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
27. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
28. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

CAMARINES SUR POLYTECHNIC
COLLEGES



ROSEL O. ONESA, MIT

Expert
(Print name and Signature)

DATE: May 5, 2025



ROSEL O. ONESA, MIT

CSPC Representative
(Print name and Signature)

DATE: May 5, 2025

APPENDIX J

HEARING FORMS

THESIS/ CAPSTONE PROJECT HEARING FORM

Date filed: May 1, 2025 [/] Title Proposal [] Pre-Oral [] Final Oral
Date of Hearing: May 5, 2025 Time: 1:30-3:00 Venue: Deans Office

Department: COLLEGE OF COMPUTER STUDIES

Research Title:

POTHOLE DETECTION AND ROAD CONDITION MAPPING WITH IMAGE

PROCESSING USING YOLOV11 IN RINCONADA

Proponent/s:

DE LAS LLAGAS, GIAN B.

NANALE, KRIZIA BELLE L.

BELANO, CHRISTINE KYLA O.

Recommended by:

ROSEL O. ONESA, OIC, DEAN

Thesis/ Capstone Project Adviser

CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research. [Please PRINT NAME and SIGN]

IAN P. BENITEZ, DIT
PANEL MEMBER 1

KAELA MARIE FORTUNO, MIT
PANEL MEMBER 2

JOSEPH JESSIE S. ONATE, MSC
PANEL CHAIR

NOTED BY:

ROSEL O. ONESA, MIT
Subject Adviser

APPROVED:

ROSEL O. ONESA, MIT
OIC, DEAN, CCS

THESIS HEARING FORM

Date Filed: October 9, 2025 ☐ Title Proposal ☒ Pre-Oral ☐ Final Oral

Date of Hearing: October 10, 2025 Time: 10:30 - 12:30 pm Venue: Conference Room

Department: COLLEGE OF COMPUTER STUDIES (CCS)

Research Title:

POTHOLE DETECTION AND ROAD CONDITION REPORTING AND
MAPPING WITH IMAGE PROCESSING USING YOLOv11

Proponent/s:

DE LAS LLAGAS, GIAN B.

NANALE, KRIZIA BELLE L.

BELANO, CHRISTINE KYLA O.

Recommended by:


ROSEL O. ONESA, MIT
Thesis Adviser

CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research.


IAN P. BENITEZ, DIT
PANEL MEMBER 1


KAELA MARIE N. FORTUNO, MIT
PANEL MEMBER 2


JOSEPH JESSIE S. ONATE, MSC
PANEL CHAIR

NOTED BY:


ROSEL O. ONESA, MIT
Subject Adviser

APPROVED:


ROSEL O. ONESA, MIT
OIC Dean, CCS



THESIS HEARING FORM

Date Filed: December 4, 2025 ☐ Title Proposal ☐ Pre-Oral ☒ Final Oral

Date of Hearing: December 5, 2025 Time: 9:00am - 10:00 am Venue: Conference Room

Department: COLLEGE OF COMPUTER STUDIES (CCS)

Research Title:

**POTHOLE DETECTION AND ROAD CONDITION REPORTING AND MAPPING WITH
IMAGE PROCESSING USING YOLOV11**

Proponent/s:

DE LAS LLAGAS, GIAN B. NANALE, KRIZIA BELLE L.
BELANO, CHRISTINE KYLA O.

Recommended by:


ROSEL O. ONESA, MIT
Thesis Adviser

CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research.


IAN P. BENITEZ, DIT
PANEL MEMBER 1


KAELA MARIE FORTUNO, MIT
PANEL MEMBER 2


JOSEPH JESSIE S. OÑATE, MSC
PANEL CHAIR

NOTED BY:


ROSEL O. ONESA, MIT
Subject Adviser

APPROVED:


ROSEL O. ONESA, MIT
OIC Dean, CCS

APPENDIX K

PANEL RSC (TD, POD, & FOD)

Title : POTHOLE DETECTION AND ROAD CONDITION MAPPING WITH IMAGE PROCESSING USING YOLOv11
 Alias : TEAM APATOSAURUS
 Date : MAY 05, 2025, 1:30 PM ☒ TD ☐ POD ☐ FOD
 Secretary : JELAND O. ABONAL

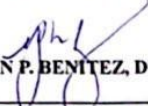
MANUSCRIPT			
CHAPTER	PAGE NO.	RECOMMENDATIONS, SUGGESTIONS AND COMMENT (RSC)	ACTION TAKEN
		Recommendation for Manuscript - In the Title remove the word "in Rinconada" - Remove hanging pages - Include additional datasets in the materials section. - In the research process, add the theoretical framework.	- Changed the title to "Pothole Detection and Road Condition Mapping with Image Processing using YOLOv11" removing the phrase "in Rinconada" - Checked the manuscript and removed all hanging pages. - A "Dataset" subsection is added in the materials section to strengthen the analysis. - Added a Research Process and gathered relevant theories for the Theoretical Framework.
1		Specific Objective - SO1- To develop... - SO2- To evaluate... - SO3- To integrate...	Rearranged the Specific Objectives in the manuscript following the suggested order.
3		Methodology - Increase the size of the images. - Insert Conceptual Framework - Apply proper formatting.	- Increased the size of all images for better readability and clarity. - Inserted the Conceptual Framework diagram and explanation after Theoretical Framework. - Revised the manuscript format, applying consistent font, spacing, and citation style across all sections.

SOFTWARE PROGRAM/ PROTOTYPE		
MODULE NO. (DFD)	RECOMMENDATIONS, SUGGESTIONS, AND COMMENTS (RSC)	ACTION TAKEN

NOTED BY:


JOSEPH JESSIE S. ONATE, MSc

PANEL, CHAIRMAN


IAN P. BENITEZ, DIT

PANEL, MEMBER 1


KAELA MARIE N. FORTUNO, MIT

PANEL, MEMBER 2

Title : **POTHOLE DETECTION AND ROAD CONDITION REPORTING AND MAPPING WITH IMAGE PROCESSING USING YOLOv11**

Alias : **APATOSAURUS**

Date : **OCTOBER 10, 2025 10:30 AM – 12:30 PM** ☐ TD

☒ **POD**

☐ **FOD**

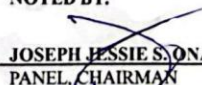
Secretary : **MARRI GRACE MORATA**

MANUSCRIPT			
CHAPTER	PAGE NO.	RECOMMENDATIONS, SUGGESTIONS AND COMMENT (RSC)	ACTION TAKEN
3	36	Include sample dataset from Kaggle.	Inserted an image of pothole dataset from Kaggle repository
	40-41	Expand discussion of the Research Process.	Enhanced and expanded the explanation of the research process providing a clear and step-by step description.
	44	Remove the Conceptual Framework.	Removed the Conceptual Framework section
4	48	- Follow the research process in your discussion. - Add a figure of the annotated dataset and discuss.	- Revised the sections in chapter 4 aligning precisely with the research process. - Provided a sample annotated dataset image with discussion.
	49	Convert the data augmentation techniques into tabular form.	Presented the data augmentation techniques used in a structured table.
	58	Format figure 10 in LaTeX—make subfigures for multiple images presented together.	Reformatted Figure 10 and provided subfigures for clearer and organized presentation of multiple images.
	60	Add figure showing the reporting of potholes in the prototype.	Added a global feature in minimap to view local/server potholes in real time; created a UI in upload manager to display all sessions uploaded in the server and remote
	-	OTHER: - Show proof that the model was used in your prototype. - Follow the PIHLS format in your discussion.	- Added a subsection in chapter 4 discussing how the deployed model is loaded, interpreted, and used. - Revised the discussion section to follow the PIHLS format.
5	-	Lacking findings	Additional findings were included or clarified based on the available data
	69-71	Present conclusions in numbered form, based on the findings.	Both the conclusion and the

		Present recommendations in numbered form as well.	recommendation were formatted and organized into numbered forms.
ALL		- Fix hanging pages and empty spaces. - Add an introduction for each tables and figures. - Be sure to mention or identify which figure/table is being discussed.	- All unnecessary hanging pages, blank spaces, and formatting inconsistencies were corrected to ensure consistent spacing across all sections. - Added introductory statements before every table and figure to provide context and guide the reader in understanding their relevance to the study. - Ensured that every table and figure referenced in the narrative includes its correct label for clarity and alignment.

SOFTWARE PROGRAM/ PROTOTYPE		
MODULE NO. (DFD)	RECOMMENDATIONS, SUGGESTIONS, AND COMMENTS (RSC)	ACTION TAKEN
	- Use faster hardware such as nano jetson instead of just a mobile phone. - Include an alert when the user is nearing a pothole. - Demonstrate the real-time functionality of the system.	- The current system is optimized for mobile implementation as intended in the scope. However, the recommendation is acknowledged for future system enhancement and is included in the Recommendations section. - Implemented a nearby pothole proximity warning notification service that listens to user movement and fetches nearby potholes in a 110meter radius, hooked to floating notifications for persistent warning

NOTED BY:


JOSEPH JESSIE S. ONATE, MSc
PANEL, CHAIRMAN


IAN P. BENITEZ, DIT
PANEL, MEMBER 1


KAELA MARIE K. FORTUNO, MIT
PANEL, MEMBER 2

Title : **POTHOLE DETECTION AND ROAD CONDITION REPORTING AND MAPPING WITH IMAGE PROCESSING USING YOLOv11**

Alias : APATOSAURUS

Date : DECEMBER 5, 2025 9:00 – 10:00 AM ☐ TD ☐ POD ☒ FOD

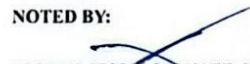
Secretary : MARRI GRACE MORATA

MANUSCRIPT			
CHAPTER	PAGE NO.	RECOMMENDATIONS, SUGGESTIONS AND COMMENT (RSC)	ACTION TAKEN
Initial pages	ii	- Input the date of your FOD - Ensure proper spacing of words - Correction: use "OIC Dean", not "Dean"	- The exact date of the Final Oral Defense was added to the manuscript. - Spacing errors throughout the document were reviewed and corrected to improve readability and consistency. - The term "OIC Dean" was used consistently instead of "Dean"
	iv	Refer to Dr. Benitez as "Dr." not "Sir"	Dr. Benitez was properly addressed as "Dr." rather than "Sir."
	v	- Revise Abstract. It should include: - Intro/Prob - Objective - Method - Results - Conclusion	The abstract was revised to include all required components: introduction/problem, objectives, methodology, results, and conclusion.
		Fix the formatting of your Table of Contents. Section titles are currently too large	The formatting of the section titles were resized to follow the prescribed format. (page
1	6	- Project dictionary - Add the term "real-time"	The term "real-time" was added to the project dictionary. (page 7)
4	66	Remove figures (<i>subfigure 1: Labeled Pothole & subfigure 2: Predicted Potholes</i>)	Subfigure images showing labeled and predicted potholes were removed as instructed. (fig. 14, page 68)
	73	Replace figure 21 with black and white or light mode image	Figure 21 was replaced with a black-and-white or light-mode version to improve clarity (page 78)
	-	Include screenshots of pothole alerts and real-time potholes	Screenshots demonstrating pothole alerts and real-time pothole detection were included to show system functionality. (fig. 20, page 77)
	-	Present the evaluation results in your manuscript	Added a dedicated section summarizing system performance, user feedback, and usability findings. (table 7, page 67)
Appendices	A	Ensure proper format	Formatting adjustments were applied across all sections.
ACM	4	Merge figures 4 and 5 into one figure, with the multiple images formatted as subfigures.	Figures 4 and 5 were merged into a single figure, with multiple images arranged as subfigures (fig. 4, page 4)

ALL		Use proper formatting for figures with multiple images (LaTeX subfigure environment)	All figures with multiple images were reformatted using the LaTeX <i>subfigure</i> environment to ensure consistency and proper alignment.
		Have manuscript checked by your grammarian	Had the entire manuscript reviewed and corrected by a grammarian.

SOFTWARE PROGRAM/ PROTOTYPE		
MODULE NO. (DFD)	RECOMMENDATIONS, SUGGESTIONS, AND COMMENTS (RSC)	ACTION TAKEN
	Have at least 10 test cases and System Usability Scale	17 test cases were conducted, including 10 iterative tests (5 for pothole detection and 5 for severity classification) and 7 system functionality tests. A System Usability Scale (SUS) survey was distributed via Google Forms to gather participant feedback on usability and performance, with results incorporated in the system assessment section. (page 44-46)

NOTED BY:


JOSEPH JESSIE S. ONATE, MSc
PANEL, CHAIRMAN


IAN P. BENITEZ, DIT
PANEL, MEMBER 1


KAELA MARIE N. FORTUNO, MIT
PANEL, MEMBER 2

APPENDIX L

CONSULTATION LOG FORMS

CONSULTATION LOGS FORM

Thesis/ Project Title		Capstone Pothole Detection and Road Condition Mapping with Image Processing using YOLOv11 in Rinconada					
Proponents:		De las Llagas, Gian B.		Belano, Christine Kyla O.		Nanale, Krizia Belle L.	
Alias:		APATOSAURUS					
Total # of Modules				as approved by the CAPSA / TSA			
PROTOTYPE	DT of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Manager's Signature	TA/TSA/ CAPA/ CAPSA/ Consultant Signature
Chapter 2 (Draft)							
Remarks:		Please check my comments on the manuscript.					
Deadline:							
Chapter 2 (Draft)							
Deadline:							

CONSULTATION LOGS FORM

Thesis/ Project Title		Capstone Pothole Detection and Road Condition Mapping with Image Processing using YOLOv11 in Rinconada					
Proponents:		De las Llagas, Gian B.		Belano, Christine Kyla O.		Nanale, Krizia Belle L.	
Alias:		APATOSAURUS					
Total # of Modules				as approved by the CAPSA / TSA			
PROTOTYPE	DT of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Manager's Signature	TA/TSA/ CAPA/ CAPSA/ Consultant Signature
chapter 2	4-4-25						
Remarks		Add PRL for BIS, remove subsections for PRL, use credible sources for references.					
Deadline:							
Deadline:							

CONSULTATION LOGS FORM

Thesis/ Project Title		Capstone Pothole Detection and Road Condition Mapping with Image Processing using YOLOv11 in Rinconada					
Proponents:		De las Llagas, Gian B.		Belano, Christine Kyla O.		Nanale, Krizia Belle L.	
Alias:		APATOSAURUS					
Total # of Modules				as approved by the CAPSA / TSA			
PROTOTYPE	DT of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Manager's Signature	TA/TSA/ CAPA/ CAPSA/ Consultant Signature
Deadline:							
	Remarks						
	Chapter 2						
	- include articles on evaluation of Yolo v11						
	- revise group						
Deadline:							
	Remarks						

CONSULTATION LOGS FORM

Thesis/ Project Title		Capstone Pothole Detection and Road Condition Mapping with Image Processing using YOLOv11 in Rinconada					
Proponents:		De las Llagas, Gian B.		Belano, Christine Kyla O.		Nanale, Krizia Belle L.	
Alias:		APATOSAURUS					
Total # of Modules						as approved by the CAPSA / TSA	
PROTOTYPE	DT of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Manager's Signature	TATSA/ CAPA/ CAPSA/ Consultant Signature
Chapter 3	1-11-23						<i>[Signature]</i>
Remarks		add log function in the evaluation, revise conceptual framework. add flask on the web framework, implement osm instead of gts add formula for every obligation metrics.					
Deadline:							
Remarks							
Deadline:							

CONSULTATION LOGS FORM

Thesis/ Project Title		Capstone Pothole Detection and Road Condition Mapping with Image Processing using YOLOv11 in Rinconada					
Proponents:		De Las Llagas, Gian B.		Belano, Christine Kyla O.		Nanale, Krizia Belle L.	
Alias:		APATOSAURUS					
Total # of Modules				as approved by the CAPSA / TSA			
PROTOTYPE	DT of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Manager's Signature	TA/TSA/ CAPA/ CAPSA/ Consultant Signature
Chapter 1 (Draft)	5-20-25						
Remarks		Remove discussion for specific iOS and Android remove discussion about Rinconada. Update the content of Chapter 3.					
Deadline:							
March 11, 2025							
Chapter 1 (Draft)							
Remarks							
Deadline:							
March 11, 2025							

CONSULTATION LOGS FORM

Thesis/ Project Title		Capstone Pothole Detection and Road Condition Reporting and Mapping with Image Processing using Yolov11					
Proponents:		De las Llagas, Gian B.		Belano, Christine Kyla O.		Nanale, Krizia Belle L.	
Alias:		APATOSAURUS					
Total # of Modules				as approved by the CAPSA / TSA			
PROTOTYPE	DT of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Manager's Signature	TA/TSA/ CAPA/ CAPSA/ Consultant Signature
	9/9/25						
Remarks		improve the model by enhancing the training process. additional dataset for improvement.					
Deadline:							
Remarks							
Deadline:							

CONSULTATION LOGS FORM

Thesis Title:	Pothole Detection and Road Condition Reporting and Mapping with Image Processing using YOLOv11						
Proponents:	De Las Llagas, Gian B.		Belano, Christine Kyla O.		Nanale, Krizia Belle L.		
Alias:	APATOSAURUS						
Total # of Modules:				as approved by the CAPSA /TSA:			
PROTOTYPE	Date of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Heads's Signature	TSA/TSA/Consultant/ Grammarians Signature
08/25 9/8 Remarks: make a recorded demo of the prototype that will use during presentation.							
Deadline:							

CONSULTATION LOGS FORM

Thesis Title:	Pothole Detection and Road Condition Reporting and Mapping with Image Processing using YOLOv11						
Proponents:	De Las Llagas, Gian B.		Belano, Christine Kyla O.		Nanale, Krizia Belle L.		
Alias:	APATOSAURUS						
Total # of Modules:				as approved by the CAPSA /TSA:			
PROTOTYPE	Date of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Heads's Signature	TA/TSA/Consultant/ Grammarian Signature
	10/9/25						<i>[Signature]</i>
	Remarks: Good to go! Good luck!!						
	Deadline:						

CONSULTATION LOGS FORM

Thesis Title:	Pothole Detection and Road Condition Reporting and Mapping with Image Processing using YOLOv11						
Proponents:	De Las Llagas, Gian B.		Belano, Christine Kyla O.		Nanale, Krizia Belle L.		
Alias:	APATOSAURUS						
Total # of Modules:				as approved by the CAPSA /TSA:			
PROTOTYPE	Date of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Head's Signature	TA/TSA/ Consultant/ Grammarian Signature
Remarks: check the comment / suggest on the manuscript minor changes good to go (FOD) are received.							
Deadline:							

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CONSULTATION LOGS FORM

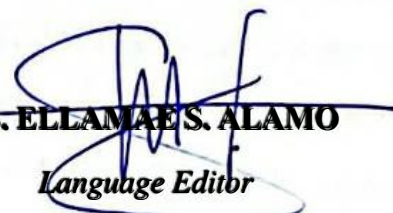
Thesis Title:	POTHOLE DETECTION AND ROAD CONDITION REPORTING AND MAPPING WITH IMAGE PROCESSING USING YOLOv11						
Proponents:	De las Llagas, Gian B. Belano, Christine Kyla O. Nanale, Krizia Belle L.						
Alias:	APATOSAURUS						
Total # of Modules:				as approved by the CAPSA /TSA:			
PROTOTYPE	Date of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Heads's Signature	TA/TSA/ Consultant/ Grammarian Signature
							
	Remarks: <i>Good to go!!</i>						
	Deadline:						

APPENDIX M

LANGUAGE EDITING CERTIFICATION

This is to certify that the undersigned has reviewed and went through all the pages of the
Bachelor of Science in Computer Science thesis manuscript titled
**"POTHOLE DETECTION AND ROAD CONDITION REPORTING AND
MAPPING WITH IMAGE PROCESSING USING YOLOv11"**

of **De las Llagas, Gian B., Belano, Christine Kyla O., Nanale, Krizia Belle L.**, as
against the set of structural rules that govern research writing in accord with the
composition of sentences, phrases, and words in the English language.



MS. ELLAMAE S. ALAMO
Language Editor

Date: January 12, 2024

APPENDIX N
SECRETARY'S CERTIFICATION

This is to certify that the undersigned has provided accurate recommendations,
suggestions, and comments unanimously agreed and approved by the panel of examiners
during the oral examination of the thesis titled

**"POTHOLE DETECTION AND ROAD CONDITION REPORTING AND
MAPPING WITH IMAGE PROCESSING USING YOLOv11"**

prepared and submitted by **De las Llagas, Gian B., Belano, Christine Kyla O., Nanale,
Krizia Belle L.**, and that the same have not been amended, modified or obliterated.


MS. MARRI GRACE MORATA

Secretary

Date: 01/23/2024

APPENDIX O

JOINT AFFIDAVIT OF UNDERTAKING

JOINT AFFIDAVIT OF UNDERTAKING

We, (1) GIAN B. DE LAS LLAGAS, of legal age, single/ married and a resident of SAN NICOLAS, IRIGA CITY, CAMARINES SUR, (2) CHRISTINE KYLA O. BELANO, of legal age, single/ married and a resident of MATAOROC, MINALABAC, CAMARINES SUR, (3) KRIZIA BELLE L. NANALE, of legal age, single/ married and a resident of SAGRADA, PILI, CAMARINES SUR, after having been sworn to in accordance with law, do hereby take oath and state:

- (i) That, we are officially enrolled for the thesis/capstone project on the topic titled POTHOLE DETECTION AND ROAD CONDITION REPORTING AND MAPPING WITH IMAGE PROCESSING USING YOLOV11 in the COLLEGE OF COMPUTER STUDIES of CAMARINES SUR POLYTECHNIC COLLEGES.
- (ii) That, the contents of our thesis/ capstone project submitted to the Camarines Sur Polytechnic Colleges, for award of BACHELOR OF SCIENCE IN COMPUTER SCIENCE Degree are original and our own work and is not plagiarized.
- (iii) That, if, after completing the thesis/ capstone project, are found copied or come under plagiarism, we will be solely responsible for it and College shall have sole right to cancel my research work ab-initio.
- (iv) That, this work has not been submitted for the award of any other Degree/Diploma in any other University/Institute.
- (v) That, we shall be responsible for any legal dispute/case(s) for violation of any provisions of the Copyright Act relating to our thesis/ capstone project.

IN WITNESS WHEREOF, I have hereunto set my name this 12th day of December, 2025 in Nabua, Camarines Sur, Philippines.

(1) GIAN B. DE LAS LLAGAS
Affiant

(2) CHRISTINE KYLA O. BELANO
Affiant

(3) KRIZIA BELLE L. NANALE
Affiant



SUBSCRIBED AND SWORN TO before me this 12th day of December, 2025 at Nabua, Camarines Sur, Philippines, affiants exhibiting to me their competent proofs of identity above stated.

Doc. No. 146 :
Page No.: 41 :
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Series of 202_


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APPENDIX P

CERTIFICATE OF UTILIZATION



Republic of the Philippines
Camarines Sur Polytechnic Colleges
Nabua, Camarines Sur

AI Research Center for Community Development

CERTIFICATION

This is to certify that the GPU server is utilized for training AI and machine learning models in the study titled "Pothole Detection and Road Condition Reporting and Mapping with Image Processing using YOLOv11" by Gian B. De las Llagas, Christine Kyla O. Belano, and Krizia Belle L. Nanale, Team APATOSAURUS, which has been reviewed and verified by the AIRCODE office.

This certification is issued on December 15, 2025 by the office of AIRCODE at Camarines Sur Polytechnic Colleges, Nabua, Camarines Sur.


JOSEPH JESSIE S. OÑATE, MSc
Head, AIRCODE

APPENDIX Q
CERTIFICATE OF TRANSFER



CERTIFICATE OF TRANSFER

This Certificate is to certify that

POTHOLE DETECTION AND ROAD CONDITION REPORTING AND MAPPING WITH IMAGE PROCESSING USING YOLOv11

was developed and completed by Gian B. De las Llagas, Christine Kyla Q. Belano,
and Krizia Belle L. Nanale, bona fide students of the College of Computer
Studies (CCS), Camarines Sur Polytechnic Colleges (CSPC), as part of the
requirements for their undergraduate program.

This certification is issued on the 5th day of December 2025 at the Office of
the Dean, College of Computer Studies, Camarines Sur Polytechnic Colleges,
Nabua, Camarines Sur.



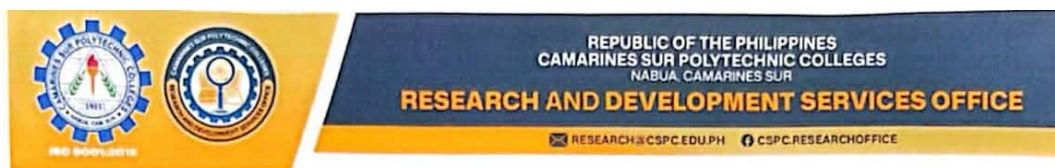
ROSEL O. ONESA, MIT

OIC, Dean, College of Computer Studies



APPENDIX R

CRD CERTIFICATE



CERTIFICATION

Date of Release: January 23, 2026
Submission ID: 23367:126763056
Output Title: POTHOLE DETECTION AND ROAD CONDITION REPORTING AND MAPPING WITH IMAGE PROCESSING USING YOLOV11
Author(s): Christine Kyla O. Belano
Gian B. De las Llagas
Krizia Belle L. Nanale
Program: Bachelor of Science in Computer Science

Authenticity Report:

Similarity Index Report: 7%

Internet Sources: 3%
Publications: 4%
Student Papers: 5%

- Interpretation: The similarity index report means that 7% of the output is similar to the sources in Turnitin's (authenticity software) online repository and databases.

AI-Generated Report: 24%

AI-generated only: 24%
AI-generated text that was AI-paraphrased: 0%

- Interpretation: The AI generated report means that 24% of the output indicates a *likely* AI-generated text and *likely* AI-paraphrased text, which is **within the acceptable AI usage** in the field of Engineering.


HAROLD JAN R. TERANO, Ph.D.
Director, Research & Development Services Office

APPENDIX S

ACM PAPER

POTHOLE DETECTION AND ROAD CONDITION REPORTING AND MAPPING WITH IMAGE PROCESSING USING YOLOV11

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Abstract

Road infrastructure faced significant challenges, particularly in maintaining road quality and ensuring the safety of motorists and pedestrians. Traditional manual inspection method of assessing the road condition, were time consuming and often inefficient particularly in rural areas where resources and manpower for infrastructure maintenance were limited. This study aimed to develop a real-time pothole detection and road condition mapping system using image processing and the YOLOv11 algorithm. A dataset of 856 road images manually annotated potholes classified as minor, medium, and major was used, with data augmentation techniques applied to improve model robustness. The YOLOv11 model was trained using a staged transfer learning approach and evaluated using mAP, precision, recall, and F1-score. The trained model was converted to TensorFlow Lite and deployed on a mobile platform, with GPS integration for road condition mapping. The results showed that the system achieved an mAP@0.5 of 39.99 and F1-score of 44.6. The successful implementation of the YOLOv11 algorithm emphasized the potential of deep learning models to achieve balance between accuracy and processing efficiency.

CCS Concepts

• Computing methodologies → Object detection.

Keywords

pothole detection, yolov11, road mapping, image processing

1 Introduction

Road infrastructure faced significant challenges, particularly in maintaining road quality and ensuring the safety of motorists and pedestrians. The World Health Organization (WHO) revealed that the leading cause of death globally are road traffic incidents [9]. The Philippine Statistic Authority (PSA) stated that in 2022 road traffic accidents caused up to 12,000 fatalities and one of the contributing factors for this was the poor road condition [7]. Aside from safety, this issue also affected the country's economy as the Department of Transportation estimates that road accidents cost the Philippine economy approximately 105 billion pesos annually [7]. Included to this are the healthcare costs, damaged property, and productivity loss. This proved that roads play a crucial role in growth of a country.

There are a combination of factors that caused these road defects, including heavy rainfall, poor drainage systems, and the constant passage of overloaded vehicles. The consequences of these road conditions were severe as potholes does not only caused discomfort to the drivers but led to damaged vehicles, higher maintenance costs

for public and private transportation but more importantly, it contributed to traffic accidents which raised a significant risk to public [2]. Despite the need for regular road maintenance, there were still challenges in addressing these issues effectively since traditional methods of assessing the road condition were time consuming and often inefficient particularly in rural areas where resources and manpower for infrastructure maintenance were limited. There was also a lack of real-time data on road conditions which made it difficult for the Department of Public Works and Highways (DPWH) to strategically prioritize and allocate resources for road repairs.

Considering the impact of poor road conditions on the public safety, the economy, and the infrastructure of the community, this study aimed to develop a real-time pothole detection that can enhanced driver safety and a road condition mapping system, which helped the authorities locate potholes quickly. By utilizing YOLOv11 for image processing and integrating GPS data, the system sought to provide a cost-effective solution to monitor road conditions, improve road safety, and assist local authorities to make a data-driven decision for road maintenance, contributing to better infrastructure and public safety.

This study's general objective is to develop a pothole detection system using YOLOv11 algorithm and generate a road condition map that marks the locations of the potholes identified by the system.

The study focused on developing a real-time pothole detection model using YOLOv11. It also aimed to evaluate the performance of the model using different evaluation metrics and integrate GPS to generate a road condition map.

2 Related Literature and Studies

Potholes were clear signs of neglected roads, revealing a deeper structural problem. When a vehicle hits a pothole, it not only disrupted the journey but also caused serious damage to your vehicle's wheels, tires, and suspension system, all at the cost of repairing it [6]. In line with this, the study by Bidve et al. [2024] stated that traditional methods of pothole detection were ineffective, and there was a need for a new solution [3]. The transformational solution proposed in this study integrated cutting-edge technologies specifically computer vision and machine learning, intending to enhance roads, increase safety, and streamline maintenance operations.

Moreover, as current techniques for identifying potholes typically rely on labor-intensive manual image processing and vibration-based sensors, which were often slow. To address this issue, Addanki and Lin [2023] proposed the use of recent advancements in deep learning particularly Convolutional Neural Network (CNN) leveraging Yolov8 for evaluation of advanced model detection. With the

focus on real-time performance and computational efficiency led to impressive results, its findings achieved a 91.1% mAP with an inference speed of 8.8 milliseconds per image, highlighting the effective transition from manual inspections to AI-powered real-time detection that marked a significant enhancement to road safety technology [1]. Consequently, Guerreiri et al. [2020], conducted a case study concerning several urban roads in Palermo, Italy, and the different distress types, extensions, and proliferation of such highways, and road areas crossed by railway lines. With the various Deep Learning techniques used, specifically the YOLOv3 to inspect the pavement distress system the model obtained a 98.7% for classifying distress on different types of pavement areas successfully implemented a real-time and efficient detection system [8].

Road mapping had become a critical component of transportation systems nowadays. Integrating the Global Positioning System (GPS) and Geographic Information System (GIS) enabled the development of an accurate and dynamic road network by combining real-time geospatial data with advanced spatial analysis. GPS provides precise geospatial data while GIS facilitates data visualization, pattern identification, and decision making for infrastructure management, which are used in several studies to address challenges such as signal obstruction in urban areas, and limited infrastructures in rural areas. Using technologies such as GPS and GIS for road mapping of potholes helped with having an accurate detection and timely repairs, which reduced the road accidents and helped the authorities enhance the road quality. Dwiandharu et. al [2024] developed a system that detected and mapped road damage using an accelerometer to record acceleration values and integrate GPS to visualize the mapping of road damage. In their study, GPS was evaluated by visualizing the reading results with an application and comparing the actual location with the position shown by the GPS based on the given data. Furthermore, their study revealed that the average distance between the actual hole position with GPS coordinates reached 5.7 meters [5].

The gathered related literature and studies provided a comprehensive foundation that highlighted the methods in developing a pothole detection system with integration of GPS for road condition mapping. The clear transition from traditional, labor intensive pothole detection methods to advanced AI driven solutions highlighted the need for real-time and automated systems. Previous studies noticed the inefficiency and inconsistency of manual inspection especially in urban road networks while emphasizing the advantage of utilizing deep learning models such as YOLO variants in acquiring accurate and real-time detection. This study was based on and improves the past works by adopting to the advancements of the latest version of YOLO algorithm, the YOLOv11, which incorporated features like anchor free detection and spatial attention mechanism to improve performance even under challenging environmental conditions. Moreover, the past studies about mapping have underscored the importance and benefits of integrating GPS technology for accurate and reliable geospatial mapping. These studies highlighted the critical need for real-time road condition maps, combining the GPS with image processing, which not only enhanced travel efficiency but also significantly improved safety for drivers. By addressing these challenges, this study introduced an innovative pothole detection system alongside a road condition mapping solution, providing a practical tool for underserved areas.

Gian B. De las Llagas, Christine Kyla O. Belano, and Krizia Belle L. Nanale

3 Methodology

3.1 Theorems, Algorithm, and Mathematical Framework

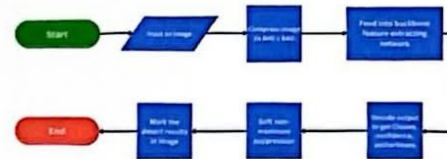


Figure 1: YOLO Inference Flow Chart [9]

The YOLOv11 model was based under the principle of single-pass object detection, where recognition of the objects is treated as a regression problem that predicted the bounding box coordinates and class probabilities directly from full images. The algorithmic implementation, as shown in Figure 1, was explained below through the following comprehensive steps: The Image Input and Preprocessing where unprocessed images or video frames were resized to a fixed input size, Backbone Network – C3k2 and CSP Modules extracts rich hierarchical features from the input image, Neck Layer – SPPF and C2f Blocks is a layer that collated and adjusted features from various scales, Attention Mechanism – C2PSA (Parallel Spatial Attention) implements a visual attention that directs the model's resources to the informative areas of the image, and lastly Detection Head – EfficientHead a layer that predicts the class labels, bounding box dimensions, and objectness confidence scores of the predicted object.

Dataset. This study utilized an annotated image dataset specifically designed for pothole detection to train the YOLOv11 model for real-time pothole identification. The dataset encompasses annotated potholes labeled with three categories namely minor pothole, medium pothole, and major pothole. These labels are used solely to improve the model's learning during training to ensure that it can recognize potholes of varying shapes and sizes. The data was split into training, validation, and testing sets, ensuring proper training and evaluation of the model.

3.2 Evaluation Methods

To evaluate the performance of the system, several performance metrics were used to assess the speed, accuracy, and efficiency of the system. The metrics include Mean Average Precision (mAP) for assessing the system's accuracy, F1-score to evaluate the detection quality, Latency evaluation, which showed how quickly the model processes every frame, and the Accuracy for the system's overall effectiveness. In addition, Confusion Matrix was used to analyze the prediction of the model by identifying the false positives and negatives as well as the true positives.

Mean Average Precision (mAP). The primary metric to evaluate the model. It measured the model's accuracy based on how well it can detect potholes across different confidence levels. A higher AP value indicated a better performing detection model [4].

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$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (1)$$

F1-score This metric was defined as the "harmonic mean of Precision and Recall" [4] as it assessed the balance between the correctly detected potholes and missed potholes. The combination of these two provided a comprehensive evaluation of the model's effectiveness, especially if there is an imbalance.

$$F1 \text{ Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \quad (2)$$

Latency Evaluation The system's real-time performance through a frame processing latency, measured how long it took to process each video frame. This metric was crucial as it evaluated the YOLOv11 model's inference speed by measuring the time it takes from frame processing each frame to showing the final mapped result. The system's speed was assessed in various conditions such as day/night and on different types of roads, to ensure consistent real-time and reliable operation.

Latency(Single Image) in Milliseconds

$$\text{Latency}_i (\text{ms}) = (t_{\text{end}} - t_{\text{start}}) \times 1000 \quad (3)$$

Average Latency

$$\text{Average Latency} = \frac{1}{N} \sum_{i=1}^N \text{Latency}_i \quad (4)$$

Confusion Matrix. The model's output was classified as true positive, false positive, and false negative to evaluate and visualize the detection performance of the system. This metric provided a detailed description of how well the model detected potholes while minimizing the incorrect detection [4].

3.3 Theoretical Framework

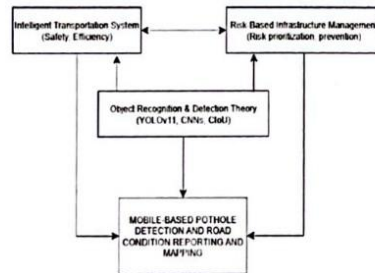


Figure 2: Theoretical Paradigm

The relationship among the three theories is interdependent, with multiple flows of influence converging into the system, as illustrated in Figure 2. First, Object Recognition and Detection Learning provides the core capability of detecting potholes in real time. With the YOLOv11 algorithm, not only does it merely detect the existence of potholes but also generates bounding box data, and

confidence scores. Of which contains characteristics such as the pothole image, class label, size, and frequency.

Second, the results of Object Recognition are also linked to the Intelligent Transportation System (ITS) Theory. ITS emphasizes the integration of technological innovations for road monitoring, safety, and efficiency. The detections generated by the system in real time, coupled with its GPS tagging, enable continuous mapping of road conditions, thereby aligning with ITS objectives of enhancing transportation safety and infrastructure management.

Third, ITS and Risk-Based Infrastructure Management (RBIM) exist in a feedback relationship. ITS ensures that road map data is collected and shared efficiently and systematically, while RBIM uses this data to determine which potholes require urgency based on severity and risk to road users. In return, risk-based prioritization can inform how ITS platforms should flag or highlight high-risk areas for local authorities and drivers.

4 Results and Discussion

4.1 Data Collection

The dataset used in this study is a combination of images retrieved from the Kaggle repository containing annotated images, categorized into minor, medium, and major potholes and manually captured images to add more localized and realistic scenarios, enhancing dataset diversity. After capturing the pothole images, each sample was manually annotated to identify and label the potholes present in each image.

Table 1: Annotated Pothole Count

Class	Training	Validation	Testing	Total
Minor	438	330	42	810
Medium	482	253	60	795
Major	438	192	45	810
Total	1,358	775	147	2,280

The model was trained on a dataset of 856 valid images, which included 2,280 annotated pothole instances. The dataset was split into training (1,358 annotations), validation (775 annotations), and testing (147 annotations) sets. Two sequential training runs were performed: the first with 100 epochs without data augmentation. This division ensures that the model is assessed on unseen data, providing an accurate measurement of its real-world performance in Table 1.

A raw image of a road surface containing multiple potholes with varying sizes and shapes are manually annotated using Roboflow annotation tool with boxes assigned with specific classes namely minor, medium, and major with different colors representing these classes.

Although manual annotation improves the model's accuracy, the process of performing this is time consuming and may still involve human inconsistencies, particularly if pothole boundaries are partially obstructed. This ensures consistencies across the dataset, correcting mislabeled and missing annotations from the existing

sources. Additionally, environmental conditions like water filled potholes or shadows may also cause uncertainties in placing bounding boxes.

4.2 Model Training

To effectively tailor the model for domain-specific pothole detection, a progressive transfer learning strategy was implemented. The process began with head-only fine-tuning, in which the backbone was frozen to preserve pretrained feature representations while allowing the detection head to adapt to the pothole dataset. In following phases, the network's increasingly deeper layers were unfrozen, which allowed controlled feature adaptation and at the same time reduced the risk of catastrophic forgetting. The learning rate was correspondingly reduced gradually at each stage to provide stability in weight updates as more parameters were trained.

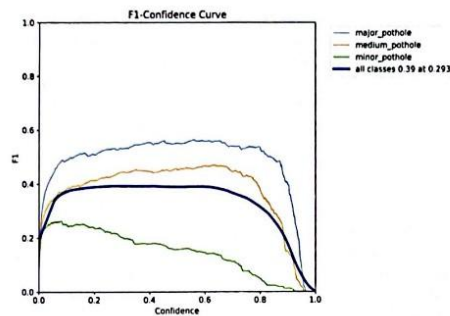


Figure 3: F1-score vs. Confidence Threshold Curve

The evaluation of the model illustrated on Figure 3 revealed an optimal F1-score of approximately 0.327 across confidence thresholds from 0.0 (0%) to 1.0 (100%). Lower thresholds yielded high recall but increased false positives, while higher thresholds reduced false positives at the cost of missed detections. Major potholes were consistently detected even at 0.1 to 0.7 confidence threshold. Medium potholes thrived at around 0.4 threshold, but minor potholes showed low F1 scores peaking at 0.2 confidence threshold. A 0.327 confidence threshold is recommended to maximize detection performance.

Overall, the model development and training phase confirmed that YOLOv11 is a suitable algorithm for real-time pothole detection. This development clears the path for its integration into a comprehensive road condition mapping platform, resulting in safer and more efficient navigation. YOLOv11's adaptability allows for potential modifications to changing environmental conditions, enhancing its reliability in various kinds of urban and rural locations. Furthermore, with steady performance improvement and reliable convergence, it established a solid foundation for evaluating the system's overall detection performance.

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Table 2: Model Performance Metrics

Metric	Result
mAP@0.5	0.399 (39.99%)
mAP@0.5:0.95	0.236 (23.6%)
Precision	0.42 (42.0%)
Recall	0.477 (47.7%)
F1-score	0.446 (44.6%)
Average Speed	2.5 ms/image (0.1 ms preprocess, 1.3 ms inference, 1.1 ms postprocess)

4.3 Model Evaluation

The overall mAP@0.5 for the model reached 39.99%, meanwhile mAP@0.5:0.95 was 23.6%, meaning that detection performance is somewhere between poor and moderate for all severity classes. Precision was 42.0%, recall 47.7%, resulting in an F1-score of 44.6%. The model demonstrated fast performance with an average processing time of 2.5 ms per image, including 0.1 ms for preprocessing, 1.3 ms for inference, and 1.1 ms for postprocessing on an NVIDIA RTX 5000 GPU at a 640×640 input size. Despite the model's moderate performance, the results presented in 2 remain acceptable for a preliminary implementation of a real-time pothole detection system.

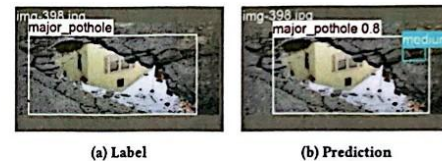


Figure 4: Validation Sample

Figure 4 demonstrated a more clear representation of the model's high speed and accuracy implying that the YOLOv11 possesses a balanced trade-off between detection accuracy and computational efficiency, making it an ideal foundation for future experimentation in model optimization and comparative studies. The observed class imbalance and variability in detection performance opens a new opportunity for developing adaptive learning techniques or hybrid models that integrate contextual information.

The performance of YOLOv11 validated its reliability for real-time pothole detection and classification. The result confirmed the model's effectiveness in recognizing significant road defects while also maintaining high processing speed. Although further improvements are necessary to detect smaller potholes better, the study successfully established a robust foundation for integrating automated detection technology into road condition reporting and mapping systems.

POTHOLE DETECTION AND ROAD CONDITION REPORTING AND MAPPING WITH IMAGE PROCESSING USING YOLOV11

5 Summary of Findings, Conclusion, and Recommendations

5.1 Summary

The study "Pothole Detection and Road Condition Reporting and Mapping with Image Processing Using YOLOv11" is designed to develop an automated system that detects, classifies, and maps the potholes based on severity using deep learning and image processing. This study is conducted to address the lack of an efficient road damage monitoring system by introducing a computer vision-based solution that is capable of classifying minor, medium, and major potholes in real-time. To tackle this, the researchers developed an automated, real-time pothole detection system using the YOLOv11 deep learning model served as the primary detection architecture. To improve the training performance, the dataset underwent annotation, preprocessing, and augmentation, the model was then evaluated using the standard metrics, which are mAP, precision, recall, F1-score, and latency. Its Results revealed that the YOLOv11 model achieved a mean average precision (mAP) of 39.99%, with inference latency of 2.5 ms per image, signifying real-time detection capability. The model was observed to performed best in identifying major potholes however showed reduced accuracy for minor ones. These demonstrates the model's ability to effectively serve as the core detection engine of an automated road monitoring system

5.2 Findings

The development phase successfully produced a working YOLOv11-based detection model trained on 865 images with 2,280 labeled pothole instances, demonstrating consistent learning progress across training epochs with smooth loss convergence and steady metric improvement. Data augmentation significantly enhanced robustness and generalization, enabling recognition of diverse pothole shapes and surface textures while effectively differentiating severity categories. Performance evaluation revealed mAP@0.5 of 39.99%, precision of 42.0%, recall of 47.7%, F1-score of 44.6%, and average inference speed of 2.5ms per image, validating real-time capability with major potholes achieving highest accuracy followed by medium, while minor potholes showed lowest performance, with moderate precision and strong computational efficiency. GPS integration successfully captured coordinates for each detection and marked potholes on digital maps, transforming the prototype into a functional road monitoring tool that provides both detection results and spatial data for efficient condition tracking and assessment.

5.3 Conclusion

The study successfully developed and trained a pothole detection model capable of identifying and classifying road surface damage with acceptable performance, confirming YOLOv11's suitability for road monitoring tasks and demonstrating that deep learning combined with image processing effectively automates detection while reducing manual inspection reliance. Data augmentation improved robustness across pothole variations, establishing YOLOv11 as an appropriate approach for pothole detection and road

condition monitoring. The model achieved study objectives by accurately detecting and classifying potholes with map visualization and 2.5ms real-time inference, showing average detection performance but high computational efficiency; despite limited minor pothole accuracy due to dataset imbalance and visual complexity, medium and major pothole reliability proves system effectiveness. GPS integration effectively transformed the prototype into a working road monitoring system, enabling clear defect visualization and enhanced maintenance planning through automated coordinate mapping.

5.4 Recommendations

Future work should prioritize improving minor pothole detection by enriching datasets with small, shallow potholes from varied angles and applying advanced augmentation techniques like blur and multi-scale transformations. Additionally, training across rainy, low-light, and wet surface conditions, along with preprocessing filters for glare and shadows, will enhance environmental robustness. Moreover, hyperparameter fine-tuning and model quantization for mobile deployment will boost precision and inference speed on low-end devices. Finally, deploying on hardware with AI accelerators like Google Pixel TPUs or NVIDIA Jetson Nano, combined with heatmaps and GIS integration, will maximize practical utility for road maintenance planning.

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